V1.8

v1.8

ClassTechStar

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:’ “ ” 357

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. ?) (

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. , “ ” “ ” .
, , . ,
“ ”, - , .

. “ ”

, , :

-] [: App, ,[ ] .[ ]
- <: , ” [ ] < ,”[ ] , .
- : ! , .
- : , .

. ,iOS-

, . : ( ) App- (iOS),) ,(
( ) (iOS). , .
, Huawei, , . ,
, .

. : 128G ?110G-

: 1TB = 1024GB, 1GB = 1024MB, 1MB = 1024KB... :
,1000 ,1024 “128G” “ ” .

:

1. : , . 10-20GB, .

2. : App- , .

.3 : , .

128GB , 256GB . , 512GB- . , , .

: ” ”

,[ ]) ,(: , , . .
, .

“ ”
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, : , .
, , ’ , , App , , , ... ,
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, , . :

- : , .
- : , .
- : + + , ” ” .

: “ , ”

, “ ” . ,
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- : App , , - ” ”
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- : , , -
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- : , , ” ” -
.

:1 ,

- . , .
, “ ” .

. “ ” :

, . , .

SoC (System on Chip - :(“ + ”

SoC, System on Chip, : “ ” . CPU) ,(
CPU, GPU) ,(NPU) ,(ISP) ,( ) (- .
, , - , .

CPU:

CPU- ,SoC- . ARM,
x86 . ARM (RISC), .
CPU : “ ”)
big.LITTLE ARM), :

- - / : , . , , .
(3GHz+), .
- : , . , ,
- .
- / : , . , , - .

: - ,
, . .
:

- **Qualcomm Snapdragon 8 Elite**: CPU Oryon) Nuvia), 2
- (4.32GHz) + 6 (3.53GHz), 3nm- TSMC.

ARM, . GPU Adreno 830 ,
40% , NPU Hexagon 45- TOPS, 45% AI. :)
Samsung Galaxy , - (.4.47GHz-

- **MediaTek Dimensity 9400**: - (Cortex-X925, 3.62GHz) + 3 (Cortex-X4, 3.3GHz) + 4 (Cortex-A720, 2.4GHz), , - - 3nm TSMC. GPU Immortalis-G925 (12 ,(41% , 40% . AI NPU 890, , LoRA .
- **Apple A18 Pro**: 2 (4.04GHz-) + 4 ,(2.4GHz-) 3nm TSMC. GPU 6 , NPU 16 35-) TOPS) AI- Apple- Intelligence. Apple , - AI.
- **Huawei Kirin 9020**: Mate 70, CPU Taishan, NPU ISP- , 5.5G- , .

GPU:

GPU (Graphics Processing Unit) .
CPU) ,(, - .
:

1. : . -
.2 : - . -
.3 : , .
.4 : .

GPU : ! , , , , AR –
.GPU- GPU , – .

GPU : Qualcomm Adreno- GPU ; MediaTek ARM- Mali GPU; Apple GPU- , .

NPU: AI

NPU (Neural Processing Unit) . CPU- ,GPU- NPU
AI.
: NPU –
. NPU, , , ,CPU-
.

AI:

1. : , , .’

.2 : .

.3 : NPU , .

.4 : AI- , , , .

:

- : , , HDR, AI.
- : , , .
- : AI, .
- : (OCR), , .
- : AI) 7B (, AI , , . Hexagon NPU Snapdragon- 8 Elite 45- TOPS, NPU- 890 Dimensity 9400) LLM ,(80%- NPU- 16 Apple A18 Pro 35- TOPS.
- **AI**: AI) ,(AI) ,(AI, .

ISP:

ISP , .

? ! Apple A18 Pro, Qualcomm Snapdragon 8 Elite, MediaTek Dimensity 9400, Huawei Kirin 9020 – . –Snapdragon 8s Gen3/7+ Gen3, Dimensity 8400. – ( ) . 3nm , - 5nm .4nm-

RAM:

RAM . , . - 12GB , 16GB , 24GB- – .

: LPDDR5 .LPDDR4X- Apple RAM, , “ ” .

RAM? , .

WeChat- , .

(ROM / :(

– , . UFS 3.1 UFS 4.0, eMMC- .

? , . ,

.

:

, .

- : OLED – , ,) , ,(; LCD – , , .OLED-
- : 1080P , 2K . ” ” Apple : , .
- : 60Hz , 90Hz/120Hz , . , (LTPO), .
- : , – , .
- : DC PWM, . PWM , .

HDR

HDR (High Dynamic Range) - , .

- : (SDR), . HDR) 1000 ,(, , .
- : HDR10) , ,(HDR10+) Samsung, - ,(Dolby Vision) .(.
- : HDR, , , . ,HDR- HDR.

HDR

HDR HDR- . HDR “ , ” .

- : , -) , , ,(.

- : -) (/ .
- : HDR”- ” – HDR, . Smart HDR Apple HDR- Android .

:

“ ” , - . -5500 7500mAh.

- : Android- 100W- 200W, ; Apple .20-30W- ? , .
- : , . , .
- : , , .

:

, :

- : , , .
- : , , .
- : , .
- : .

!) – ,((f/1.8 ,(f/2.4- (OIS – (– .) AI) , .

, . 90 , , 5 10 .

- : 90- .
  .
- : (5x-10x), .
  , .
- : , , – .

OIS) (EIS) (

, :

- **OIS**: . ,
. – , – , .
- **EIS) (:** .
, . – , –) ,(
.
- : – OIS, – .
”Sensor-Shift”- – , .

- : , .
) 2x, 3x, 5x, 10x), .
- : , + . , .
100x , .
- : , - .
, .

:

:

- : , , . . -
- ( ): , . ,
, . .
- : ” ,” . ,
.

:

, :

- : 30fps- 60fps- 120fps , .
, GPU- .SoC-
- : AI, .
- : , SoC- ,
, .

GPS:

GPS .

- **GPS** : L1, 3-5- . , ,
  .
- **GPS** : - –L1 .L5- L5 - ,
  , 1-3- . .
- : ,GPS- BeiDou- ,( ) GLONASS ,( )
  Galileo ( ) . - , .

:

, :

- \+ : , , .
- : .
- : , – .
- : . -
- : Face ID Apple , - ; Android , -
  .
- ” ” , , . : , ,
  , .
- : , .
- : .

. : Android iOS HarmonyOS

, . :

- **HarmonyOS**: , , . ,
  , . , , .
- **:** : , Apple , . ,
  , – Apple.
- **iOS**: Huawei, , . .

, . , .

•

, :

Micro USB

- : ,) – .(,
.
- : USB 2.0, 480Mbps, 10W.
- : , , , .

Lightning) Apple, (

- : 8 Apple, . iPhone- 14 ,
.
- : Lightning USB 2.0, 480Mbps. USB-C ,Lightning- Lightning- .
- : 2024- USB-C , Apple- iPhone 15 .Type-C- Lightning .
- : MFi Apple , .
iPhone- , Lightning.

USB Type-C

- : 24 , . .
2024- , , USB-C ,
.
- : Type-C , . USB- 2.0 (480Mbps), USB 3.0/3.1/3.2 (5-20Gbps) USB4 (40Gbps).
- : Type-C USB- 3.0! Type-C, USB 2.0, . iPhone 16/17 USB 2.0, Pro USB- 3.0) USB-C to USB-C).
- : USB PD, 240W .
- : DisplayPort- Alt Mode, .

3.5mm

- : 3 4 , 3) ,(4 .
- : , , - . , .
- : ,3.5mm- Type-C Bluetooth. .

•

, .

:

- :iPhone) (/ iPhone Plus) (/ iPhone Pro) (/ iPhone Pro Max) (/ iPhone SE) (/ iPhone 16e) AI , A18 C1 .(2026- iPhone 17 A19 Pro, AI .
- **OnePlus**: Mate) (/ P/Pura) (/ Mate X) (/ nova) (/ Changxiang ( )
- **Huawei**: Xiaomi ( ) / Xiaomi MIX) (/ / Redmi K) (/ Redmi Note ( ) / Redmi ( )
- **OPPO**: Find X ( ) / Reno ( ) / A ( ) / OnePlus) (
- **vivo**: X) (/ iQOO) (/ S) (/ Y ( )
- **Apple**: Galaxy S ( ) / Galaxy Z Fold/Flip) (/ Galaxy A ( )

:

- **Pro**: Professional, . , , .
- **Max**: . , , .Pro-
- **Ultra**: . , . Samsung S24 Ultra, Xiaomi 14 Ultra.
- **SE**: Special Edition , . , . iPhone SE.
- **Plus**: . , , .
- **Lite/** : . , .
- **Note**: , . Redmi Note .
- **Neo**: . .

:

.

:

, ,
. , . Qualcomm ,
MediaTek- Samsung- . Apple C1 iPhone- 16e,
Apple , C1 4nm ; iPhone 16/16 Plus/16
Pro/16 Pro Max Qualcomm (Snapdragon X75 5G), .
iPhone XS/11 Intel, , .

:

, . :

- **MIMO**: Multiple Input Multiple Output, - ,
. 4×4 MIMO 2×2- MIMO.
- : ,) Wi-Fi, Bluetooth, NFC, GPS - .(
- : ,) .(
.

:

- **5G SA NSA**: 1Gbps/ 500Mbps (Cat.18), -50
150Mbps. , .
- **5G NR**: 10Gbps+. Sub-6GHz , mmWave) (
.
- **5G-A (5G Advanced)** : 5G, 5.5G. 10Gbps, 1Gbps,
. 2024- , 5G-A .
Huawei Mate 70/80, Xiaomi 15/16 .5G-A- 5G-A - ”
,” ,) (.

:

:

- **FDD-LTE Band) B)**: B1, B3, B5 ,’ 4G. ,
.B1/B3/B5/B8/B34/B38/B39/B40/B41-

- **TDD-LTE Band**: 4G , .B38/B39/B40/B41-
- **5G NR Band) N)**: n1, n3, n28, n41, n78, n79 .' n78 (3.5GHz) 5G , n41 (2.5GHz) China Mobile.
- : , " .
, . "

- ?

:

.1 : , SMS, WeChat, ... "

.2 : , , ...

.3 : , , , ...

.4 : , , , ...

.5 : AAA...

6. : , , , ...

.7 : , , , ...

.8 : , , , ...

.9 : , , ...

.10 : , , , ...

, ! , .

:

,[ ]- , (RAM) (ROM).
. Android "AnTuTu" "DevCheck" .

: Android

, *"12GB+8GB "* - *8GB* .
? *?* " ".

“ ” Android) , , RAM Plus (’ ,
: (ROM/ UFS),
(RAM).
: RAM “ ”, “ ”. ,
, . ,
.

:

zRAM

,RAM- , .2:1 ,RAM-
, , CPU .

swap

UFS , . , /
.
“12GB+8GB ” 12GB RAM + 8GB .
zRAM- , zRAM , ,
.

. , UFS 4.0 ,4200MB/s-
1/10–1/15 LPDDR5X . zRAM RAM- ,
, .

, :

- : 6GB RAM + UFS 3.1 , ≥30GB, 3GB,
,3-4- ,68%- , ’ - .
- : WeChat, , ,’
.
- : , 4K .

- : 8GB + UFS 2.2 , ,
, .
- : 16GB - , , 12%-
.℃2-3- RAM 12GB, .

: Xiaomi “ ”, Huawei “ ”, Samsung “RAM Plus”, vivo “Extended RAM”, OnePlus “Memory Extension”.
, Android.
:][ ] [- [ ] [ ] “ ”)
2GB–7GB), . “8GB+4GB”.

- , 20-30GB, .
- , / UFS ,
swap- .
-) 4GB ,(,
.

,
, , “ ”. ,
, , .

:

.1 ,[ ] ” ,” ” ,” ”RAM Plus” ”Extended RAM”.

2. , , .

.3 : ? ? ?

.4 , , .

:1 HarmonyOS

HarmonyOS Huawei, 2019- . “
Android” , . .

HarmonyOS? ?

HarmonyOS “ ”. , HarmonyOS
.

HarmonyOS. , “ - ” . , ,
, , - , “ ” , .
: , , ; ,HarmonyOS- “ ”
, . , .

HarmonyOS 4.3: AOSP

HarmonyOS 4.3 Huawei . HarmonyOS- 4.3
Android Open Source Project (AOSP), Android -
Huawei- AppGallery Android, HarmonyOS. HarmonyOS
4.3 ,Android- AOSP Huawei ,
Ark .’

:

- **Android**: AOSP, HarmonyOS 4.3 Android,
.
- **Ark**: Huawei, . ,
.
- **-** : , HarmonyOS , .
, - .
- : ’ iOS, . ,
.
- : , .
- : , .

:

- : , .
- : .
- **-** : , , , , .
.
- : , , , , .

HarmonyOS NEXT (HarmonyOS 6): Huawei

HarmonyOS NEXT (HarmonyOS NEXT 5.0) HarmonyOS- 4.3 - -
HarmonyOS Huawei, Linux AOSP,
Android. HarmonyOS “ Android” “ ” .

:

- : HarmonyOS NEXT Huawei, AOSP.
, , .
- **Android**: . HarmonyOS NEXT HarmonyOS
, APK. HarmonyOS .
- : Huawei HarmonyOS . ,
WeChat, Alipay, Taobao ' HarmonyOS ,
.
- **Star Shield**: , . ,
" " - , .
- **AI**: AI , AI. AI , AI
.
- : . ,
.
- : . QR ,
.

HarmonyOS- NEXT

HarmonyOS- NEXT, Android. : (1)
HarmonyOS ; (2) Android.
HarmonyOS- 4.3 .

:

- : / , , .
- : , , .
- **Changlian**: Huawei, , , .

HarmonyOS:

HarmonyOS , .

:

- : , .
- : .
- ( ): .

- : .
- : HarmonyOS, .
- : , , .’

:

HarmonyOS , . Huawei
.HarmonyOS- ,2026 WeChat, Alipay, Taobao, Douyin, Meituan, JD
,HarmonyOS- .
, .

HarmonyOS-

- **-** : , ,” - ” , .
- **-** : , / / ” ,” , .
- : , .
- : , , .
- : < , , .’

HarmonyOS , . Har-
monyOS 4.3 .6-
.

: HarmonyOS

.1 , ,” - ” HarmonyOS .

.2 ” :” - , .

.3 , .

.4 < , .

:2 —

, . ,
. , .

: -

.1 : , , . , .

.2 **SD**) (: , , . :
SD.

3. : + . , . iCloud, Baidu Netdisk, OneDrive
.

“ ”

[ ]] ,[iOS- [ ] . “ ”
.

:

- **DCIM**: .
- **Pictures**: , , .
- **Download**: .
- **Music**: .
- **Movies**: .
- **Documents**: .
- **Android**: , , .

:

Linux, . ,
.

:

- **/system**: , . , root.
- **/data**: , . .
- **/sdcard /storage/emulated/0**: , .

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- : , . .
- : , , . .

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.1 : , , . iCloud, Huawei Cloud, Xiaomi Cloud .

.2 : , . , , .

.3 : + , .

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- : , .
- : .
- ’ **WeChat**: WeChat.
- : .

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- : ” ” ” ,” ,
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- : , . ,
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- : , , - .

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- : , . <.

- : , ” ” .

- : , , .

- : , .

- **WeChat**: WeChat , ’ .

, :

JPG/JPEG		, , . .	/
PNG		, . . .JPG-	/
MP3		, , . .	
FLAC		, ,CD- -3 5 .MP3- .	/
WAV		, . .	
M4A		AAC, MP3- . .	Apple Music /
MP4		, . .	/
MKV		, . .	VLC / nPlayer
DOC/DOCX		Word. DOCX , . DOC .	WPS / Office
PPT/PPTX		PowerPoint. PPTX .	WPS / Office
XLS/XLSX		Excel. XLSX .	WPS / Office
PDF		, . , .	Adobe Reader / WPS
ZIP		, .	/
RAR		,ZIP- .	
ISO		, . .	/
HTML		, . .	

: ” ”

< “ ” . ? , ,)
“ ” , “ ” .(, .

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“ ” , .
, , , , .

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- ” + ” : , . 202606 _
,2025 _ ,2024 _ . .
- - :
– :) / / / (
– :) / / (
– :) ”2026Q2 _ A”)
- :) 1- ,(:” ”
, , . 12 .

:

.1 2026Q2A _;

2. ,” ” , ;

.3 2025, 2026 .

“ ” __ .

- **72** : WeChat QQ- ,) , ’ (
– . ” ” [WeChat]- > [ ] < [ ] < ,[ ]
72- . , .
- : [ ] < [ ] <] ;[iOS [ ] < [ ] <] iPhone].
” / / / .” : (1) ; (2)
) ;((3)) .(

- : : , , , . ,
” < ’ ,” .” ” ,
” ” .

“ ” — , .
“ ” :

/

1. :) Meituxiuxiu, Snapseed)
, , , ;
.2 - : , ” ” ” ”; , ” ” .
, GPS- ;
.3 :( ) ” XX ” , .

(EXIF)) ; 6 “ ” :(

- : iOS [ ] <] [< [ ] — ” ;” ” ”
EXIF- .
- : ” ” ” ” — .
- : ExifTool ,( ) Metapho (iOS), Photo Metadata Remover ( )
- .

) , , , ,(:

- iPhone: [ ] < <) Face ID);
- Huawei/Xiaomi: [ ] < < ” ;”
- : **360** ,**Tencent Gallery Manager**) (,**KeepSafe** —
.

3: “ ” ——

, “ ” . , , –
.

—— “ ” “ ” Wi-Fi

. “ ” ,

” “

Wi-Fi. , ( ) . **.1** **:**
Wi-Fi , , .
, “ ” , . ,
) (4G/5G , ,
.

:

Wi-Fi -) ,(, ,
, . , , ,
. **:** [ ] <] [<]
[, . , .

Wi-Fi: “ ”

, Wi-Fi- . , Wi-Fi-
“ ” . : HTTPS ,

) ,(

HTTP

; WeChat, ,HTTPS- ——
SSL-stripping . , ,

. , , , Wi-Fi-)
“Free_WiFi_2024”). , . **.4 : “ ”**
? Huawei, ,
,) ,(-
WLAN- , ; , -
, . -

iPhone : Wi-Fi- , ,Wi-Fi- –
, . “ ” “4G” ,
.

. (4G/5G) **. 2G- 5G,**
, . :
4G LTE

. (4G/5G)

5G

:) 10Gbps), , . HD.

- **5G SA NSA**:SA , NSA 4G. SA ,
.
-
- **: . , .**

: “ ” , HD .Wi-Fi-

- ≠ : , .
- **:] [! ”** ” ! ”
, SIM / .
- **. Wi-Fi:** , Wi-Fi , . Wi-Fi-
, .

:Wi-Fi-
[ ]- < [WLAN] . QR , .

. Wi-Fi: ,

iOS . QR, iOS .

Wi-Fi:

Wi-Fi IEEE, :

Wi-Fi 4	802.11n	2.4/5GHz	600Mbps	,
Wi-Fi 5	802.11ac	5GHz	3.5Gbps	,
Wi-Fi 6	802.11ax	2.4/5GHz	9.6Gbps	,
Wi-Fi 6E	802.11ax	2.4/5/6GHz	9.6Gbps	6GHz,
Wi-Fi 7	802.11be	2.4/5/6GHz	46Gbps	,

- **Wi-Fi 6**:OFDMA) (, - MU-MIMO) , (, TWT) (.
- **Wi-Fi 6E** : 6GHz Wi-Fi 6, . .
- **Wi-Fi 7** : MLO (Multi-Link Operation), , . 320MHz . Snapdragon 8 Elite Gen5 Dimensity- 9400 Wi-Fi- 7. Xiaomi 17, Huawei Mate 80, Samsung S26 Wi-Fi 7) Wi-Fi 7).
- **2.4GHz 5GHz**:2.4GHz ; 5GHz . 6GHz Wi-Fi 6E/7 , , .

Wi-Fi :

:Wi-Fi . :

- .
- VPN .
- Wi-Fi- .
- HTTPS) .(
- . Bluetooth: -
- Bluetooth —— , , , , .
- :

- Bluetooth ISM 2.4GHz, (FHSS) 79 , . Bluetooth Class: 1 100- , Class 2 10-) ,(Class 3 1- .

. Bluetooth: -

Bluetooth 4.0

:

Bluetooth 4.2

: , **IPv6.**

- **Bluetooth 5.0** : (2Mbps), 4) 300 ,(.8) .(-
- **Bluetooth 5.1** : (AoA/AoD), .
- **Bluetooth 5.2** : LE Audio , LC3) , ,(, - .
- **Bluetooth 5.3/5.4** : , .
- **LE Audio** : 2024-2025, LE Audio . True Wireless) Huawei FreeBuds Pro 4, Xiaomi Buds 5 Pro, Samsung Galaxy Buds3 (' LE- Audio LC3, . , LE- Audio.
- :

: , . **LE Audio Bluetooth 5.2** . -2025- : 2026, LE Audio .) Huawei FreeBuds Pro 5, Xiaomi Buds 6 Pro, Samsung Galaxy Buds3 Pro) LE- Audio LC3, . , LE- Audio.

: , - .

- : , , , , .
- :Beacons Bluetooth) Apple AirTag).
- : Bluetooth Mesh , .'
- . : - - , .) Xiaomi (Huawei- - , , , , .

- - -) 38kHz), - . , NEC, RC5, RC6, SONY .’ - , .

. - :

:

: **5-8 .**

- iPhone- : , - .
- **.5 NFC:** **-** NFC(Near Field Communication, ( ) 10 ,(” 13.56MHz, 424Kbps.
- **:** NFC RFID, :

NFC: -

: NFC, NFC .

-

: NFC , Android Beam( ).

- : , —— , , .
- **:**
- **: 10 ”** **,** **.**

: (SE))HCE- ,(, root .

- / : , .
- :Apple Pay, Huawei Pay ’ , , - .
- **:** /
- **:** **,** **.**

:Apple Pay, Huawei Pay ,’ , .

- :NFC - . ,’
- : ,NFC- .
- , . ,
- : < / . ,

: Wi-Fi

: , , .

- **USB** : , .
- ,[ ] ” ” ” , ,” .
-) (**Wi-Fi-** . , , .

:4 —— ” ”

.1 , - . ,
.

.2 1. ”: ”

.3 Android

:4 — “ ”

, - . ,

.

: “ ”

- : Google Play (”)) Huawei AppGallery, Xiaomi App Store).
- **iOS**: App Store.
- **HarmonyOS**: Huawei AppGallery.

: , . : !

/ : “ ”

(Huawei, Xiaomi, OPPO, vivo, Honor) —

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) “ ” (: ,

; , QR, ’ — .

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- : ” ” ” — .
- : , .
- : — ; , .

) Huawei/Xiaomi)

- **Huawei/HarmonyOS**: [ ] <] [<] [, .” ”
- **Xiaomi**: [ ] <] [<] [, [ ] <] [<] [— ” .”
- **OPPO/vivo**: [ ] < [ ] <] [— . -

“ ” — :

- : , , - .
- : ” ” — ? .
- : , . — ; ” .”
- : ; .

“ ” : !

, . : “ ” .

“ ” ?

“ ” — “ ”, “ ”. :

- ” ,” ” ” ;”
- ” ,” ” ” ’ ” ;”
- ” ” , ” ” ’ .

, **, SMS,** . — .

:

- :) Huawei Phone Manager, Xiaomi Security Center) .
- : , SMS, , , — ” ” .
- : , , — ” ” .

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.1 .

.2 .

.3 .

: , “ ” .

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- : ” ” ” ”; .
- : ! . .
- : . ? [ ]- <] ,[.

: “ ”

) , (—) ,( ) .(

- : ” ,” ” ,” ,” ” ” .”
- :

1. : .

.2 : ” .”

.3 : , .

- :SMS/ .
- : .
- : / .
- : .

APK Android

APK(Android Package Kit) ZIP , :

AndroidManifest.xml

: “ ” “ ” . , , , Activity, . .

- **classes.dex** : . Java/Kotlin Dalvik- bytecode, . dex(classes2.dex, classes3.dex).
- **resources.arsc** : . .
- **res/** : . , , , . drawable) drawable-hdpi, drawable-xhdpi).
- **assets/** : . , , , . AssetManager.
- **lib/** : . .so) C/C++), CPU(arm64-v8a, armeabi-v7a). .
- **META-INF/** : . MANIFEST.MF, CERT.SF, CERT.RSA, APK.
- **2. IPA iOS**IPA(iOS App Store Package) , :

Payload/

: . .app, .

- **Info.plist** : , AndroidManifest.xml- Android. , , , .
- **-** : .app, Mach-O.
- **Frameworks/** : .
- , , : , .app.
- **_CodeSignature/** : , .
- **3.** ? .

Android

: APK , . APK) ,(, Google. Play App Signing, Google , .

- **iOS** : . IPA , . iOS- App- Store . , .

- :

  1. :
  .2 :
  .3 :

. :

- : APK/IPA- .
- : , .
- : , .

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- **App Bundle ( )**: Google Play.
) ,(.
- **App Thinning (iOS)**: ; .
- : , .

: ?

, HD , - .
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3) , , ,(,[ ]- ,
“ ” .

:5 –

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.1

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- : .
- , , : , .
- : , .
- .2

.2

- / : , .
- :) OLED), . .
- : ,) 120Hz) .
- : 15- 30 , .
- : ? .
-) ,(. ,

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, .3 ,

: , , – . iOS- ,

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.4 : “ ” “ ” ,

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: “ ” .4 : “ ”

/ ’ :’

- :
- · : ” ” ;
  - : , ;
  - / / : .

: + / . : iOS- ”
.”

- **.2 : ” ”**

: “ ”

- : - ,) SMS, (; , , ” .”
- : , .
- **.3 :**

:

- / : , ” .”
- : ” ” / .
- **(iPhone)**: [ ]- < [ ] < [ ] <] [” ” ” .”
- : Siri / XiaoAi / XiaoYi / XiaoBu, .
- **.4 :**

:

- · **Sogou / Baidu**: , ;
  - **iFlytek**: , ;
  - Gboard :( ) , , ;
  - iOS: , .

: , , , .

- : ” ” ” ,” .

- **.5 : /**

:

- / : WeChat / Douyin ” ,” .
- : 4 Dock-) ,(.
- ” ” , 10 :

(1) + (2) + (3) / /

(4) / / “ (5) / (6)
.5 :
” “, . ,
– , , , ...
,

: .5 :

: . **.1** “ ” -
.

“ ” –

, . - “VoiceOver” iOS TalkBack- . **iOS VoiceOver**

: [ ] < [ ] < [VoiceOver], .) (/
.

:
: / ;

.1 :) ;(” ”

.2 : ;

.3 :) .(

.4

.5 : [VoiceOver] > ,[ ]

) ,50%- 70-80%- ,(, . iOS “ ”
“ ” , .

: [ ] < [ ] < [TalkBack], . 3
) .(

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: / ;

.1 : / / / ;

.2 :) ;(

.3 : / .

.4

.5 : [TalkBack] >] ,[

,) Google .(**.2** ,
-

, -

- **:** . ,) .(**(1.5x 15x)**, . : [ ] < [ ] < ,[ ] ,
;
- **HarmonyOS**) .(8 / iOS
/
- **:** : [ ] < [ ] <] ,[: , ”, , .” Huawei/Xiaomi
” ” ” .”
- **HarmonyOS**iOS
) (
- **:** , ” ” ” .”) (
. : [ ] < [ ] <] ,[
- **HarmonyOS**+ ” .” **3.** **AI:** **AI-** ” ” ,
.” ” AI

AI: **AI-** “ ”

, , . **AI** iOS 15+

: [ ]- < [ ] < [VoiceOver] >] [” .”

- **AI**) iPhone 12 .(**TalkBack** : Google TalkBack
- , ” ” AI-) , TalkBack .(: ”Seeing AI” (iOS), ”Lookout” ,( ) ”XiaoYi BangBang” (Huawei) AI , , .
- ” ? ,”? AI- –

“ ? ? ” , AI- — “ ” .

“ ” –

, , . iOS

: [ ] < [ ] <] ,[. ,) , , **FaceTime,** (.

: :

“ ” ; [ ]- < [ ] <] .[: , ;

.1 ;

.2 — ” ;”

.3 , , ; ;

.4 .

: +

) , , ,(“ ” . **(TTY / RTT)**iOS

: [ ] < [ ] < [RTT/TTY], , ,

: . : [ ] < [ ] <] [+] .[“ ” ; “ ” .

HarmonyOS:

:

- **TTY ( ):** TTY , RTT;

-
- WeChat/QQ- : ” .” **.6** ” ”

“ ”

- / , , . . / : iOS [ ] < [ ] <] ,[[ ] <] .[,
  .
- : , , ,
- . iOS [ ] <] ,[[ ] <] .[(iOS): [ ] <] ,[” ” , , , ,
- - + , . **LED**: iOS [ ] < [/] <] LED ,[[ ] < [ ] <] .[,
- : [ ] < [ ] < [ ] <] [— .
  .
- , .
  . iOS ,
  .

30

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VoiceOver / TalkBack

1. , WeChat- ;

.2

.3 ,200%- ; , ”
, **.”** :6 -

.4 = + . , .

:6 –

= + . , .

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/ PIN

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1. : .

.2 : ,) 3D , 2D .(

.3 : .

.4 **:**) , .(123456

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- ” ” .
- 2. ” :”
- !

“ ” :

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- : , .
- : .
- : ,) .(
- ,[ ] ” **iPhone- ,”** ” ” ,
  .( )3. : ,

, . , , .

: **,**

.4

AI, “ ” . AI Deepfake,
“ ” “ ” .

(1) AI

AI-

, **,**

.’ , .**(2)** **/** : AI-

, ,

. , .

, . **(3)** **AI** AI

“ ”, “ ”,

,

.2 – **“ ”** : AI- , —

:

– **“ ”**

: , , “ ”, “
” “ 5 ” . ,AI- ,

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- **,** . : , ”
”? ” ”? – .

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- : ” ” — , , , ;

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- ” – , , ,

. , . ,

” 24 ” . **.3**)

:(/

) (:

- ”: ,” .
- :) .(AI-
- **.4 :** .

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- SMS.
- 5. :
- , , , , – !

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.5 “Doxing” : “Doxing”) (

“Doxing” :

. . - **.1**

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EXIF 2 . , **,**
, **,** , . **iOS:**[ ] >]
[< [ ] “ ” **;** **,** “ ” **.** **-** :
) Google Photos) .

- ,ExifTool: Metapho ’ / EXIF.
- 2.
-

4 . ,
“ ” **,** **:** **?**
– . , .

- 3.
- , , IoT –

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2022- :) ,(

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, , “ ” . — ,

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:7

, . , , , *Star Flash...* . .

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Apple

:

- iPhone 14 SOS . iPhone 17 iMessage SMS-
, .
- **OnePlus** : ,2024 BeiDou.
- : , .
BeiDou .
- **:** , .

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- :
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.1 .

.2 .

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4. , . , ,
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.2 :

“ ” Wi-Fi.

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, -) .(

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- : .
- : .
- **:** Apple

:

- : ChangLian .
- **OnePlus** : Xiaomi Miaoxiang Center.
- **Huawei** .

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.1 .

.2 :

.3

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- : .
- **HarmonyOS** .

:

- , ,100%- .
- 4. SparkLink:
- SparkLink (SparkLink) Huawei, Bluetooth Wi-Fi- . : 900Mbps) 50Mbps- Bluetooth), 20 μ s (Bluetooth ,(15ms- 4096 - (Bluetooth .(8

SparkLink:

:

- : 900Mbps , Bluetooth- , HD.
- : ,Bluetooth- , .
- + : 50%- ,Bluetooth- - .
- **:**

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- : .
- : . -
- : .
- **:** , Huawei, Huawei Mate 60, Huawei. .

:

SparkLink . ,
SparkLink .

:

“ ” .

” ” ”SOS ,” .

.1 Huawei, SparkLink.

2. , .

.3 2.) 8-14)

4. , . , ,
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.) 8-14)

:8 —,

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) :(

- : — , , .
- : , , .
- - : - , ” ”. , , — . , ” ” ; .
- : , ’ - ; , .
- : Android- ” ” ; iOS- , Bluetooth .

- , .
-) .(
- : ” ” — .

:

- : , ” ,” ” ” .
- : Android- — < < ; iOS- — < < . , 13 ” .”

- : —Tencent Mobile Manager, 360 Mobile Guard Phone- Bang (Tapir). ” ” ” ” (China Mobile/Unicom/Telecom) . (Huawei, Xiaomi, OPPO, vivo) .
- : ” ,” ” ” ” ,” : Phone Bang: (haoma.tk.cn) .
- 360: ” ” 360 Mobile Guard.
- Tencent: ” ” - ”Tencent Mobile Manager”.
- : China Mobile SMS ”KTFSR” 10086- ; Telecom ”Tianyi Anti-Harassment”; Unicom Unicom”- Mobile Manager”.

: ” ”) China Mobile) AI.

: / : , .

/ : ; .

: ,” ” ,” ” ” ” .

:iOS- — ,

.

” :” , — .

: , ;

Qingzhuo, — .13

VoLTE/VoNR :

- VoLTE (Voice over LTE): 4G — ,2G/3G- , .
- VoNR (Voice over NR): 5G — VoLTE. , , HD. : 5G, 5G 5G .
- : iOS- — ” < < < ” ”VoLTE”; Android- — ” < < < ” ” VoLTE HD”. ; — VoLTE- .

— VoLTE/VoNR .2G/3G- “ ” .

SMS iMessage/RCS-

SMS- — , .

- iMessage: Apple , . ,
  — .
- RCS: .Google- iMessage- WeChat,
  .
- iMessage :RCS- iOS- 18, Apple RCS — iPhone -
  Android HD, , ” .” : RCS,
  .

RCS :

- China Mobile: , ” 5G” .
  .
- China Telecom: ” 5G Telecom”, ; SMS
  .
- China Unicom: 5G , — Android
  ” ” ” 5G” .
- : RCS , WeChat .
  ,WeChat- RCS- ” ”) .(

- : ” ” ” SMS” — ”
  ,” ” ” ” .” = .
- : ” ETC- ,” ” ,” ” —.”
  .SMS-
- : 95- 1069- . ” ”
  11 .
- : ; — ,
  SMS.

. WeChat/QQ:

WeChat QQ- , - “Super App” , .

:

.1 WeChat- PC Mac .

.2 <] [<] .[

.3 ” ,” .

.4 : WeChat [ ] < [ ] < [ ] <] [< [ ] <] ,[/ .

:

- : WeChat [ ] < [ ] < [ ] <] [— .
- : < ”•••” <] [<] [— , , .
- / : — ” ”] [.
- : [ ] < [ ] < ,] , [— ” ” .WiFi-
- : - [ ] < [-] - — .
- : ” ” — GB 30%- –50% .

:

- : ”- •••” >] [— , . .
- : ” ” — , .
  .
- : ” #” — . .
- : ”+” >] [— , , -
  .
- : < [ ] — ” ,” . @
- : @- 500 , everyone@- — .

Moments:

- : ” ,” ” ” ” ” —) ,” ” ,” ” .(” ”
- : [ ] < [ ] <] [< [Moments] >] [— ” / / / / .” ” ” .social-

- Moments: <] Moments [/] [— .
- Moments: [ ] < [ ] <] WeChat] >] [<] [— .
- :] [< [Moments] >] [— .

— .

- WeChat Video: . . 20) - ”Group Video”); — .
- FaceTime: ,Apple- . iOS- 15 ” ” Android/Web, Apple. .
- Zoom: , 100 40- . , , — . / -
- Tencent Meeting: , WeChat, . 300) ,(, — - .
- : —WeChat; Apple —FaceTime; —Zoom Tencent Meeting) —Tencent, - —Zoom).

:

- :WeChat- 500Kbps- –1Mbps, 3- –5MB , HD- 8- –12MB.
- FaceTime: SD ,1Mbps- 720p ,2Mbps- 1080p 4Mbps . 5- –15MB .
- Zoom/Tencent Meeting: 720p 1.5- –3Mbps, 1080p 3–5Mbps. 300- –800MB ; .
- : ;WiFi-) ” HD” (.

:

- : ,) ,(.
- : — ; .
- : ; .” ”
- : WiFi; 4G/5G- ; .
- : Tencent Meeting Zoom- AI — . , .

— , . “ ” Tencent-Meeting.

“:

- : ” iOS — < ” < < ; Android- .” ” ”
- : ; , .

: IMAP POP3 Exchange:

- POP3: , . . : — .
- IMAP: , . — , . .
- Exchange) ActiveSync): Microsoft, , . ” . ,Outlook- NetEase Enterprise Mail Tencent- Enterprise Mail.
- : ” —IMAP; ” —Exchange , IMAP; POP3- .

“ :

.1 iOS: [ ] < [”] < [ ] <] [< ”Microsoft Exchange” .” ”

.2 Android: ” ” ” Outlook- ” ,” .

.3) :(Exchange- ”mail. - .com” ”outlook.office365.com”, — .

4. ” ,” : (IMAP) ,993 SSL; (SMTP) 465 ,587 TLS.

5. ”Push” , ” ” ” .

“ ” “ ” IMAP/SMTP “— . — .

: WeChat

.1 : WeChat < <] [<] .[

.2 : ” .”

.3 , .

:9

. , , .

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WPS Office

: , Microsoft (docx/xlsx/pptx), PDF .
“ ” – , ,
. , WPS , .

- **WPS Office**: Office- , . ,
.WPS- OneDrive,
.
- **Shimo Docs/Tencent Docs**: ” .”
, - . Shimo- Docs , , -
.
- : – ,OneDrive- ,WPS- Shimo- Docs.
, ” .”USB-

.2 : , , –
. – “ ” , “ ” .

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: iOS , , , . ,
. , , . , ,
.

- – : ” ” WeChat . ,
” ” , , . ”
.” , Cubox Pocket .

- -**Notion Feishu-** : , .Notion-
  _ " " , , ; " "
  . " " - Feishu , . Notion- ,
  .
- : " " –) ,(
  WeChat ,( ) Notion- .( ) , .

.3 :OCR- , , .
, .

iPhone

: iOS . " " ,
. _ " - " , "
" .

- : " " . , Microsoft Lens
  ,) OneDrive).
- : , , OCR. " " WeChat
  _ , . , . - ,
  OCR- Baidu ; WPS .
- : , , . ,
  . PDF- , .
- **.4 : ,** , "
  " .

iCloud

: . , . .
: Windows , 5GB, .

- **Windows OneDrive**: Windows + Office 365, OneDrive .
  ,Word- . 1TB Office .
  .
- **Baidu Pan**: , Baidu Pan ,
  . Baidu Pan. ,
  . , , .
- **Aliyun Drive/Quark Drive**: . Aliyun Drive
) ,(Quark Drive Quark, , .

- : iCloud) ,(Baidu Pan)
  ,(OneDrive) Office .(, , .

.5 . , .
:

WeChat) + .(

OneDrive, , ,WPS- WeChat- .

.1 , PDF- ” ”PDF- WPS, WeChat- .

.2 , WPS PDF- , .

.3 , ” ” WPS , .

.4 PDF- Baidu- Pan ,iCloud- .Notion-

.5 , . , , .
. , .

.6 :

(WPS Microsoft 365), .

, , PDF- .

.1 ,) ” WiFi- ” .(

.2 Notion- Feishu, ” .”

.3 :10

.4 , ” .” ” , ”? ,
: . , ,
.

:10

, “ ”. “ , ? ” ,
: . , ,
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. , .

.1 SMS- . , 6 .

.2 , – , .

.3 : , II III , .
, , .

.4 2. : ” – ”

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: “ – ”

: + QR . “ ”, .

- . , . :
. , , .
- , . ” ,” . 3.
–5

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–5

- : Alipay- WeChat- ” .” , ” ” . : Alipay .
- : . ” ” Alipay- ,WeChat- ” / ,” , .
- : . , . : Alipay .
- **.4 :** , , . , :

:

: Apple, Huawei, Xiaomi. : , , , . , .

- **Alipay/WeChat**: ” ” , . : , . , .
- . **10-20 ,** **,** **,** **.** 5. , ,

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, **,**

- : Meituan- Ele.me- . , , . , ” ” Meituan, , .
- : Didi- Gaode, ” ” ” .” , . , ” ” ” ,” , .
- **.6 ,** . , .

,

: “ ” Alipay- WeChat. , , . , .

- : ”Zhangshang 12333” ” ” .Alipay-
, , .
- : ” .”
) , , (– .
- , , ” .”

– .

- , ” ETC” ” .”
- , .
- :
- Alipay WeChat- / .

” ” .

.1 ” / ” .

.2 ” ” .

.3 – .

.4 :11

.5 , . ,
. – .

:11

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: . , – .
: , .

- : , , , .
  .
- : LED – . ,
  . , .
- : – , .
  , (0.5x/1x/2x/3x) – .
- **.2** : , ,
  . , (0.5x/1x/2x/3x). 0.5x) (1x-) (
  ; 2x/3x " " - ,
  . 3x .

+

: “ ”) .(, –
“ ” . , – .

- ” ” : , –, , , –
  . , .
- : , . ,
  – . –
  .

:) .(, , AI –
, AI- . 1-2 – . ,
, , AI- “ ” . 65

- : , . ”
” – , , . ,
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- : , . –
. –, –, – .
2 4 , .

- **.3**

:) .(,
, AI – , AI- .
1-2 – . ,
, , AI- “ ” .

- : , . : ISO-
, ; , ; / –
. ,
.

- – . ,
, , .

Snapseed

: , . : “ ”
“ ” “ ” “ ” . “ ” Snapseed –
.

- **Xingtu/Meitu**: . : 30% , ,
. .

- **Lightroom** : , . , HSL
. .

- : , iPad .
. , - .

- **.5 ?** , –
. :

: iPhone- . “ ”, “ ”, “ ”, “ ” – AI . , “ ”, “ , ” . , .

- : , , – . iOS ” ;” ”screenshots” . . - ,

- , : , ” ,” ” ,” ” ” – . ” ” Apple – , .

- : – , , , , – , . . iCloud Photos , , Baidu- Netdisk/Aliyun Drive . : , .

- **.6**

: - + – .

, .) (

– , “ ”

, , .

- : ” ” , , . , – , ; , . .

- **:** + + ” ” ,) .(

Snapseed- ,) , .(

.1 ” .”

.2 .

.3 :11

.4 , .

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:11

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, :

720P (HD)

: 1280×720 , .

- **1080P (Full HD)**: 1920×1080 , .
- **2K (QHD)**: 2560×1440 , .1080P-
- **4K (UHD)**: 3840×2160 , ,HD- .
- **8K**: 7680×4320 , , .
- , , . (Frame Rate):

, (fps).

24fps

: , “ ” .

- **30fps**: , .
- **60fps**: , , .
- **120fps/240fps**: , - .
- , , . (Bitrate):

, (Mbps).

, , .

.

- .
- 2. : H.264, H.265 ProRes-
- H.264 (AVC):

: H.264, H.265 ProRes-

: , .

- : , .
- : , , .
- : , .
- **.3 HDR:**

: H.264, .50%-

- : , .
- : , .
- : 4K, .
- **.3 HDR:**

: Apple, .

- : , , .
- : , , .
- : , .
- **.3 HDR:**

: ProRes RAW.

- : , .
- : , .
- : , .
- : . iPhone 15 Pro Apple ProRes) 4K 60fps), ProRes RAW , .
- **.3 HDR:** : . iPhone 15 Pro Apple ProRes) 4K 60fps), iPhone 17 Pro ProRes- RAW .

HDR:

: HDR , .

- **HDR10+**: Samsung, - .
- **Dolby Vision**: HDR Dolby, , .
- **HLG**: HDR , .
- **HDR,** **HDR** **.** 4.

:

: , .

- **Display P3**: Apple, .
- **Rec.2020**: HDR, .
- **:** 8-bit

: 16.7 , .

- **10-bit**: 1.07 , , .
- **12-bit**: 68 , .
- : 1080P/4K + 30fps + H.265, .

: 1080P/4K + 60fps, .

- : 4K + ProRes) ,(.
- : , .
- :
- .

, .

.1 , HDR, ProRes.

2. .

.3 :12 -

.4 ,MP3- , Kindle- . ,

, .

.

. RAW HEIF/JPEG

RAW:

RAW “ ” - . , “ ” .
JPEG HEIF “ ”, RAW .

RAW, HEIF :JPEG-

- **JPEG** : ,) .(8-bit, , (2-5MB .(
, .
- **HEIF/HEIC** : , JPEG- (1-3MB). bit-10- .
: Windows- .
- **RAW** : , (20-100MB .(12-16bit, . :
(Lightroom, Snapseed) .

RAW:

- JPEG- :) ,(; RAW-
.
- JPEG- : 2 , ; RAW ” ”
3-4 . .
- : RAW ,JPEG- .
- , – : ,” - ” RAW ,
.

RAW :

- **iPhone ProRAW** : iPhone 12 Pro . ”Apple ProRAW” [ ] < [ ] < .[ ]
”RAW”, . 25MB- , .
- **Android Motion Raw / Ultra RAW** : . Xiaomi/OPPO/vivo ” ”
” ” ; Huawei ” ;” Samsung Galaxy S ” .” 20-50MB.
- : ,RAW- ; ,) , .(

.

“ ” - .
.

: -

. .

- : 1-3- , - , . : .
- **HDR**) (: , , .
  .
- **Deep Fusion ( ) / Photonic Engine** : , - , ,
  .

:

AI , “ ” - , , , , - .

- : , , .
- : , , .
- / : , .
- : , AI , .

:

- **AI-** : ” ” ” ” ” ” ” .
  .
- **AI-** : DSLR,
  . .
- **AI- / AI-** : , AI ” ” . AI .

:

- : , , .” ”
- : ” ” - , , ; ; ” ”
  .
- : + , , ; ” ” .
- : ” + ” - , . .

.

3- 4 , . .

:

- **(13-24mm (** : , , , .)
,(. ” ,” .” ”
- **(24-28mm (** : , . . ”1x”
.
- **(70-200mm (** : .) ,(
.
- **(200mm+ (** : , . , ,
. .

:

“ ”, . “ ” - , =
= = .

- : , , . :) ,(.
- , : , , . , .
- : 1/1.3 1 ’) 1 ,(’ 50- - .

(Pixel Binning):

, “ ” - .

- **(Quad Bayer)** : 50 , 12.5 . 4- , ,4
.
- **(Nonacell)** : 100 , 1100-1200- . , .
- : , - , . ,) ,(”
.”

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- , , - .
- , . , .
- - ” ” ” .”
- - , .

:

- : 120fps- 240fps, . , , . , .
- : . , , , . , .
- : ,timelapse- , 15-30 , 10- .

/ :

- , . .
- , . .
- ,) .(.
- ” ” iPhone) 30 ;(” ” ” ” Android- . , .

:

- , , , . 1-2- . ”
-) 2 5 ,(; , .
- , .
- , , . , .

- : HEIF/JPEG , , .
- ,) , :(RAW .
- : - , - , - .
- : , .
- 1) (’ , .

:

.1 RAW-) < < ,(, JPEG RAW.

.2 , , , .

.3 , .

.4 , .

:12 -

, – :

NetEase Cloud Music

: . “ ” “
”, . . ,
.

- **NetEase Music** : , ’ ’ QQ- Music.
, . ” ,” QQ
Music .
- **QQ Music** : . , , , Lossless- Dolby- Atmos.
, NetEase
Cloud. .
- **Spotify** : , . ”Discover Weekly” Daily”- Mix”
, . , .
.
- : ,’ ” ” ,” ”
, . , , ” Lossless ”WiFi-
, .

: .2 “ ” ,
. “ Netflix ” — , , , .

XiaoYu

: , , . “ ” “ ”
, . “ ”) (“ ”)
,(.

- : ,iPhone- . CarPlay , .

- : 1.2x 1.5x, .
  1.2x- , 1.5x- . ” ” ” ,”
  .

- **.3 :** ,
  . .

VLC

: , —MKV, AVI, FLAC, ISO.
, , VLC . , .srt ,
VLC .

- **nPlayer** : , . , — /
  , , / . / DTS- Dolby- Audio,
  .

- **Bilibili (Bilibili)** : , . , , , ,
  . ” ”) (Bilibili, , .
  , .

- : , , . WiFi, ”TV”-
  , . ,
  . iPhone AirPlay- .

- **.4 :** , .
  — , . ,
  .

WeChat Reading

: . , . “
” , , . “ ”
, . “ ” , “
” , .

- **Kindle** : Kindle, Kindle .
  Kindle- PDF- , Kindle ,
  .

- **Apple Books ( )** : ,iPhone- EPUB .PDF- , .
  , ” ” .

- : ,) ,(. ” ” — , .
, .

- **.5 : ,** : , . ,
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: . , , —
, , . ,E-sports/ ,
.

- : . ( ) , .
60fps ,30fps- 120fps , .
) Honor of Kings, PUBG), — .

- : , . : ;
70%-) ;( ) , .(,
.

- ”: ” iOS- ” ”
, . ” . , .

- : ,) Tencent
START, NetEase Cloud Gaming). , .
. , ,RPG- FPS.

- **:** 3 , .

XiaoYu- , 3- , .

.1 VLC, .

.2 WeChat- Reading, , .

.3 ,) , .(’

.4 :13

.5 , .

:13

, , : -
Qingzhuo, , ; *Sunlogin/ToDesk,* *AI*
Trae, , *LocalSend,* - , -
Aegis/Microsoft Authenticator, *TickTick* .

:

“ ” — , . .

) (

- TickTick — : , ; .
: , , ;Web- + + ; —
” 3- ,” ” ” — ; , - , ,
. ” + + .”
- TickTick- : : / ” ,” ,
.
- : — .
- : , — .
- : , ” ” .
- : ” ...;” . -

Todoist : . : ,)
- ,(- - karma .
Microsoft To Do : Outlook. : , , ” ”
, .
/ : . : , ,
Siri/XiaoAi, ’ .

) (

- Forest : . : , — ” ,” .
- Tide — : . : ,) , (+ + , .
- : 25 + 5 , 50/10- 90/20 .

) (

- : . : , , Siri/XiaoAi.
- Google Calendar : , . : ” ” , / Gmail.

) (

- Notion — : , , ; , . : - (Blocks), , , .
- Obsidian : , . : Markdown , - + — ” ,” .
- flomo : . : , Moments, ,# .
- Apple Notes : iPhone/Mac . : , / / / iCloud, .
- Evernote : . : , OCR , , .

PDF

- PDF Expert : iPad/iPhone, , . : PDF- ,iOS- , , , / .
- WPS : Word/Excel/PPT/PDF. : , - WeChat, . -
- Adobe Acrobat : PDF ,) , .(: , PDF/A- , .

) (

, , , , ... :
PDF/ OCR .

- CamScanner : , . : , ,
  OCR, PDF .
- Microsoft Lens : , / .Word- : ,
  ,OneNote/Word- .
- Apple- Notes : iPhone, . : ,
  , , , .
- Baimiao / CamScanner / " " : Huawei, Xiaomi,
  OPPO, vivo ;" / " Android- Google Camera .
  .
- : , , ; ;PDF-
  watermark- , " " .

" " — , , -
- .

- Keep : , , . : , HIIT- ,
  AI, .
- Nike Training Club (NTC) : . : Nike,
  , , . /
- Apple Fitness : Apple Watch, . :
  ,(/ /) , .

- AutoSleep : Apple Watch, / . :
  , / , .
- Sleep Cycle : . : / ,
  , .
- Mi Fitness : / Xiaomi. : Xiaomi, /
  , / - .

- Calm : , . : ” ” +) ,(, .
- Tide — : . : , + + , .
- Xiao Sleep — : . :) + + ,(, — .
- Flo : , . : AI , / , .
- Meiyou — : . : , —, , .
- Yongyao Zhushou) (— : , . : , , .
- Apple Health / Google Fit : . : , ,) , , (— ” ” .

, . .

- Umetrip — : . : — , ; .
- 12306 : . : ,) ,(. /
- VariFlight — : . : , .

- Amap — : , . :
  , , .
- Baidu Maps : . : / AR,
  , / .
- Google Maps : . ” : , , Street View
  — .
- Google Translate : , ” . : ” ” - , ” ”
  — , .
- (Jijian Huilü) — : , . :
  , 160- , , .
- / : , . :
  ” ” — , , , / .
- Caiyun Weather — : . : ”
  ,” , .
- Carrot Weather : ’ . : AI ,
  ’ , ” , .
- : ; —, , , —
  , . 7 ” / /
  .”
- Mafengwo — : . :
  , —, —
  . -
- Xiaohongshu — : .
  : , ” 3 X”, .

“ , ” — . ;
:23 .

- ” ” :iOS- Apple, — ” : WiFi
  + + .”
- :Android-) ” ” Xiaomi, ” ” Huawei) —
  ; Tasker , .
- : ” ,” ” + .”
  .23

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, . .

:

: (1) DNS IP- ; (2) HTTP/HTTPS
; (3) (HTML, CSS, JavaScript); (4) .

:

- Microsoft Edge: Chromium, . Copilot AI , .
  — .
- Safari: Apple, , . ,
  .
- Chrome: , . , .

:

- Bing: Microsoft, Edge. AI —
  . .
- Yandex: , . .
- Baidu: , — .
- Google: , .

:

- AI: AI Bing Copilot- Edge —
.
- : ” ” ” ” , .
— .
- : ” + ,” ”WeChat ,” .
— .
- : Windows 11 ,microsoft.com- Chrome .google.com-
— .

:

“ ” — :

- : .com — ,
- .cn —
- .org —
- .net —
- .edu —
- .gov —
- .xyz/.tech/.info — , —

: / .jp —
.kr —
.uk —
.de —
.tw —
.hk —
: .gov edu.- , . .xyz top.-
— .

:

- AdGuard : , . :)
,(, ; .

:

- Brave : . : Chromium, , Tor , Chrome- .
- DuckDuckGo : ” .” :) ,(”) ,(” .

:

Microsoft Edge	Chromium + Copilot AI	, AI
Safari	, / / Apple, ,	) (iPhone/Mac,
Chrome	,	,
Brave	,	
DuckDuckGo	,	

” ”

.1 :) .(22:30

2. : < ” ” < < .

.3 .

: ,

, : “ , , ”
. , —
.

- Qingzhuo — : Android , , ” ” . : , , ; ” ” , . : — WeChat, ; , .

- : , . : Android [ ] < [ ] <] ;[iOS [ ] < [ ] <] iPhone]. —
, .

- WeChat/QQ- : . ’ : WeChat [ ] < [ ] < [ ] <] ,[’ ; ” WeChat”.

- : / — ; / — ” .”
, .

- : , , , / . : ,
, SMS, ; — ”
.”

- / : . : Huawei Phone Manager, Xiaomi Security Center, OPPO Phone Manager, vivo Phone Manager — ,
, , . —
.

- : AI. : China- Mobile KTFSR ;10086- Telecom- ”Tianyi Anti-Harassment”; Unicom- Unicom”-Mobile Manager”. — .

- AdGuard : , . :)
,(, ; .

- :” / ” —
, . , .

- : , , —
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- : — .

- : — .

- : , , ” ” .

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. — “ ” , “ ” , “ ” .
8 “ ” : , .

iPhone / iOS	[ ] < [ ] / [ ]	; ; ” ”
Xiaomi / Redmi	< < ;	, , SMS
Huawei / Honor	< ; <	, , SMS
OPPO / realme / OnePlus	< ; <	, , SMS
vivo / iQOO	< ; iManager	, ,
Samsung	< < /	,) (

- “ + + ” : , , . — , — .

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.1 , ” / ,” .

.2 : + China-) Mobile KTFSR .(10086-

.3 ” ,” .

.4) (.

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“ ” , “ ” .

, — , ,

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: ,

- : / / — . WeChat ” ,” Weibo, ” .”
- : Huawei Weather, Xiaomi Weather, OPPO Weather, vivo Weather, ” ” Apple — ; ” / ” .
- Caiyun Weather : — ” ;”?

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- :( ) : , .
- : , .
- : , , .
- : , / — , .

: , , , , , , , , , , , .

: “ ”

- : ; , ;
, SMS, 12379 .
- : + .
- WeChat- ” ” ” ” .
- 12379 (SMS/ , .(
- , , / — .

: ; , - ;
” ” — .

: , , — 32 , / 7
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(EEW)) S- (—
, P- . , “
, , ” . , — .

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- : ” ” ’) —
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- :) , , ,(; ,
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- : ” / ;” , .0-

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— “ ”, “ ”, “ ”. iOS-
— , .

Xiaomi / Redmi	/ < ; ” ”	,
Huawei / Honor	< < /	” ,”
OPPO / realme / OnePlus	< (SOS) >	
vivo / iQOO	< / <	
iPhone / iOS	;	; ” ”
Samsung	< /	” ”

- ≠ : ” ” — .
- : , .
- = : , , .
- : ” ” — .

: “ + / +) (”. —
; , , “ ”. ,
.

: 10-

.1 , , ” / .”

.2 ” ” ” ” / (Android); iPhone-
.

.3 / .

.4 ” ;” .

: ,

“ ” . : , AI
, , , - , 2FA, .
.17

: Sunlogin / ToDesk

- Sunlogin Remote Control : Windows Mac / ;
) .(27 : , , , /)
;(.
- ToDesk : / . : + ,
,) ,(— .
- : / +) (
. /
-) :(” / / ” Sunlogin/ToDesk —
—) .(27
- : , - , ;
.
- — .

AI: Trae / WorkBuddy / Dumate

- Trae : , . : IDE/ AI ByteDance,
SOLO ; . ,
41 AI.
- WorkBuddy : AI , , . : AI ,
” — —;” , .
- Dumate — : - . :
AI, ; ,
.
- ” :” AI ; / Sunlogin/ToDesk ,
/ , ” ” AI . : ;
.
- : , — ;
.

: miHoYo) + PC)

- Artificial Desktop : miHoYo . : miHoYo, ;PC- Live2D/ , / .
- : GPU- ; ,) .(
- : , ” ” ; .

: LocalSend / Feige / AirDrop

- LocalSend : Android/Windows/iOS/Mac/Linux . : , , Wi-Fi, — .
- Feige — : Windows . : , , ; , .IT-
- AirDrop : iPhone, iPad, Mac. : ,Apple- Bluetooth + Wi-Fi , ; ” ” .
- : Xiaomi Share, Huawei Share, vivo Share — ; ” ” WeChat- — LocalSend/AirDrop.
- : Wi-Fi; Wi-Fi — ; .

- : KDE Connect / -

- ” ” Apple : iPhone + Mac. : Apple ID Handoff, Mac- ,( ) .
- KDE Connect : Android + Windows/Linux/Mac, - , . : — , - , / ; .
- ” ” - : Huawei Multi-Screen Collaboration, Xiaomi MiMo, vivo Share, OPPO Connect, Honor Connect — - , .
- Windows + Android: ”Phone Link / Link to Windows”) Your Phone) — , - ; Samsung- .
- : - ; , ; - — .

2FA: Aegis / Microsoft Authenticator

- :SMS- SMS SIM; TOTP , —) .(
- Aegis Authenticator : Android . : , , / , . /
- Microsoft Authenticator : Microsoft, Office /365 ; iOS/Android. : ,) ,(Windows/ .
- Google Authenticator / 2FAS : ; — .
- : ” / ” ; — .
- : TOTP — ; ” + .” .14

:

- : TickTick / Trae/ , .
- : ,PDF- LocalSend/AirDrop; AI.
- : ToDesk/Sunlogin ; Aegis/Microsoft Authenticator.
- : ;TickTick- , Artificial- Desktop .

- , .
- , ” ” ” .”
- AI — IT- .
- — .

: ” ”

.1 Wi-Fi, LocalSend) AirDrop).

2. TickTick- ” :20:00 ” .

.3 2FA WeChat-) ,(Aegis Microsoft Authenticator, .

.4 Sunlogin ToDesk , , — .

:14

. , , .
, .

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.2

.1 : . .

.2 : , .

.3 : , .

.4 **.3** :

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. .

AES-256

AES (Advanced Encryption Standard) NIST,
, . “256” - 256 , .
(1Password, Bitwarden, KeePass) AES-256- .

Zero-Knowledge

“ ” , , . :

.1 .

.2) .(

.3 .

4. , , .

: , . , - .

”Zero-Knowledge” ?

, . -
. Bitwarden 1Password- “ ” -
, .

: PBKDF2 Argon2-

- “ ” . : ,
.

- **PBKDF2** : , iCloud, Bitwarden, LastPass. .(+100,000)
- **Argon2** : ,2015 .PBKDF2- + - .GPU/ASIC- Bitwarden, KeePassXC .

= . .

: LastPass 2022

,2022 LastPass ; , . :

- **-** : , - .
- : , / .
- **-** : , URL , .

LastPass

(1 ,+16) ;(/ / / (2 Argon2id, ; (3 2FA ,)
“,(.

.

: . **Safari ,** , .

.

- **Google Password Manager** : . Chrome , .
- **Huawei Password Vault** : HarmonyOS, Huawei.
- **:** Bitwarden

: , , . .

- **1Password** : , , . .
- **LastPass** : , .
- **KeePass** : , . .
- **.4**

.

“ ” ,(/ /) . “ ” “ ” .

:

“ ” , - . = .

- **”123456”**: , .
- **”P@ssw0rd”** : , , .
- **”correct horse battery staple”**:4 , ~44 , .”P@ssw0rd”-
- **16**) (+ + + :~100 , .

: “ + ” “ ” . .

HaveIBeenPwned:

Troy Hunthaveibeenpwned.com . :

.1 , ” .

.2 Pwned Passwords -) .(

, , – .

k-Anonymity

HaveIBeenPwned- ? . “k- ” : 5 hash- , hashes 5- , . , hash- . “ ” 1Password Bitwarden- .

- **1Password Watchtower** : / / / / 2FA, . / /
- **Bitwarden Vault Health**: : ; / : (HaveIBeenPwned).
- ” ” **iCloud- Keychain**:iOS > < , . /
- ” ” **Google- Password Manager**:Chrome , . / /

:

.1 haveibeenpwned.com, , ” .

.2 iPhone: < < , .

.3 Chrome: < < , .

.4 1Password/Bitwarden: Watchtower/Health, ” ” ” .

. Passkey:

,2024-2026 -**Passkey**) (. Apple, Google, Microsoft, FIDO, W3C, .

Passkey?

Passkey FIDO2/WebAuthn, . :

- :) (+) .(
- , ,” ” , .
- , (Face ID/ (PIN.
- , .

Passkey

		Passkey
	, / /	, , Face ID/ , -

- **Apple iCloud Passkey**:iOS 16+/macOS Ventura+, Apple, .
- **Passkey Google- Password Manager**:Android+Chrome, Passkey ,Google- ; 2023 Google .
- **Microsoft**:Windows 11/Edge ; Microsoft, Outlook, Office, .Authenticator-
- **1Password / Bitwarden**, : , , .

-

Passkey “ ”- . ,2023- Apple, Google Microsoft- Passkey: iCloud Keychain Apple, Google Password Manager Android. 1Password . - : Passkey- .

Passkey-

iOS (iPhone):

.1 (Google, Microsoft, Amazon, eBay, PayPal) > ”Add passkey”.

2. Face ID/Touch ID, Passkey .iCloud-

.3 , ” iPhone”, Face ID.

4. : < < .

Android:

.1 < ”Add passkey”.

2. Google Password Manager > <) .(PIN/ /

.3 Passkey ,Google- Android.

.4 , ”Passkey”, .

.5 : < < Password Manager > google.com.

Passkey -

Windows- Passkey, QR :iPhone-

.1 ” .”Passkey-

.2 ” .”

.3 QR, .iPhone-

.4 iPhone Face ID, .

Bluetooth+ , .iPhone-

:

.1 Google > < Passkey > , Face ID/ .

.2 Microsoft > < < Passkey.

3. Taobao, JD, eBay, PayPal.

4. , .Passkey-

Passkey - “Passkey+ ” . : , , ; . Passkey = “ ” “ ” .

•

- , , / / - .SMS-

?

) USB), , .NFC/ :

- **YubiKey**(Yubico): , FIDO2/WebAuthn/U2F/OTP/PIV/OpenPGP, USB-C/USB-A/NFC. 5 Series 50-70 USD.
- **Google Titan**:Google, USB-C+NFC/Bluetooth, 30-40 USD.
- **OnlyKey**: , .
- :Feitian ePasskey, .

SMS?

- : Passkey, , .
- **SIM**:SMS ,SIM-swap- .
- **-** :FIDO2 challenge, .
- : , .

- , .
- SSO , VPN, SSH.
- , ” GitHub, ” ” - = .
- , , – .

-

Google + YubiKey:

1. YubiKey :) + , .(
.2 myaccount.google.com > < 2FA.
3. ” ” < USB-C NFC.
4. .
.5 .
.6 , SMS .

: Google, GitHub, Microsoft, Apple, Facebook, Twitter, Dropbox, GitLab, AWS, Coinbase, Binance.

: “ ”. 1) ; (2 ,( ) ; (3 . = .

, - backdoor. , Yubico .

• /

: Netflix, Wi-Fi, . : , , . **WeChat-**
— ’ , .

“Vault” 1Password-

1Password . :

- : , , ” .
- : : Netflix, Wi-Fi, .
- : : ” ” ” .”

, Vault . = .

“ ” Bitwarden-

Bitwarden : Send) ;(: , . self-hosted, .

: Netflix, Wi-Fi

- **Netflix /** :1Password Family Vault, , . ’
- **Wi-Fi** :AirDrop Wi-Fi (iOS); Android 12+ ”Nearby Share”.
- **,** ” ”: ,App- , .
- : ,1Password- Wi-Fi SSID .

- 1Password

1Password Families 4.99- USD , 5 , + 4 .
1- USD - . Bitwarden Family 10 USD
6- , - , .

: /

- **1Password Teams / Business**:Vault , / , .
- **Bitwarden Teams / Enterprise** : , self-hosted, SSO.
- : - 2-3 ” ,” .
- : = , .

Emergency Access:

: , / - . “ ” - ,
.

:Bitwarden-

.1 Bitwarden Web Vault > < Emergency Access.

2. ” .

.3 ” ” 7-14) :(.

4. , .

.5 .

:1Password-

.1 1Password: ” / ” - Vault- .

.2 : , .

.3 1Password: ” ,” .

: ” ”

.1 ” .”

.2 Emergency Access (Bitwarden) Family Vault (1Password).

3. + , , .

.4 , .

•

- **-** : - , .
- : ” ,” .
- : , . .
- : , , .
- , : , .

, . , .

:

) Bitwarden .(

, .

.1) (.

.2 , .

.3 , .

.4 .) 15-20)

5. , . ,

” .”

.) 15-20)

:15

-) (- :) ,( )
.(:

, ?

: , , .

- : .
- : , , .
- **:** . , 60% , 40% , 100% .

. **80%-** .

- ?
- iPhone: < < ; : ” ” .
- **,** **?** **,** ?

, **?** **,**

- : , . .
- : 20% ,80%- (0%) (100%) .
- **,** **:** **?** : ,
. ” ,” , ” ” -
.

iPhone ”80% ” -

iPhone 15/16 (iOS 17+) - “80%-20%” . ,80%- . :
< < < < 80%

: (1) - ;100% (2) — , 100%- ; (3 **80%**— ,80%- . : / ;80% : .

GaN -

? : **GaN**) (. GaN :

- : , 65W 5W- Apple.
- **,** : , +95% , .
- :65W/100W , - , + + .
- :PD GaN (65W+), Baseus, Anker, Nubia, Xiaomi. E-Marker.

= × (P = U × I). , , , .

.

:

: , .

.1 : .

.2 : , .

.3 : , .

.4 **.2** USB PD (Power Delivery)

-

: USB-IF, . PD 3.0 100W (20V/5A), PD 3.1 240W- (48V/5A).

- : 5V, 9V, 15V, 20V, PD 3.1 ,28V- 36V, 48V.
- : iPhone, Google Pixel, , , Switch - PD.
- **240W**: , ; PPS-) ,(.
- : Type-C , .
- **Qualcomm QC (Quick Charge)-**

QC 3.0: 3.6V-20V 0.2V, 18W.

- **QC 4.0/4+**: USB PD, 27W, . -
- **QC 5.0**: 100W, 0-50% 5 , .
- : Snapdragon , .
- **240W**-

:) ,(5V/4A- , .

- **VOOC 1.0**: 20W (5V/4A), ,2014- .
- **SuperVOOC**: 50W-65W, .
- **SuperVOOC 2.0**: 65W, 4000mAh 30 .
- **150W/240W**: .
- :) ,(.
- -

FCP (Fast Charge Protocol): 9V/2A, 18W, QC 2.0.

- **SCP (Super Charge Protocol)**: , 22.5W- 66W 100W.
- : , , .98%
- : Huawei/Honor, PD.
- **240W**-

Surge P1: Xiaomi, 120W.

- **Surge G1**: , P1 .
- : + , .
- -

PD: USB PD.

- : iPhone 8 , 27W- (iPhone 15 Pro Max .(+27W-
- : , , .
- : USB-C to Lightning) USB-C to USB-C) PD.
- **.3**

		VOOC 1.0	**240W**
/	OPPO	QC 5.0	100W+
	Qualcomm	VOOC/Dart	240W
	OPPO	SuperCharge	100W
	Huawei/Honor	Apple	~27W
PD	iPhone/iPad	4.	
			:

Qi

: WPC) ,(. Qi .5W-15W-

- **Qi2**: ,2023-) Apple- MagSafe), .+15W-
- : , 50W Xiaomi, 50W Huawei ,' .
- **.5**

: , . **.**

- : . 60W.
- : Apple MFi; .
- : , , .
- , .

- , .
- , .
- ,
-) (

. : -

,2024-2026- " - " = . , :

- , : , 372mAh/g.
- : 10x (4200mAh/g). - = .

- : , **~15%-20%**. 2024-2026 7-9mm- 5500-6000mAh.

?

- **Honor**:Magic , Magic6 Pro/7/V3.
- **Huawei**:Xiaomi 14 Ultra/15.
- **vivo**:vivo X100 Ultra/X200 Pro.
- **Huawei/OPPO** 2025-2026: , - = .

: - “ , ” -
; , 4500mAh- .5500mAh- , “ ”
. : - ,
80% .

•

- : , , , .
- **5G**: , ,4G- .
- : , ” .”
- : iPhone < < ; .
- **:** : .

: , .

.1 : .

.2 :16

.3 , . ” ,” ” .”
, . , - , .
.

:16

. , :

:

- : , .
- : , , ,) .(
- : .
- : , , , .
- , **Apple- ID .** , **Huawei/Xiaomi/OPPO/vivo , Google .** ” ” , , . , ,

-

: Apple- , “ - ” **“ - ” . - , .**

, .

.1 : - , - ? ” ” / ” .” , , .

.2 ” ” : , , , . . , - .

.3 **, iCloud: ” ” Apple** , : ?iCloud-

Apple 5GB , , 5GB .
: “ iCloud- ” . :

: iCloud , . , .
.

- : 50GB 6- , 200GB 21- . 50GB , .
- : , , iCloud. .
- **: iCloud ” ” - .” ” , .iCloud-** **” ,” iCloud , iCloud,** . , -

, :

iCloud+:

(50/200/2TB) =**iCloud+**, :

- **Private Relay**) (:”VPN” Apple. Safari Apple,) Apple) . Wi-Fi- . < < iCloud > Private Relay.
- **Hide My Email**) (” : , - ” . = . ” . < < iCloud > Hide Email.
- ” : , . /

: Private Relay .

iCloud ADP - E2EE

2022 Apple **iCloud Advanced Data Protection (ADP)**, E2EE / - - **iCloud**—, , , - 23 .

- **E2EE** : , .Apple- , / Apple .
- : < < iCloud > ADP, /) , .(
- :1) ; (2 E2EE; 3) .

: , ADP - .

: - , ; , . iCloud, , - .
: . : “ = ” .

- : - ” ” . , .
  iCloud, - . : . : , .
- , ,Apple- :

(Huawei/Xiaomi/OPPO/vivo): , , , . ,
+ .

- **Google**: ,Gmail- Google , Google , Google.
  Google 15GB , ,Apple- Gmail .Driver-
- **WeChat/QQ**: . ’ WeChat
  Google, WeChat .
- , **:** , **Google, ’ WeChat -**
  . , :

“ ” Apple , , , iCloud, . 6 .
: . Huawei- Xiaomi- , “ ” “ ”
.

: ,)
,(, .

, : , , .
, , . , - - .

:
, , - .

” ” ” ” , .

.1 iCloud , , .

.2 ” ” .

.3 , , Google, WeChat - .

.4 :17

.5 . , , , , , ,
. ” .”

:17

, , , , .
, .

Apple (AirDrop + (

: iPhone- Mac, AirDrop . AirDrop Mac-
, , Wi-Fi, Bluetooth- Wi-Fi- Direct. “
” - , Mac- , .

- + **Windows (Link to Windows)**: Microsoft- ”Link to Windows”,
Samsung, . , ,SMS-
, . , , .
- **-** : ”Multi-Screen Collaboration” Huawei Xiaomi”- Smart Connect”
Xiaomi - , , Douyin-
WeChat- . - , - .
- **, :** , .
, :

iPhone

: ,) ,(,
. : Wi-Fi, AirPlay-) .(

- : , Huawei ” ,” Xiaomi ,” ” OPPO/vivo
” .” - , ” ”.
- : iQiyi, Bilibili, YouTube, -
Wi-Fi, , , . , ,
.
- : 90% Wi-Fi. 10% ,
” ” - ”AirPlay” .

- , / : , ” , .” , . / - .

: , - WeChat ,Pinduoduo- , - .

- : 24 , , , , . - , .
- / : , . , , ” .”
- : ” ,” . - .
- , : ” ” , : , , , .

, :

: , . , .

- : , , ” Siri/Xiao Ai, .”
- : ” ,” . .
- : , , .
- ? : iPhone- Apple- HomeKit) .(, Xiaomi Mi Home , , , .’
- Huawei- HarmonyOS Connect, Huawei.
- , Bluetooth:
- Bluetooth- . , .

:) ,(Bluetooth , .
- , .

- : AirPods iPhone , Apple. Huawei FreeBuds Huawei, Xiaomi Xiaomi - .
- **Bluetooth**: , , Bluetooth . , DJ- .
- : Bluetooth) .(, , Bluetooth ” .”
- **, :** , . , , . CarPlay (Apple) Android- Auto .( )

UWB - ” ” Bluetooth-

Bluetooth, UWB. : — , . Bluetooth “ ”, UWB “1.2 ’ ” . :

- **AirTag/** :iPhone AirTag ”7 ,’ ” = UWB. Bluetooth .” ”
- :UWB = , , / relay.
- : ” ” , / .GPS-
- :iPhone 11+ (U1/U2); Xiaomi, Samsung, vivo, OPPO .

LE Audio Auracast- -

A2DP **LE Audio**(LE Audio), Bluetooth 5.2+:

- **,** :LC3, ,SBC- , .
- **-** : , .
- **Auracast**:Auracast. - - , , , .
- :2024-2026 , . LE Audio/Auracast.

:) ,(USB- ,
CarPlay Android Auto. , .

- :) ,( ) ,( ) ,(
WeChat (Siri Xiao Ai .(

- : , CarPlay , , .
- , , . :

, , •

, - , “ ” “ ” .

- - , . .
- : ,
- Wi-Fi.

” ” ”AirPlay” .

1. : iPhone ” ;” ” .”
.2 , .
.3 , ” ” : , , .
.4 :18
.5 , . , -
. , .
.

:1

- : Xiaomi 14, , .
- :1) < ; (2 App ; (3 5G4G; 4) ; (5 ; (6 .
- 30-50%: , .

:18

“ ” , :

,

:) ,(, SMS, , .

- , : ,WeChat- ,Taobao- . , .
- , : ’ WeChat) WeChat , ,(, -) Google Authenticator).
- . , :

“ ” - , , , , - .

:’) , (

iPhone

: , iTunes) Windows (Finder (Mac), “ ” . “ ” , .

- : , Huawei Phone Assistant, Xiaomi Assistant, Samsung Smart Switch. .
- **HarmonyOS** :’) , (
- iPhone

: iCloud iCloud . , ,Wi-Fi- . 5GB , 6 .50GB-

- : . Huawei ” ,” Xiaomi ”- Xiaomi”, OPPO/vivo . Google) ,(.
- **HarmonyOS**, :
- , , :

) (

: iCloud/Google , .

, , .

- : Google 15GB , ” ” .
  .
-) , (iPhone- iCloud,
  . , vCard iCloud.com- .

Google . .vcf, .

- ’ WeChat) , (
- , ’ WeChat , .

:’

. WeChat ’ . / Wi-Fi,
. : , WeChat- .

- **:’** . WeChat PC ,
  ’ . .
- **:’** . WeChat iCloud- , : ,
  . .
- **, :** , . ,
  :

) (

iPhone iPhone

: , “ ”, . , ,
, **Wi-Fi. 20-30- .**

- : . Huawei ” ,” Xiaomi ” Xiaomi”, OPPO ” ,” vivo ” .” : , , , .
-) , ( )
.(: , , SMS, , . :
) ,(’ WeChat) WeChat .(

) ,iPhone- (

iPhone

: Apple- “Move to iOS” , , iPhone- “ ” . , , SMS , . ’ WeChat WeChat.

- **iPhone** : , , .
: , , ’ WeChat. .
- **,** **:** **3-2-1** , :

3

:) , .(

- **2**:) , .(
- **1**:) , ’ .(
- , ” , ” . , . ’ WeChat

’ WeChat ! . “
’ ” WeChat, WeChat- . ’ ,
- .

:

iCloud , .

: , DCIM Pictures- .

.1 : iPhone vCard ;iCloud.com- .vcf .

.2 ’ WeChat: WeChat ’ .

.3 vcf- USB, ” ”

.3-2-1

4. :19

.5 , , .” ” , , , ... ,
. ” ” - , , .

:19

, , . , “ ” - ,
, .

, :

: . ,10%- , . :
20%-15% , .

.1 : 24- .
, . .

.2 : ?) , .(
?) .(

.3 : , , - . CPU- ,
. .

.4) (:) 60% ( ) (
, : .

, , - , . ,
:

: , . .

.1 : . .

.2 : , . .
, .

.3 : . : ” ” ” !”
, .

.4 : . - WeChat- , ’
.

5. , - ” ” , , .

: , . - , ,
.

- : , . ,
.
- : , ,) .(,
- .
- : (iPhone PD 20W),)
3A+),) .(
- , , **Wi-Fi** , ;Wi-Fi- .
:

: , 5- . , .

.1 ” ” **Wi-Fi** : Wi-Fi,) ,(” ”
,” ” .

2. : , . 30- , Wi-Fi-
.

.3 : Wi-Fi ? ,
) , , .(Wi-Fi-

.4 , : , ,

: - .

. , .

- **/ (burn-in)**:) ,(Douyin- ”
” OLED. , . .

- : , PWM dimming, OLED. ”DC dimming” ”anti-flicker”, . .
- / : , , . ,
, .
- , , - , , :

: . , 30 , .
.

- : iPhone full-screen: , ,
Apple) 10-15 .(: 10 , .
- : ”Apple ” (iPhone) ” ” .( ) ,
. iPhone iTunes/Finder, ” ”) ,(” ”
. Fastboot- Recovery, , +
” .”
- ” : . , .
- , : ,

:) iPhone), , . ,
40% . Bluetooth .

- : - . .
, .
- / : - ! , .
) ,(.
- , , , . ,
:

, “ ”

: (Apple.com.cn, Huawei.com.cn (’ “ ” . Apple ,
, .

- :) ,((Apple ID, WeChat,
Alipay), ” .” , - .

- : ” ”) ,(” ”) ,(” ”) .(, . , .
- : , . Apple- , .
- : , / ” ,” - . , .
- : - , , , - . , , 60% . , , , , . ” ,” .

:

: iPhone full-screen: . : 10-15 .

.1 :) SIM), .

.2 **Wi-Fi** : Wi-Fi ” ” , .

3. : , , 5 , , .

.4 , . , . :20

.5 ” ” . , Root, BL... , .

:20

“ ” . , *Root, BL...* , .

, :

: < < **7** “ ” , “ ” .

- **HarmonyOS**: < < 7 ” HarmonyOS”.
- **iOS**: , Apple Developer .
- : USB

:

- **USB Debugging** : (ADB). "ADB".
- : ,( ) .

:

- (/ /) : App . 1x, 0.5x- " ") , ,(. 0.5x - ; , , ." "
- **Force GPU Rendering**:2D GPU. / = , . = , . .
- **Profile GPU Rendering** : , 16.6ms) 60fps). = . , App - App , .

:

- **GPU** : " " " "1 " 2 " .' , Android , App) .(, , " ."
- : . , , .

Bootloader . BL ,
.

,' OEM Unlock: " "

Developer- Options "OEM Unlock" . BL, " " : , fastboot/ .

- ": ." = .
- **BL**:OEM = ," " BL = ." " , + fastboot (`fastboot flashing unlock`), .
- . : OEM , BL - `fastboot flashing lock` .
- : ,Root/ . USB"- Debugging".

:

, . Google “OEM Unlock” “ ”. BL , OEM , .

,’ BL Lock:

:

- : , .
- : BL .
- **BL :** Xiaomi

: , **7-30) .(**

- **Huawei**: 2018 , .
- **OnePlus**: , .
- **OPPO/vivo**: , .
- **Samsung**: , Knox fuse, Samsung Pay .’
- **Apple**: iOS BL lock , .
- Root Administrator- ,Windows- . Root, , , Root.

. ADB :

ADB = Android Debug Bridge, Google Android USB Wi-Fi. Root, “USB Debugging” . “ ” - ROM, App, , - .

USB:

1. : Developer Options > ”USB Debugging”.
2. : Platform Tools (`adb`. , ; . , AI ” ,” . `fastboot` ,(.

.3 , ” USB Debugging?” > ” ” < .

.4 :cmd/PowerShell- `adb devices`, - `List of devices attached /xxxxxxxx device`, .

Windows `adb devices` “unauthorized” ,
“ ” . Samsung/Xiaomi “USB Driver” “Phone Assistant” ; Mac
Linux- . , .

ADB :

- `adb install app.apk`: APK)`-r` - ; `-s` SD- card).
- `adb uninstall com.xxx.xxx`: App .
- `adb push PC /sdcard/target`: .
- `adb pull /sdcard/file PC` :) .(WeChat-
- `adb shell`:Linux Shell, `ls pm am` .
- `adb reboot bootloader` : Fastboot- .( )
- `adb reboot recovery`: Recovery.
- `adb shell pm list packages` : App. `-s` App- , `-3` App-
  .
- `adb shell pm uninstall --user 0 package` : App , Root.

App Root :

ADB: `pm uninstall --user 0` App “ ” ,
Root, . App- “ ” , .

.1 - `adb shell pm list packages -s` App , . Xiaomi:`com.android.browser`, Huawei AppGallery `com.vmall.client`.

2. ? , `adb shell pm list packages | findstr keyword` ”Package Viewer” .

.3 : `adb shell pm uninstall --user 0 package`, `adb shell pm uninstall --user 0 com.android.browser`. `Success` .

.4 ? (1 ; (2 : `adb shell cmd package install-existing package`.

!

“pm uninstall” Root, App)`com.android.systemui` , `com.android.phone`, `com.android.settings`(= , . Google . .

: –user 0

Android 4.2+ , - ID) ,0 .(+10`--user 0` “ ” - APK-
, . Root - , “ ” . :Root- Root

, ; ADB “ ” .

. Root: “ - ” Android

Root - Linux-) /- , / ,(.Android- - Administrator :Windows- Windows) TrustedInstaller), Administrator root Linux- `sudo rm -rf /*`. Root: , App, App .

) Titanium Backup, Greenify).

- .
- Recovery ROM- .
- Root:
-

: Root.

- : Root, .
- : .
- : Root .
- **OTA**: .
- Root , bricking-) .(, , . , .

,

,’ SafetyNet/Play Integrity: Root =

Root : , App - “ , ” , Google Pay , Netflix .HD- : SafetyNet Play- Integrity API .

SafetyNet / Play Integrity:

- **SafetyNet Attestation ,) (2017**:API Google , App Google ” CTS .” Basic- ( ) Strong- ,) .(Google SafetyNet- .2024
- **Play Integrity API ,) (2022**, : . + App + . App .
- , : (Google Pay, Alipay, WeChat), Netflix HD, , MDM.

Play Integrity 3 :

- **MEETS_DEVICE_INTEGRITY**:Android , BL , Root, CTS. / .
- **MEETS_BASIC_INTEGRITY**, : . Netflix , .
- **MEETS_STRONG_INTEGRITY**, : (Titan M), .
- **(NO_INTEGRITY)**:Root/BL , /ROM/ .

Root = App :

App Google Play Integrity API, Google - BL, , su, SELinux permissive. , App “ ” .

: Magisk Hide / Shamiko / Zygisk:

Root- , “ Root” . : “ ” .

- **Magisk**:Systemless Root - , . OTA , . BL .
- **Magisk Hide ( )** : , Root. 2021 , Google , ” .”
- **Zygisk**:Magisk 24.0+, Riru, Zygote, . .
- **Shamiko** : , Zygisk, ” ,” Root/Magisk/BL. LSPosed/GitHub.
- **Play Integrity Fix(osm0sis)** : Play- Integrity, fingerprint, Google ” ” .

Google , App Play Integrity. .
App- , / : Root, BL. “ Root + Root” .

.

- **BL** : , ,GPU/ .
- **OEM BL** : ,Root/ .
- **ADB** : Root - `pm uninstall --user 0` App. .
- **Root** : Root, < . : ADB, Shizuku.

7 ” ,” .

1. , ” ”) .(” ”

.2 ”Window animation scale”, ”Transition animation scale”, ”Animator duration scale”, 0.5x- .

.3 .

.4 :20 Termux — Android

5. ,” ” , Termux
. root, Android ,
.

:20 Termux — Android

“ ” , , *Termux*
. *root,* *Android* ,
.

, Termux

,

,

Termux**Bash** Shell. ,Zsh/fish- : `pkg install zsh`.`pkg install fish`. Termux: Vim/Nano/Emacs. “pkg install” 2000+ , GCC, OpenSSH, SQLMap.

SSH

- : OpenSSL, .
- : Nginx Apache .
-

- ’ , .
- ,
- Termux , :

,

Termux:API	Android API
Termux: Boot Termux: Float Termux: Styling Termux: Tasker Termux: Widget	Termux Tasker , Termux , sharedUserId.) F- Droid (GitHub- .

,

,

.1 : ”Installing bootstrap packages”.

2. :

.3 **termux-setup-storage** . :

.4 **pkg update && pkg upgrade** . `Python:`

- **termux-change-repo**) Tsinghua). :
  Ctrl, Esc, .
- : Zsh + Oh My Zsh, .
- ,

,

SSH, , , .

, , ,Shell-
.

Linux , Shell)`chmod 755` rwxr-xr-x,`chown user: group` , `ls -l` owner/group/other 3 9 (. Unix/Linux (owner / group / other × / / ,(ACL- NTFS .Windows-

Termux v0.118.2, Android ≥ 7. Termux- Android- 12 , .

,

Termux Android, . , , Termux , “ ” .

:

Termux ,F-Droid- .

.1 pkg update && pkg upgrade

2. `pkg update && pkg upgrade` .

.3 Python:`pkg install python`.

4. `python3 -c "print('Hello from Termux!')"` .

.5 , . ,) -2126) . AI, , .

.) 21-26)

:21

, , , . , , ,
... “ ”, ? , , —
— . “ ”.

, VPN: “ ”

: . , “ ” .
Wi-Fi- , , “ ”.

- **VPN-** : VPN VPN . , ” ” . ’ ,
IP- VPN- IP- .
- **VPN ? : ,** Wi-Fi-

; ,) ;(, **(ISP)**
. : **VPN- !** “ ” “ ”. VPN- ,
, VPN- . VPN,
) NordVPN, ExpressVPN), . **, VPN .** “ ”,
, , VPN . , “ ” Wi-Fi —

, Wi-Fi- ? : ,
“Starbucks_Free” , . , ,

, “ ” Wi-Fi —

— HTTP , App TLS — .
: WeChat, , Alipay , HTTPS. ,
, , ’ ,

. ,) ,(App HTTP ,
. “ ” (Man-in-the-Middle), “ ” .
, “ Wi-Fi- ”, Wi-Fi
) “ChinaNet” “CMCC”), . **? :** HTTPS

: https://, - “s” Secure. , http://)
s), ! “ ” HTTPS, .

- ” ” : iPhone Android, Wi-Fi ” ” ” .”
  .
- : , , ...
  4G/5G- , .
- ” ” **Wi-Fi** : , ” ” Wi-Fi-
  . .
- **, HTTPS ? — SSL/TLS** HTTPS, ?
  HTTPS, ” SSL/TLS”, :

, HTTPS ? — SSL/TLS

: “, , , ,
CA . ”

.1 : CA ? ? ?

.2 , ” ,” .

.3 : ,( ) , ,
. , ’ .

.4 : HTTPS ” ” , . HTTPS- —
” ,” . .

, DNS: “ ” ?

DNS) (: “baidu.com” , DNS-
IP) ,(110.242.68.66 . DNS

, DNS: “ ” ?

, ! , , ,Wi-Fi- , . “ **DNS”**
: ,DNS- “baidu.com” . ,
, —. , DNS :

DNS over HTTPS (DoH)

: DNS HTTPS, , DNS.

- **DNS over TLS (DoT)**: TLS DNS, ,
  DoT- .

- **1.1.1.1- Cloudflare 8.8.8.8- Google DNS, DNS-**
  . ”DNS ” Android ”one.one.one.one” ”dns.google”, .
  iPhone- , App, . , ”
  ” ?

, :

, “ ” ?

. , . **:**

- , ,IP- .
- ,) .(HTTPS- /
- . **,**
- , . **:** ” ,” ” .” **,**
  VPN + Tor, , . , :

, . , , , , ...
. :

, :

: , , . “ ” — —
. . - ’

- **” ”** : ” , ”’ ” ” .
  , . .
- : Firefox Focus, DuckDuckGo, Brave —
  , Chrome .Safari-
- : DuckDuckGo- Baidu/Google ,
  . , ,Google- .
- **App**: . , , ?
  .
- **, :** , , , :

, :

: iPhone - Wi-Fi, Android Wi-Fi.

1. **DNS-** : Android DNS ,"dns.google"- iPhone App DoH .1.1.1.1

.2 **- (2FA)**: WeChat, Alipay, Apple ID, Google, 2FA —. SMS 2FA, TOTP-) Google Authenticator, Microsoft Authenticator).

.3 **App**: App, , , SMS, , . — .

.4 : , , .

.5 : USB ,) "Juice Jacking") . .

.6 **:** " " DNS

: Android: < < DNS < "one.one.one.one" "dns.google". iPhone: Safari- "1.1.1.1 DNS profile" , 1.1.1.1 Cloudflare.

1. **Wi-Fi**: ,Wi-Fi- " " , , .

.2 : < / , App. App) ,(.

.3 : Apple ID Google, , 2FA- , .

.4 **:22 AI** " " , : . " ," " ," " " — .

:22 AI

AI . , *AI*
: , “ ”, “ ”, “ ”.

, AI: “ ”

) (

, : “ ” ?

.1) (: , .

.2 : App- , ” ” .

.3 , **AI** **AI** AI , , , . AI ,
.

, AI AI

,2024-2025- AI) DeepSeek, ChatGPT, Wenxin Yiyan, Tongyi Qianwen
(’ . , , .

, AI :

,2025-2026 AI (DeepSeek R1, ChatGPT-5, Wenxin 5.0, Qwen 3) . , .
NPU 80+ TOPS = . “AI Phone” . NPU .

AI :

: **DeepSeek, ChatGPT, Doubao, Kimi,** , , ,
. — , .

- : AI- , , , .
- : AI , , , .
- : , , — . ”
- : AI , , , .
- **:** **AI** AI) 7B- ,(
:

: AI , , .

- : , .
- : , AI .
- : , . AI ,
.
- **AI :** Apple Intelligence

: **AI iOS- 18,** , , **Siri ,** . ,
.

- **Pangu Huawei**: HarmonyOS, Xiaoyi , ,
, AI .
- **Pengpai AI Xiaomi**: AI , AI, AI, AI,
.
- **Galaxy AI Samsung**: , AI, , Circle to Search
.
- **,** ” - ” AI , ,
AI ” ” :

: **?** **“** **”** **iPhone** **-** **,**
, **.** **-** **Android,** **Galaxy AI** **Samsung**
, **.**

: (ASR) (NMT) (TTS) ,
.

, **:** **“** **”** **AI-**

: **,** **?** **“** **”** **,**
“ **”** **,** **,** **.**

: (Inpainting) , **“** **”**
. AI- .

, **:** **“** **”** **AI-**

: **?** **“** **”** **AI** **,**
. **AI** **,** **“** **”** **.**

: (Diffusion Model) AI (Generative AI),
, . **“** **”** AI.

, **:** **“** **”** **AI-**

: **,** **AI-** **,** **,** **,** **.**
PDF **10** **,** **AI** **10** **.**

: LLM (Long Context LLM) **“** **”** ,
. Apple Intelligence, Xiaoyi Huawei, WPS AI, Tencent Yuanbao .

, **:** **“** **”** **AI-**

: **,** **,** **AI-** **“** **”** **.** **,**
“ **”** **, AI-** **,** **.**

: (CV) (Embedding) , . **“**
- ”
.

, **:** **“** **”** **AI-** AI **“** **”** .
, , .

: “ **” AI-

, . **?** AI- Baidu- — . AI ,
“ **”** .

, , , . “ ” AI- . **.1 (Role):** **AI-** AI- “ ” , .

,

:

200- ,

- .”) (”
- .”) (” , ,
- .”) (”
- .”) (” , ,
- .”) (**?**

.5 : 5 , . , ; . , AI “ ” , . **.2 (Background):** **AI-** “ , , ” , AI-

, . **:** “

200- ,

- .”) (” ,60 ,
- .”) (” CET-6 ,2026
- .”) (” , 30 ,
- .”) (” ,
- .”) (**?** AI-

.5 : 5 “ ” . , ; . , AI “ ” , . **.3 (Task):** **AI-**

,) , , , , ,(... . . **:**

“

200- ,

- , , , ".) ("
- , , , ".) (" 3000
- , , , ".) (" ' 5G'
- , , , ".) ("
- , , , ".) (? " " AI.

.5 : 5 . " ", " ", " " AI- —
, . **.4 (Constraints): AI-** " " " ",

AI

. : , , , , . **:** "

200- ,

- , ". ( )" ,
- , ". ( )"
- , ". ( )" 3-
- , ". ( )" : , ,
- , ". ( )"
- , ". ( ) ? , AI " " — , , .
= " " , , .

.5 : 5 , -

,

: : " ", " 800 , 3 , ".

1. : AI- ", ," " ."

.2 : , , , JSON? .

.3 : , , , .

.4 **AI**: , AI- ; , .

.5 **6.**

,	) / (
/	+ ” ”
/	, ” ”
,	) 200 (
AI	” , ’ ”’ +
	, ” ”...
.7	:1) (

:1) (

:

)). , 15 , 6 60 ( ). **7-**)). : (1) 15- (2) + +) ((3) (4))).

: , , ,

.

:2) CET-6 (

:

CET-6.( ) , **480 ,CET-4- 500 ,CET-6- 3** .( ) **12-** .( ) : (1) , , (2) (3) 3) , ((4) .( ) : , , “ ”, “ ” “ ” .

: AI “ ”

:3) (

:

/ / / .( ) , **Q2**.( ) / .( ) : (1) “ / / ” (2)) — “ ” ((3) 600 (4) markdown .( ) : , , , 4 .

: AI “ ”

, AI:

- : AI, ” , .”
- **(CoT)**: AI- ” ,” : ” .”

- **AI**: A- , B , C — .
- : AI , , ’ , .
- : AI , . AI) .(

:AI-

AI , “ ” . :

- ” ”: AI) : Hallucination). , . .
- : AI) ,(2024 .
- : , , —AI .
- : AI , . AI- .
- : AI , .
- **AI-** AI-) ,(” ” .

AI-) , , .(’

- : , AI .
- AI , .
- ,) , ,(AI.
- : AI ” ”
- , AI:

: : ” 7- , 15:00- ”.16:00-
.

1. **:2** **AI**: , ” ” ” AI”, . ” ” AI .

.2 **:3** : ” ” ,AI- CRISPE. .

.3 **:23** ” ” , ” ”AI- . , : ” ” , . : ” , ”? , – .

:2

- :iPhone 13 > Xiaomi 15 .
- :1) iCloud/WeChat ; (2 Apple ID + ;" " (3 Xiaomi App; 4) ; (5 WeChat; 6)) ;(WeChat/ / (7 .
- : , .

:23

“ ”, “ *AI-* ” . ,
: “ ” , . *:* “
, *?* ” , –
.

?

, – . **:**
, **’ ,’ WeChat ,** **,**
NetEase Cloud Music . **,** **“**
7:00” , **, . , ?** **“ …**
…” , if-else- . , , .
:

: ? “ 7:00”, “ Wi-Fi- ”, “ ”, “
20%- ” .

.1 : ? WeChat, , , ,
, **…** , .

.2) :(A X, Y. ” ,
,” .

.3 **.2** **” ” iOS-** – ” ” (Shortcuts) ,
. . , :

“ ” iOS- –

: , .Siri- , “ Siri, ” ,
.

- **” ”**: , . ” Bluetooth- ,
.” ,” ” ” .”

- ” ”: . , , ,
. .

- , : :1 ” ”

) “ ” ,(Siri- :
. : “ ” “
” “ ” “ ” “ ” NetEase Cloud Music. , .

:1 “ ”

“ ” , “ ” “ 30%- ” “ ” ,
“ 30%- ” , “ Bluetooth” . 30%- , .

:2 “ ”

,WeChat/Alipay- ,
Numbers. “ SMS/ ” (iOS 17+ ,(
“ Numbers- ” . , .

:3

.3 –Tasker
, , . , :

–Tasker

“ ” . : -
, , . ,
, , . : ,)
,(.

:

Tasker , . ,
, , . - Tasker “ ” :

: Tasker –

: “ ” . , , Wi-Fi, Bluetooth, , ,
··· Tasker- , .

- **(Task)**: ” .” , . :
) , , Wi-Fi, Bluetooth, ,(SMS, HTTP, Shell,
, - .

- **Tasker:** Wi-Fi

: “ ”) (“ Wi-Fi” . GPS,
, Wi-Fi .

- : ” ”’ , ” , , WeChat-)
.”(.

- : ” ”22:00-06:00 ” , ,20%- -)
.”(- – , .

- : ” / Bluetooth ” ”
XiaoYuzhou .” , .

- **Tasker- , :** MacroDroid

. , , . , (MIUI
Xiaomi, HarmonyOS Huawei, ColorOS OPPO) “ ” “ ” ,
Tasker, . **MacroDroid** ,

. “ X , Y ” , .
: **,** • ,
, WeChat.

- SMS , ” ” .

- ,) .(

- ,) ,(” .”

- , ” + ,” Tasker,
.

- ” ”

– , . **.5** **:**
, :

:

: “ / ” iOS- Geofencing, GPS .
. Wi-Fi ,
.

- ” ” : ,iOS- ” ”)
  iOS). Wi-Fi, Bluetooth, – , .
- **Tasker** :) (,
  . , .
- : , 2-3” .” - 20
  – .
- **:** 5 : 3 ? ,
  , Bluetooth , ••• ,
  . 5 , . ,
  – .

:

, :

:1

: iOS: “ ” “ ” “ ” NetEase Cloud Music. Siri- “ ” . : Tasker- “ ” “ ” “ ” .7:00

.1 **:2** : iOS: ” ” 25%- ” ” + ” .”25% : Tasker Profile 0-25% ” ” + ” .”

2. **:3** **Bluetooth** : iOS: ”Bluetooth” Bluetooth ” ” Gaode Maps + ” .”80% : Tasker Profile Bluetooth Bluetooth ” Gaode” + ” .”

.3 **:23** **GitHub-** - ” ,”
, -GitHub.
, , , GitHub
. , .

:3 ” ”

- :7:00 + + .
- :iOS: Shortcuts > Automation > Time > Weather/Volume/App; Android: MacroDroid > Trigger (7:00) > Volume/URL/TTS.

- : , .

:23 GitHub- –

“ ” , , –
GitHub. , ,
, *GitHub* . , .

GitHub ?

Git. ,2008- ,2018- ,
, . **:** Git , GitHub

– , , . “ ” ,
“ ” – , , , . **, GitHub**
. **-** **,** **,** **,** **.**
“ ” **, GitHub** **.** .2 Git GitHub-) (!
, :

Git GitHub-) (!

Git

GitHub		( )
( )) (Git , , GitLab, Gitee (Gitee), Bitbucket	:	) , , (, , Linus Torvalds) Linux) SVN, Mercurial Git ” ,” GitHub ” ”

. Git , ; GitHub , Git , . Git-GitHub) ,(GitHub- Git-) GitHub Git)..**3**

GitHub

GitHub

: , , . “ ” .

- **(Commit)**: . Commit , ” .” ” ” Git/GitHub.
- **(Branch)**: . , , , , .
- **(Clone)**: , .
- **(Push) / (Pull)**: Push ,GitHub- Pull GitHub- . .
- **(Pull Request, PR)**: , PR, . .GitHub-
- **(Issue)**: , , . Issue , .
- **Fork**: . , . PR , .
- GitHub , :

: , .

- : , - .
- : Issue- , .
- : , (Projects) ,' Trello- .
- :
- , , , ”CI/CD) / ”(. : README , Wiki , Pages .
- : , , ” ” .
- : Star ( ) , Watch ( ) , Follow ( ) , .
- **.4 :**

:

.1 , ”Sign up”. ,) , , github.com/yourname

.2). . (Free), .

.3

.4 , ”+” , ”New repository”.

1. my-first-project
2. (Description), .

.3 (Public) (Private). , .
.

.4 ”Add a README file” –README ” ” , .

.5 ”Create repository”, GitHub !

.6 : README

7. README.md Markdown, . README
: , , , , . README ,
.

.5 :

GitHub, :

:

. : : .

.1 **git add**: ” ,” Git- ” .”

.2 **git commit**: ,” ” , ” .” .

.3 **git push**: GitHub, .

.4 **git pull**: , , .

.5 **,** **:** ” pull, , add, commit , push ”

. , . **.6** **:** **Pull Request**
GitHub . :

: Pull Request

:) main master) ,

.1 **feature/login**, . : , commit .

.2 **GitHub-** : .

.3 **Pull Request**: GitHub PR, . PR- .

.4 **(Code Review)**: , . commit
, PR- .

.5 : , ”Merge” . !

.6

, . – PR
, . **.7** **GitHub**

GitHub

- .GitHub- ,) GitHub Pages
,(. Markdown + Git + , .

GitHub . Fork , PR ;
Star . .

GitHub Actions, “ , + ” .
, , push, .

CI/CD-

, , git pull ,
.

-

- , : commit, commit .
.
- , : , , .
- **gitignore.-** : Git-) , ,
,(.
- : GitHub- Trending, , README,
, .
- **GitHub**: (Copilot, ,
,(’ .
- GitHub- , API, , –
, , .

,) ,(-

- git filter-branch
- , . , , . GitHub-
- GitHub- , - . : (1)) GitClone, ghproxy) ; (2) Hosts ; (3) ,

Gitee (Gitee)

, . GitHub- ,Gitee- - .
: **GitHub Pages**GitHub Pages ,
Jekyll, Hexo, Hugo, . Markdown, , ,
HTTPS . , , , .

9. : GitHub

GitHub “ ” ,

: GitHub

: , , .
, “, , ” “ + ” –
, . , **GitHub** : ,
, , “ ” . , ,
, . : GitHub
GitHub,) , , .(

.1 hello-github

2. , ”Add a README file”. , `README.md`) ,(
`- Markdown,` . GitHub)

.3 python-tutorial

4.), Star . ( ) `Git` , `Clone` , `README`
, `" ."` :24 Root BL

5. ,20 ,
ROM. , , .

:24 Root BL

,20 ,
ROM. , , .

Bootloader

: BL , .

- : .
- : , .
- **.2 BL) Xiaomi)**

BL (Locked/Unlocked) , (devinfo/eFuse/RPMB). , Bootloader
.

. , Bootloader (boot.img, recovery.img). :

.1 hash .

.2 = hash.

3. hash, = . .

BL = . kernel .

: - BL

BL, 5- :

	) (	
	ROM , : , , : , OTA BL = : = : BL / :	: ROM ,' , Root = : ROM ,' ,brick/ , OTA : BL = , : , , = : Google Pay/ , DRM

BL) Xiaomi)

.1 : Xiaomi.

2. : , 7-30 .

3. : Xiaomi .

.4 **.** **Root** Magisk

. Root

- **Magisk**: Root , - , .
- **SuperSU**: Root , .
- **.** **ROM**) ROM) . :

. ROM

: , .

- **Recovery -** : TWRP, , ROM . -
- **ROM -** : LineageOS, Pixel Experience, .
- , . :

. Samsung Knox: - BL

Samsung: “ ” : **Samsung: Knox , Samsung Pay , , . .** :

- **-** : eFuse, BL- , .
- :Samsung Pay) ,(/ Knox , MDM , - .
- : ,) .(
- :Huawei Knox .

Samsung Knox . Root, Root.

“ ”) (

-
-
- , , .
-

XDA Developers

: , ROM.

- **Coolapk (CoolAPK)**: , .
- : Xiaomi, OnePlus, .
- **:25** , , ,” ” ” ” . _ . ” ” , ?

. (Q& A)

Q1: BL?

A: .) Xiaomi)`flash_all_lock.bat`) . ,) Knox Samsung).

Q2: Root ?

A: BL . Root :

.1 Recovery) TWRP).

2. Recovery Magisk boot Root.

:25

: “ ”, “ ”.

“ ”

: “ ” “ ”. , ,
, , , - “ ” .

.1 ” ” : , (iCloud, Huawei/Xiaomi/Samsung, Google),
” ” - .

.2 ” ” : - , . ’

.3 ” ”) (: (iOS 8+, Android 6+) (FBE),
, , .) ,(2015
.

: , “ ” , :

OnePlus	< < <	Huawei ” ,” ; .
Xiaomi / Redmi	< <	” SD ” ” ,” ; Xi- aomi ” .”
OPPO / OnePlus	< < <	ColorOS ” ,” ; .
vivo	< < <	” ” ;SD- vivo, .” ”
Apple	< < <	Samsung, ;” ” ” ” .
	< < / <	Apple ID, ; iOS 15+ ” iCloud”.

- : , ” .
- **SD card** : SD, . /
- **USB/OTG** : , .

: , iCloud.

“ ” . , , , “ ” . , ? - :
, , .

: 49

,2021 :49 , ; / / / , .
:

- .
- ” ” = , , .
- / ,( ) .
- ,” ” .

WeChat/Alipay: ,

App , :

- **WeChat** : , ,Tencent- , .
, , / / .
- **Alipay** : , / / . Alipay/ , ; /
.
- ’, : . , . /

) ” ;(”

- ” ” — , . /
- ” ” _ .
- : = , .

Apple “Legacy Contact”

iOS 15.2 “Legacy Contact” - , + iCloud-
,) , .(:

.1 iPhone- [ ] < [ ] <] [<] [.

.2 ," " Face ID.

3. , .

4. , / " /SMS- .

.5 , .

Apple, + . 7 .

Google "Inactive Account Manager"

Google: Inactive Account Manager, - .

.1 , "Sign up".**myaccount.google.com/inactive**, Google.

2. 3/6/12/18 .

.3 10 , .Drive/ /Gmail-

.4 .

.5 : .

6. , .

:

: 2 : " ") (. , ;
. , AI " ", . " ") (. .

- **Weibo**" ": - , , " ." .
- **Douyin** ": < < < < " , , . /
- **Bilibili**, : , ; / .
- **Xiaohongshu**, : , .

- , .
- , .
- - , - , .

, , :

- **(Bitcoin, Ethereum (’** : ” = ,” = .
” ” . / ,(+) .
- : , (Binance, OKX), .
- , : (Aliyun, GoDaddy).
- **,) ,** **, (** :) ,(50 .
- : ; , .

“ ”

Chainalysis, **3** **Bitcoin** = , ~1/6. , -
.blockchain-

“ ”

. **“ ”** , . :
“ ” ( )
:) , (

- : WeChat, Weibo, Douyin, Bilibili, Xiaohongshu
- : Alipay, WeChat Pay, App
- : iCloud, Huawei, Xiaomi, Google, Microsoft
- : ” QQ Mail, NetEase Mail, Gmail ’
- : , , ,

:

- : Bitwarden/1Password, , .
- : / , 2 .(+)
- : , ,” , .

:

- .
- iCloud/Google: Legacy/Inactive.

- , .
- .

:

- , , .
- .
- : _____/___/___, : _____/___/___.

” ”

= , . **:** = + + , . + .

: , , .

: , , , . = . 3 :

			.3 HDR:
(Huawei/Apple/Xiaomi (’ - (Aihuishou, Zhuanzhuan (’) , ( ) (	, ,	, , ,	, ,

- , : , . Apple/Huawei/Xiaomi ”trade-in”.
- : , SF, . ” ” , .
- : , / . / **:** + **SIM/SD**.
- **:** ” ” , : Smart Home, , , . .

, , , , = x10. = , / / . = + .

.

.1 : , , . ’

.2 : iCloud/Huawei/Xiaomi/Google, .” ”

.3 SIM/SD, OTG.

4. , .” ”

.5 ( ) , .

.6 : trade-in, , , .

7. ” .”

;

.1 Legacy Contact / Inactive Manager.

2. , .

.3 +2 , .

:26

, *2024-2026 = : AI* , , “ ”, .
“*?* ” “*?* ”.

,’ AI Phone: ?

,2024 Android “AI Phone”, Apple Intelligence iOS 18. “AI Phone”? 3 :

- : 7B (Qwen, Llama, DeepSeek), : , , . : .
- **NPU TOPS**:NPU .TOPS- 2024 (Snapdragon 8 Gen3, Dimensity 9300) 40-70 TOPS, 2025-2026 100+ TOPS - .
- **AI Agent** : ,” / ” App - ” .” ” AI”.

,’ AI: ?

AI , :

- **Apple Intelligence (iOS 18+)** : - , ,” , , . + Private Cloud. , .
- **Google Gemini Nano**:Pixel 8/9, ”Circle Search”, Magic Compose, . , .
- **Pengpai AI Xiaomi**:Xiaoyi, . / / / HarmonyOS NEXT 5.0 .
- **Xiaomi MiLM**:HyperOS, AI . / / / .” + ”
- **OPPO (Andes) / vivo (Blue Heart)**:ColorOS 15/OriginOS 5, / / / ”) .(”

:AI = + ; AI = .

,’ :

= . 2022 Mate 60/iPhone 14 , 2025 2000-3000.

- **BeiDou SMS** :BeiDou SMS (1000 + (,Redmi/realme/iQOO- . . /
- **Tiantong** :Huawei Mate 60/70, Honor Magic6, . .(/10-200) .
- **iPhone Satellite SOS**:iPhone 14+ SOS, + . 2 .
- : **1080P/4K + 60fps,** . : = ; / = , .

,’ : ?

: “ ” “ ” . 3 : , App, .

- : +500,000 (13 ,(, IPX8. UTG + = .
- **App**:WeChat/Taobao/Douyin/Bilibili/Office/WPS , 30% . .
- ,90-120°: ” / ,” . / / = .
- **Split-screen**:2 App split, 3 . HyperOS/ColorOS/OriginOS , iOS .
- : ,5000-7000 4000-5000, 2025 3000 = .

/ / : = ; / = . 30 .

,’ 6G Wi-Fi- 7

6G , :

- **6G**:3GPP 2027 , 2030 . 1 Tbps (50x 5G), 0.1ms, .” + ” , .
- **Wi-Fi 7(802.11be)**:2024 , 46 Gbps (4x Wi-Fi 6), 320MHz, 4K-QAM, MLO. 2-5 Gbps, 2-3x Wi-Fi 6E. Wi-Fi 7.
- **Bluetooth 5.4 / LE Audio**:2024-2026 Bluetooth 5.4 + LE Audio, 2 , Auracast.

,’ : -

“ ” , 2024 - = .

- : 372 mAh/g, 4200 (10x), 300% . - = , .15-30%+
- 2024-2025: (Xiaomi 14 Ultra, Honor Magic 6, vivo X100 Pro), 7-8mm, 5000-6000 mAh, +800-1000, .

- **100W+** :100/120W 17 . 2025 150-240W, 10 .80%
- : = - + 100W+, 2024+.

,’

- AI Phone: , NPU TOPS, AI Agent.
- : BeiDou , Tiantong .
- : , App 70%, .split/
- 6G , Wi-Fi 7 , LE Audio.
- - + 100W: + .

: ” ”

“ / / ”, AI/ / /Wi-Fi . / /7 , .

- : 5000mAh+, 80W+, NFC, IP68.
- : AI , Wi-Fi 7, BeiDou.
- : , , 240W.

- AI Phone: , NPU TOPS, AI Agent
- AI: Apple , Google , Huawei/Xiaomi/OPPO/vivo
- : BeiDou , Tiantong
- : , App 70%, split/
- 6G 2027, Wi-Fi 7, LE Audio
- : - 5000-6000mAh

.) 27-32)

:27

, . “ ”, *2* . “ ” - .
.

,0 : “ ” , “ ”

, : . , , - , . , . =
.

3 : (1 —10 , 3 . (2 — “Wi-Fi ” . 3) — , .

,; “ ”

) “ / / ” ,(:

- — .
- —, SMS, , , WeChat. .
- —) , , (, .
- —, , - .

) :(

	) , (	< < , . ” < ” ” ” ” ”
~27W		< < <
Xiaomi/Redmi		< < <
OPPO/OnePlus		< <
vivo/iQOO		< , <
Apple		< <

, “ ” “ ” . , “ ” .

,’ , - 5

, “ ” . 5 :

.1 : < < , . ” ” , .

.2 : <) ,( ) 3x5), .

.3 ” ” : , .” ”

.4 : < , , SMS, WeChat. .

.5 : < Widget > , / . / /

: ” ”

.1) .(

.2) < < ,(.

.3 , Bluetooth, NFC, .

.4 4 : / , , , .

,’ -

= . , . .

6 :(!)

- **(1 ” , ” — ” , .” : ” ” = . .**
- **(2 ” ” —SMS + .” ” : SMS . .12333**
- **(3 ” , ” —WeChat/SMS ,” / ” . : +5% ” ” = 99.9% .**
- **(4 ” , / ” —. : , ” ” = .**
- **(5 ” / ” — . : = .**
- **(6 , ” ” —App (Sunflower, TeamViewer). : ! /**
  .

3 = 90% :

.1 **:1** —Android ” ;” iPhone ” ”) < .(

.2 **:2** —SMS ”KTFSR” ;10086- . .

.3 **:3** — ” ” App, .

” ” ( )

- + !
- **SMS +** !
- + !
- !
- !
- **?** .

,’

. “ ” = . :

- **~27W** ”: ” App, ” .” < < .
- **Xiaomi/Redmi**:Xiaomi ” ,” + .
- **Samsung** ”: ,” / .” ”
- :”Sunflower” App ,( ) . = .
- :FaceTime ” .” , .” ”

,’ WeChat

WeChat . :

- **WeChat**:WeChat > < < < , . WeChat.
- ” ” :WeChat > < < . / , , .” ”
- ”: ” - .” - ”
- : (Android) / (iPhone) = . , ” ” . -
- **”Discover”**:WeChat > < < < Discover, .

,’ -

, :

.1 ” ” :iPhone: < < ; Android: ” .” .

.2 :Apple ID/Huawei/Xiaomi , + . = . /

.3 : ” ? ”. iCloud/Android- Find ”Lost Mode” - .

,’

. iPhone/Android MFi, .

- :iPhone > < < , Bluetooth, .
- : < / < ” ” + .” ”
- **HarmonyOS** : < ” ,” .” ”

,’

- **SOS** :5 Power, + . iPhone: ”SOS”, Android: .” ” .
- :Apple Watch, + = + . .
- :iPhone ” ” . Android: / App.

- / =
- 3 : , ,
- 6 + 3 = 90%
-
- WeChat: ” ” +
- SOS + : 5

:28 –

: , , .5000-6000- - “ ” .

,0 3 :

“3000 ” - . :

3

.1 ? : ? ? ? .

.2 ,” ”: .” ”

.3 : + .

,’ 5

, :5

SoC ? :	. :2026 Snapdragon 8 Elite Gen5/Dimensity ; =9500 8s Gen4/8400= - ; +7 Gen4/7350=3 . :6> . OLED > LCD; 120Hz > 60Hz; 1080P+; PWM 1920Hz+. , - < . - ; 3x+; = . 4500mAh , 5000mAh+ . 60W+ , 100W+ . . Apple 5-6 ; Samsung 4+5; Pixel 7; 2-3.	SoC = OLED+120Hz+PWM = + < 5000mAh + 60W = ?

,’ (2026)

2026 , . .

:1000> + WeChat

, : , .

:Redmi Note, Honor X, realme. Snapdragon 6/Dimensity 6000, LCD.

: 5000mAh+, 6GB+128GB. Redmi.

:1000-2000 -

, : , .

:Redmi Turbo, iQOO Z, Honor X, OnePlus Ace.

:Snapdragon 7+ Gen3/Dimensity 7300+, 8GB+256GB. 3-4 .

:2000-3500 -

, : , . .

:Xiaomi, vivo S, OPPO Reno, Honor, Redmi K.

: “ ” - OIS, OLED, 80W+, 12GB+256/512. 4-5 .

:3500-5000 -

, : .

:Xiaomi, vivo X, OPPO Find X, iPhone 16.

: . : < . < . “ ” .

:+5000

, : , , .

:iPhone 17 Pro, Samsung S26 Ultra, Huawei Pura 80, Xiaomi 17 Ultra, vivo X300 Pro.

: . : vivo/Huawei; : iPhone; :S-Pen+ Samsung; : Xiaomi.

— .

,’

:

” ”		
	,”1 OIS, 3x+, ISP	vivo X, Huawei Pura/Mate, Xiaomi Ultra
	SoC , , , +	iQOO, ROG Phone, Red Magic, Redmi K
	5500mAh+, SoC , LTPO	Honor Magic, OnePlus, Redmi Note
	, , , ,	Huawei, Honor Play, OPPO A, Redmi Note
	, , ,	Redmi Note/Turbo, realme GT, iQOO Neo
Apple	Mac/iPad/AirPods, iPhone	iPhone 17 ( ) Pro) (
”	eSIM, , Google	iPhone, Samsung S, Pixel, OnePlus

,’ -

1. **:1 4-5** — , 2- ” ” ,(+) . , + + . .

.2 **:2 100-** — . , , . 50-100 .

.3 **:3** — ” ” , , , . = .” ”

.4 **:4 ” ” -** — , : , , . 2026- ” ” .” ” ” .”

.5 **:5 ”AI ”** —2026- ”AI ”, . AI : , , , , . .

,’ ?

- **618**) (: . .800-1500- .
- **Double 11**) (: , .618- , .
- : , ’ .200-500-
- : , .
- : - , .

,’ - ?

- **Tmall/JD/** : , , .
- **Pinduoduo ” ”** : , . . .
- : . : . 2-3 .
- **(Xianyu/Zhuanzhuan)** : . : , IMI, **,** .

:

1. :) , .(
.2) .(3500
3. JD/Tmall, , 3-5 .
4. Bilibili- ”XX ,” 2-3- .
.5 , .

- : < <
- 5 : , , , , /
- 1500-3500
-
- 5 : , , , , AI
- 618 Double- 11

:29 –

5 ? ?2- .
“ ” “ ” .

,’ “ ” -

, . : “ ” (iPhone) “ ” (Android),
.

- : < < . , , .
- **HarmonyOS** : < . Huawei ” ,” Xiaomi ” .” .

80% . 4 ,Douyin- 28 - . “
” .

,’ “ ” -

, . , “ ” :

1. : < < < .) Douyin), 30 . , .

.2 **HarmonyOS** : < < . , , .

.3 : . , ” ” .

,’ “ ” -

, . “ ”, 15-20 .
“ ” “ ” .

- : .22:00-07:00 .
- , : ” ” ” ”) 3- .(, .
- ”: ” iPhone- - ” ” , ” ” .

,’ - 20-20-20

.) (, :

20-20-20

20 , **20** **6-**) (’ **20** . .

- : < < , , . .
- : . . ” .”
- : 30 ” . , .

,’ “ ”

5-6 . , , . :

- : , . .
- **30** : , 3 , ; , 3 . 5 . ” ” .
- **3** , : , , , 3 , .

,’ –

, “ ” . “ ” , , . :

- : . , ” ” / .
- : , . , - .
- ” ” ” ”: iPhone- ; Android- .
- : , , ” .”

,’ – “ ”

. , , , . : “ ” “ ” :

- : , ” ”? , . .
- : ,WeChat- , SMS, - . .
- ” ” :) ,(. , .

:

.1 : , / .

.2 : 30-60) .(

3. : ,(22:00-07:00) .

4. :4 , .

.5 :5 5 .

.6 :6 ” ” .

.7 :7 , .

- ,
- , ,
- 20-20-20
- ,5-6 +
- ,
- ,” ” ” ”

:30 ,

7 ,’ 55 .100 . , .

,’ -

, :

AirPlay	iPhone/iPad/Mac	, . Apple TV AirPlay
Miracast	Android	Wi-Fi Direct, . Android-
DLNA		,(/ /) .

:

- : < < Apple/ TV. , Apple TV .
- **HarmonyOS** : < ” ”/” ” < . Wi-Fi.
- : , Chromecast , .HDMI-

,’ - ,

, . :

- : Lightning/USB-C HDMI- .(100-200) .HDMI-
- **HarmonyOS**:USB-C HDMI-) DP Alt Mode, .(USB-C to HDMI.
- : , HDMI . HDMI.

,’ -

Android “ ” - :

- **Samsung DeX** : Samsung. , , . Bluetooth, .
- **Huawei**:Huawei Mate/Pura , . , , .
- **Motorola Ready For**:Moto , .DeX-
- **3. HDR:** : ,) ,(PPT, . Excel- .

,’ -

, , , . :

- **Apple CarPlay** : iPhone. USB , iPhone - , , . Siri .
- **Android Auto** : .iPhone- Google Maps, Spotify. .
- **Huawei HiCar** : Huawei, . , , , .
- **Baidu CarLife** : . , .

,) ,(-

, . , .

,’ -

. :

- : . 10 , . Wi-Fi .
- **Plan- B** : HDMI. , .
- , : WeChat . ” ” ” .”
- ” ” ” ” : PPT, ,WPS- . .

:

.1 : iPhone ” ;” ” .”

.2 .

.3 (iPhone: , Android: .(

.4 , DLNA.

5. , , .

- 3 : AirPlay, Miracast, DLNA
- ,
- DeX/Huawei =
- : 4
- : , Plan B,

:31

,2026- “ ”, . - ,
Premiere. .

,’ : ?

: “ ”. 3 :

.1 **?** —, , - .

.2 **?** —, , , . .

.3 **?** —Douyin/Kuaishou, Xiaohongshu, Bilibili, Video Account. .

,’ 5

.1 **1080P, 30fps** : . 4K , 60fps .

.2 : . .

.3 : 2 , ”AE/AF Lock” .

.4 **?** —Douyin/Kuaishou/Xiaohongshu (9:16), Bilibili/YouTube (16:9).
.

.5 : < < . .

,’ 5

: “ ”. 5 :

- **Push**. : .
- **Pull**. : .
- **Pan**, : . , .

- **Track** : , . .
- **Follow** / : . .Vlog-

: . , , (DJI OM, 500-800).

,' - JianYing

JianYing (CapCut) . , , . :

.1 : < < .

.2 : , . , .

.3 : < ." " " ." " 3D.

4. ": " . 90% , .

.5 : . BGM , 1/3 .

.6 :1080P, 30fps, ." "

,' - 70%

" ", . :

- : . , , .
- : "Clickbait". , , .
- : , . .
- **JianYing** : , , .

,' 3

- **3** 3- : . " " - , , .
- : : . , .
- < : ." " 20 , . 2-3 , .

: 30

1.) " .("

.2 3-5 , 5-10 .

3. JianYing: , , , BGM.

4. .

.5 , , .

- : , ,
- 5 : 1080P/30fps, , , ,
- 5 : Push, Pull, Pan, Track, Follow
- JianYing: < < < BGM >
- 3 , ,

:32

“ ” . , , —
; , , , . ,
— , .

:0 / —

, . 13 “ , ” ;
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.1 : , ,) , , ,(; ,
.

.2 : , ; — .

.3 — : , ; , , .

.4 : ”) ,XX- , / (” ;
SMS/WeChat — .

.5 : 7 SOS /) ;(; Bluetooth
.

.6 : / , Bluetooth, —
; ” ” .

/ / : —

- : , , - , .
- : ,) ;(119/110-
.
- : ; . /
- : ” — ”WeChat/SMS- — .

4

.1 / , .

.2 2-3 ” ” ,) / .(iOS/Android-

.3 ; — .

.4 ” ”) (— .

:1 / —30

30 “ ” — . , :

30-

.1) :(1 , —.

.2) :(2-5 :

3. iPhone: iCloud.com > < <

.4 Android : Huawei cloud.huawei.com / Xiaomi i.mi.com / OPPO cloud.oppo.com

.5 SIM) :(5-10 (China Mobile 10086/Unicom 10010/Telecom 10000), SIM. — SIM- .

.6) :(10-20 Alipay: 95188 , Alipay > < < <

7. WeChat: 110 WeChat, WeChat >

.8 : ,

.9) :(20-25 Moments- — .

) :(25-30 110 , . .

:2 —72

.1 , . . :

.2 : — . ” ” — .

.3 : ! ! , . .

.4 :96110 . —) 24 ,(.

.5 110: , .

.6 / : , . — .

: Apple ID/ , WeChat, Alipay,

, . -

:3 / — ?

- — , , , — :
- :1 —Android- , , ” ” .( ) — .
- :2 — < , , .
- :3 — ”Tencent Mobile Manager” 360” Mobile Guard” .
- :4) (— , . : — , .

iPhone — sandbox- iOS- . — , . 99% .

.1 :4 —

.2 : — . .

.3 : ” ” — .

.4 : 7- , — .

.5 : - —. .

.6 : ” / . — .

: WeChat/Alipay — , ,

.

:5 — ,

- : ”(1) / , ;” ”(2) , .” :
- 99.9% : . — .
- , : = ” ” — ” ” .
- : .
- : — , .

: **.** **.**

:6 AI) (/ —

,2024- AI “ ”, . “, ,
” — , . .

- :1 — AI
- : 15-20 , AI- .
- : .
- : , —” ” .

: , ” ” .

- :2
- : , , . / AI —
.
- : ” ”? — .

: , . — .

.1 :3

.2 : ” ” —. ”, ” — .

.3 : , , , , — .

.4 96110: ”AI ” / — .

.5 110: , AI- — .

: **— Moments-**

.

:7 SIM swap) SIM) — “ ”

. SIM swap: SIM . —
, —Alipay, , WeChat — .

- ?
- ” ”/” SIM” , — .
- ” ”/” ” — .

, — .

- — SIM-
- PIN :SIM- < < PIN SIM. , / PIN. — .
- - : App- China Mobile/Unicom/Telecom, ” ”/” ” — - . .
- SMS: , Apple ID — (Google Authenticator, Microsoft Authenticator) , SMS.

watermark: “ XX” — .

1. —
.2 : 10086 / 10010 / 10000 — .
.3 : , , Alipay, WeChat, Apple ID, .
.4 : Alipay 95188, WeChat, .
.5 : .
.6 : SIM swap ” ” —) , , .(

: , — .

.1 —5
.2 + - —
.3 ” ” —
.4 PIN SIM-) < < PIN SIM) — ,
.5 ,

,) , (

.1 : ” ”
.2 : ,(10086/10010/10000) , ,96110 110, .
.3 ” Apple ID/ .
.4 — , .

: ” ” ? PIN SIM- ?

-
- 30 : SIM

- 72 : 96110
- : ( )
- : -
- 99.9% — , ,
- AI: / / / — , ,
- SIM swap: ” ” = ; PIN, ,

.) **33-34**)

:33 –

, . *3 :* , “ ”, .

,’ AR/VR: “ ”

. AR/VR:

- **Vision Pro**:Apple Vision Pro = .” ” . / 2026-2027 . / iPhone = .
- **-** ”: .” Meta Ray-Ban, Rokid, XREAL, INMO. 2025-2026: .HUD/AI/ 3-5 .
- / :Galaxy Ring, Oura. = + .

,’ BCI

, :

- **Neuralink**:2024 , . .
- **BCI -** :Kernel, OpenBCI, , .
- 10-20: . . / . /

,’ “ ”

: “ ” - :

- : .
- ” ” ” ” : .
- . **:** **?** : = , = .

,’

, “ ” ?

- **AI**:AI > ? ?
- ? : GDPR = .
- :AI/ / = . ” ” 20 .
- . , –
- , .

,’

10 :

.1 ” ” .

.2 ” ” ” .”

.3 .

.4 : ” ”

.5 ” ” , . :

:1 . / , ” .”

“ ” :

- :3 . ,
  .
- .
- , , ! ” ”
  . .

- Vision/AR = ,
- BCI 10-20 ,
- ” ”
- AI/ / = 10
- +

:34 –

= “ + ”. + =. *3*, , .

,’ 3

3 :

- : =, + . CarPlay, Android Auto, HiCar, CarLife. : ; : .
- : + . HarmonyOS, DiLink, NIO OS, Li Auto.
- **CarPlay 2.0**:2025, - . /AC/ Ferrari, Aston Martin, Porsche.

,’

:

Apple CarPlay Android Auto Huawei HiCar Baidu CarLife	iPhone Android Huawei iOS + Android	/		

,’ HarmonyOS

CarPlay = “ ”, HarmonyOS = “ ”. :

- - :App . , / .
- : .
- :Huawei ID, .

,’

“ + ” :

- **NIO OS**:NOMI, . - NIO Phone = . +
- **Li Auto**, : .” ”
- **Xiaomi HyperOS**. + + : AC.
- **BYD DiLink** : , App Market.

,’

, / ? , .

- **Ready For / Desktop**:Motorola/Samsung/Huawei, . .
- **vs** . + : . / /
- : = ; = CarPlay/HiCar.

,) ,(-

+ = . = “ ” . . /

,’

- **CarPlay** :Bluetooth ; ; .
- :Hotspot; Wi-Fi; .
- :iOS Focus; Android Tasker; Huawei.

: CarPlay/HiCar, , App, OTA.

1. ?

.2 Bluetooth.

3. Focus.

4. App.

- 3 : CarPlay, HarmonyOS, CarPlay 2.0
- CarPlay > HiCar > Auto > CarLife
- HarmonyOS: / / -
- NIO/Li/Xiaomi/BYD
- =
- CarPlay/Hotspot/Bluetooth

) (35

:35 –

“ ” “ = ” . .

,’ ?

, . = ; = .

- **3-6**. : , 15 ,’ , . . +
- **7-9**,” ”: , .
- **10-12**, / : .
- **13+**, ” : + .

” ”

. .1500-2500 .

< . , + : /

- **Xiaotiantong**.” ”:
- **Huawei**, : .
- **Xiaomi**. :

: **4G**, 2G/3G.

,’

, .

iOS: Screen Time

- **App Limits** : < Screen Time > Limits.
- **Downtime**:22:00-7:00.
- **Content**, : IAP.
- **Family Sharing** : .

Android: Family Link

- **Google Family Link**:Google, .
- **Xiaomi Kids** : < Kids.
- **Huawei Student** : < < Student.
- **OPPO/vivo Kids**.App/ :

	iOS Screen Time	**Android Family Link**
App Limits	,	, App
Downtime		
Content		
	Family Sharing	Family Link
App		Family Link

” ”

. < . .

,’ Teen Mode

, :

- **Douyin/Kuaishou**:40 ,’ ,22-6 .
- **WeChat**. / :
- **Gaming**, : / / ,20-21 .

“ ” “

“ . .

,’

90% , 10% . “ ” < “ ” :

- : . / /
- **Bullying**. / / : .
- , / : Clickbait.
- : ;QR/App/ ; = .

” ”

+ :

- : 19:30-20:30.
- : . / /
- : App/ .
- : .
- : .” ”

,’

: / /

- :3-6: 1h; 6-12: 1h ; :+12 2h.
- ,20-20-20: 6 .’ 1 .
- : , 30 .’
- ”: .”

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. .

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: / /

- : . Legacy Contact.
- :<14, . .
- / ” : .

. “ ” = .

- :3-6 ; :13> /
- :” ” ,1000-2000
- Screen Time + Family Link
- Teen Mode + Gaming,
- <
- =
- + ” ”

:36 —

— “ ” . , — “ ” “ ” . , , , — “ ” .

:

— , “ ” .
, :

- Matter: 2022- CSA) ,(Apple, Google, Amazon, Samsung . — . 2023-2026 , Apple- HomeKit, Google Home, Amazon Alexa Mi- Home — .
- Thread: ” ” Matter. ,) — ,(, . Thread (HomePod mini, Apple TV 4K, Xiaomi) — .
- Wi-Fi: , , — , .
- Bluetooth (BLE): , , — , .
- Zigbee: , , , — Matter+Thread.

2026- “Matter over Thread” — . -Wi-Fi/Bluetooth —, .Matter-

“ ” , “ ” . ,Matter- - . :

Apple HomeKit	) ,(iPhone/Siri/	, HomePod Apple TV	Apple,
Mi Home (Xiaomi)	, - , XiaoAi	, ,	,
Huawei Smart Life (HarmonyOS)	HarmonyOS , / /	, ”	Huawei, ”Super Device” -

: iPhone HomeKit, Huawei Smart Life, Android Mi Home. Matter, — “ ” .

, “ ” — , , , , “ ” :

- : HomeKit/Mi Home/Smart Life — / , , , , .
- : ,” ” : : + + +26° + .
- : /) (+ + + + .
- : 23:00 + + +26° .

NFC: NFC —) , .(iOS- ” ,” Android- ” ” .

: ” ”

- “ ” : “ ” ,( ) “ ” “ 26°” “ ” . , — ? .

— 80% :

- : XiaoAi (Mi Home, (/ HomePod mini (Apple, Thread) / Alexa) .(: — .
- : Yeelight (Mi Home/Matter, , / (/ Philips Hue) , , Hue, .((E27/B22) .

- : Xiaomi Smart Plug ,) (/ TP-Link Kasa (Matter .(”) , (— .
- : Loock/Xiaomi ,) Mi Home) / Kaadas) , .(/Matter- , .

) ( ) , ( ) —.(, — “ ” .

— . , :

.1 : , — , .

.2 : — ” ,” .

.3 : IoT Wi-Fi) ,(. — .

4. : ” ”) , ,(;24/7 .

- ” ” — .
- — .
- ” ” — , .

- Matter+Thread 2023-2026 —
- : Apple , Mi Home , Huawei -
- , ” / / ”
- : , , , —
- , , , —

. AI) 37-47)

:37 — “ ” “ AI ”

! .

.2025- ? .

. Vibe- Coding:

,2025 Andrej Karpathy — , —
: “ Vibe Coding: (vibes), , AI- .
, , —. ”

: , -
, AI- . Karpathy , : AI- ,
“ ”, “ ”.

: Vibe?

, Vibe “, ”. Vibe Coding “ AI ” — ,
AI- . : .

“ ”: ; .
, .

Vibe Coding . 2026 —
, . “ ” ,
AI . .

. :2026- Agentic- Engineering

2026 AI . Vibe Coding , :
AI- “ ”, — , . :
Agent .

Karpathy — 2026 Agentic Engineering. “ — ” Vibe Coding,
Agentic Engineering : AI- , , , . AI-
, (Agent) , .

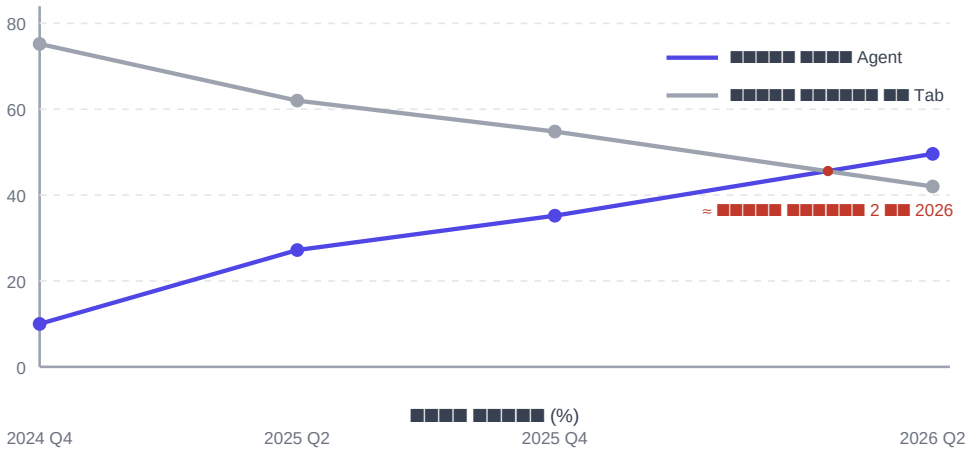
:

1. : ” - ” ” .”

.2 : ” ” ” .”

.3 : ” ” - ” .”

: Cursor 3 ,2026 Agent Tab- —
; Claude- Code, AI , 2026
2.5 . :



:1 AI) ; 2026 Agent
(Tab-

: . Cursor Claude Code,
GPT-5 Claude 4.5 — AI .
.

. : Coder- Conductor-

Coder — , API . AI
: , “ ” (Conductor) — , ,
.

“ ” , “ ” .
, , . , AI
, , .
— :

	Coder	**AI (Conductor)**
	, API, - — ,	, , AI- , ,

Coder AI

. : AI- , ;
, AI- . “ ” —
“ ” “ ” .

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, . ;
.
AI “ ” “ ” . ,
. 2026 2025 —Trae, Qoder
, Cursor IDE- Cursor- 3, Claude- Code .
, .
— . ,
: , AI, .
; .

, “ ” . ; “
” . : -) ,(“ ” “
” .( ) , , “ ”
“ ” .

: ” ”

: (1) AI-) ?((2) -
“Coder” “ ” ? (3)
?
, 47 — .

- 2025- Karpathy AI —”Vibe Coding”: -
- 2026- Agentic- Engineering: , , AI- ,

- Cursor- 3 Agent ;Tab- Claude Code 2.5-

- Conductor :( ) , ,
- , — ,

:38 AI

, . , —, ,
— . : , “ ”
.

: Andrej- Karpathy , CLAUDE.md 70
, AI. 70,000 GitHub-
AI . .
— — .

70 ?

70,000 ? GitHub- . 70
: AI- , “ ” .
“ ” .

. : Think Before Coding) , (

- . AI , “ ” ? :
AI- . , AI- “
” — “AI ” .
“ ” AI- , “ ” :

.1 ? ( )

.2 ?) (

.3 ? ( )

	( )	( )
	) (	AI , , , , HTML+CSS+JS , , -

: ≠

“ ” “ PRD ” . —
. ;
) .(40

. : Simplicity First) (

AI “ ” , “ ” . AI- “
” , —, , , —
. - AI.
“ ” “ - ” : ,
. ’ Karpathy- : “ ,
. ”
, “ ” :

- AI- , ;
- ;
- 50- , AI- ;500
- AI- , ” .”

	( )	( )
-	MySQL + ORM OAuth2.0 + JWT + React + Redux + Router Docker + CI/CD + K8s	JSON SQLite HTML + vanilla JS ,

. : Surgical Changes) (

, — .
AI, AI- “ ” : , .
“ ” AI- . “
calculate_discount” , “ ” , “ ” . AI- ,
.

Karpathy diff) .(AI- “ ”
: ; — “ ” ,
. AI, “ ” .
) (

) (

```
calculate\_discount cart.py: 1. .2
\10% .3 .4
```

. : Goal-Driven Execution) (

: AI- “ ” “ ” ? Karpathy: .
AI- “ , , ” , “ CSV
” . AI- , .
“ ” . AI- ,
; “ ” “ ” — .

: ≠

AI- : ? ? - ?
, — “ ” , “ IndexError; ”
.

) (	” Python Data-Parser, parse filename csv⋯ ”	AI- ,
) (	” : CSV, JSON. ”.	AI- ,

. : Iterate and Verify) (

Karpathy. : AI- .
AI “ - - ” , “ - ” . -
“ ” “ ? ” , . AI- , :

- — : ? ?

- — : , ?
- — : ? (hardcoded)? ?

AI-

,

:2 AI

“ ”. , “ , ”, “ IndexError
; 15”. AI- ; .

. Harness Engineering: Agent = Model + Harness

2026 Harness Engineering AI. :
Agent- = × Harness-
“ ” ;AI- Harness- “ ” — ,
, , .
: , GPT-4 Claude,
? .Harness- Harness : , ,
. , Harness- .

— , ,
, . Harness Engineering “
” .

, , Harness : 39) ,(40
(PRD), 43) Prompt), 45) .(Harness-
.

: ” ”

) , .(3 : (1)
? (2) ? (3) ,) , ?(. —
- “Think Before Coding” , .

- CLAUDE.md- 70 Karpathy) 70 ;(—
” ”
- Think Before Coding: — ? ? ?
- Simplicity First: AI ” ” —
- Surgical Changes: , ” ” AI-
- Goal-Driven Execution: ” ” ” ,” AI-
- Iterate and Verify: (/ /) + ,
- Harness Engineering: = × Harness — AI- ” ”

:39 — AI-

: , AI- ? AI- — “ ” . AI- “ ” , 2026 — (Spec-Driven Development).

. : “ ” AI-

AI : “ ” . AI- .
“ React”, “ - Node.js”, “ PostgreSQL”, “ — ” . , — .
“ ” — , AI- .
:

Cursor	`.cursorrules`
Claude Code	`CLAUDE.md`
GitHub Copilot	`copilot-instructions.md`
	`AGENTS.md`

, : AI- “ ” . :
) ,( ) ,(,) , ,( ) (
) , , .(CLAUDE.md- Karpathy 70 ,
— .

```
#    :
##
     full-stack                                    .
##
- Frontend: React + TypeScript + Tailwind CSS
- Backend: Node.js + Express + TypeScript
-       : SQLite (   ( / PostgreSQL (   (
-    : Vitest
##
-
```

```
-      2
-          -camelCase
-           -PascalCase
-                   JSDoc
##
/src
  /frontend  -
  /backend   -
  /shared    -
  /tests     -
##
-     commit
-
-      -API        } success, data, error }
```

: ” ”

— . “ AI- ”,
“ , ” . , , “ AI- ” .

. (SDD)

Spec-Driven Development (SDD, (,2026-
: , . TDD-) ,(“ ” SDD- ,
, , . AI- , .
SDD :

.1) .md);

2.) ;(

.3 AI- ;

.4 ;

.5 — AI- .

“ ” — , .

?

AI-

/

;

:3 (SDD)

SDD: . , , AI
. “ ” — .

•

, “ ” . AI) (-
. , AI- “ ” —
.
“ ” :

:	, ,	CLAUDE.md,
:		,@
:		,
:		, ,

•

- . :

.1 . — ,
- .

.2 @- . AI @file @folder,
AI- — .

.3 ” .” . .
, .AI-

.4 AI- ” ” ” ” . , @- ,AI-
” .” AI- .

.5 . .
AI- .

: 70

) ,(CLAUDE.md: 2-3) ,(
,( ) (3-5 ,( ) ((2-3 .(70- —
.

- AI- :" "
- (CLAUDE.md (" " ,AI-
- = + + + + —70
- SDD — : ; " "
- : @-
- : , ,@- , ,

:40 — AI-

“ *AI-* ”, “ *AI-* ”. — ,
. , .

•

— “ ”, “ ”, “ ”.
, ,AI- .
AI- .
: PRD . , “ ”:

		:
Who	?	,
What	?	,
Why	?	
How	, ?	- ,React- , -
Constraint	?	, ,

; . “ ”.
, .

•

, . - . “ ”, : - , .

“ ” — :

1. ,) ;(

.2) , ;(

.3) ;(

.4) .(

.

: ” ,” .

” - .”

. , — ’ ,

) .(, AI “ ” —

, AI- .

. PRD

) - - ,(.

(PRD) . : PRD- AI- — ,

, AI- .

PRD , :) ,(’)

,( ) ,( ) ,( )

( ) .(

```
# PRD
##
                    :          ,     ,                .
##
.1      :   ,   )  /  (,    ,   ,
.2       :   ,    ,   ,   ,   -
.3      :      ,
.4      :
.5       :     -CSV
##
- Frontend: React + TypeScript + Tailwind CSS
-    : localStorage (    (
-    : Chart.js
- Build: Vite
##
Record {
  id: string
  amount: number
  type: 'income' | 'expense'
```

```
  category: string
  note: string
  date: string  // YYYY-MM-DD
}
##
Step 1:         )CRUD)
Step 2:
Step 3:
Step 4:
Step 5:
```

PRD : , . ,
. PRD ; ,AI- .

. :

— . : “
” .

: .

- Who: .
- What: , .
- Why: .
- How: , .
- Constraint: , .

: .

.1 Step 1: ” ” ,) , ;(

.2 Step 2: ” ” ,) ;(

.3 Step 3: ” ”) GitHub);

4. Step 4: ” ”) , .(

: PRD . — .PRD- .md
, .AI-

: . AI- :

```
                    :          .
                ,          )  (.
        -localStorage.
```

```
-React + TypeScript.
        .
```

:) ,(’ ;
(React, TypeScript, localStorage); AI- “ ” “ ” ,
.AI- AI , .

: ” ”

, 5 : Who / What / Why / How / Constraint —
. 3-5- “ ” . : , “ ”
“ ” .

- : Who / What / Why / How / Constraint —
- : ” ”
- PRD- : , , ’ , , ,
- = + + ” ” ” ”
- : PRD ,

:41 AI 2026-

, : “ ” .2026 —
. “ ” , “ ” .
, .

.

AI 2026- , . :

AI IDE-	Cursor 3, Trae, Windsurf, Code-Buddy	AI	, ,
Agent	Claude Code, Codex CLI, Aider	, AI-	, , CI/CD
Builder	v0, Bolt, Replit Agent		, ,

AI

“ ” “ ” . :

- ? —Builder ; —AI ;IDE- —Agent .
- ? —Builder ; —AI ;IDE- —Agent .
- ? —Builder ; —AI ;IDE- —Agent .

. AI- IDE-

AI IDE- : AI , AI .
:2026- Cursor 3, Trae, Windsurf .CodeBuddy-

Cursor 3

Cursor IDE- AI . Cursor 3 2026
AI: Agent Tab- — “AI ” “AI ”
. Cursor 3: AI-) - ,( )
(AI-) .(

Trae

Trae IDE AI ByteDance, . SOLO-)
Builder) “ ” — , AI-
. Trae: , .

Windsurf) Codeium) , “ ” . Tencent
Cloud CodeBuddy, VS- Code, Tencent-
Cloud - WeChat. Tongyi Lingma Alibaba Baidu- Comate
,IDE- .

. Agent-

Agent .2026- : AI- ; ,
, — , , . Claude Code .

Claude Code

Claude Code AI Anthropic, 2026 2.5
. IDE- : , Claude- Code
, , , — .
Claude Code “ ” : ,
. CLAUDE.md, AI- .
, , Agent IDE.

Codex CLI Aider-

Codex CLI OpenAI , /import — AI-
. Aider AI , (GPT, Claude, DeepSeek ,(
. , : AI- .

. Builder-

Builder . — , , AI-
.

- v0) Vercel): UI - , .
- Bolt: - , , - - .
- Replit Agent: IDE- Replit, .

“ ” — . : ,
IDE- . “0- 1- ”: Builder ,
IDE- .

• ,

AI- : , , .
, . , .
2024- :2026 2024- GitHub Copilot ; 2025- Cursor ; 2026-
Claude Code . , “ ”, :
. , .
:) ,(; ,
; — Cursor ,
Claude Code , v0- . , .

: ” ”

) , , :(Bolt v0 ;
Cursor- ;Trae- Claude Code . :
?

- : AI IDE-) ,(Agent) ,(Builder) (
- : ? ? ?
- Cursor- 3 Agent ;Tab- Claude Code 2.5-
- Builder 0- ” ;”1- IDE-
- : ,

:42
API

“ AI” ; : “ AI ? ? ” .
2026 — *GPT, Claude ,Gemini- DeepSeek, Qwen (Tongyi Qianwen), GLM, Kimi .Doubao-* : ,
. , *API,*
Token-) .(

. :

2026 : OpenAI) GPT), Anthropic) Claude) Google-) Gemini). — .

OpenAI GPT-5.6: -

OpenAI , GPT . GPT-5.6 2026 : GPT-5.6 Sol , GPT-5.6 Terra , GPT-5.6 Luna . 1.05 Token, ,
.

		) To-ken)	**) To-ken)**	
GPT-5.6 Sol		5.00 36-) (	30.00 217-) (	1.05 Token
GPT-5.6 Terra		2.50 18-) (	15.00 109-) (	1.05 Token
GPT-5.6 Luna		1.00 7-) (	6.00 44-) (	1.05 Token

OpenAI GPT-5.6) (2026

GPT “ ” —, , , - .
OpenAI , . “ ” , GPT -

. : , “ ”, API-
. -

2026 OpenAI GPT-5.6- Luna, ’
. : GPT , API.

Anthropic Claude:

GPT , - Claude “ ” . Claude Anthropic
SWE-bench-) ’ ,(— .
, Claude) (—1%- —
.
Claude : Token, “ ”
. . Claude Code, AI ,
2026 2.5 — Claude.

		) **Token)**	) **Token)**	
Claude Fable 5		10.00 72-) (	50.00 362-) (	1 Token
Claude Opus 4.8		5.00 36-) (	25.00 181-) (	1 Token
Claude Sonnet 5		2.00 14-) (	10.00 72-) (	1 Token

Anthropic Claude) ;2026 Sonnet- 5 (

Claude — Fable 5 : 50 Token.
-) (. , Sonnet 5 .

Google Gemini: -

Gemini Google 2026- 3.1 Pro. Gemini - —
, OCR- , , .
Google ,) , ,(. Gemini 2
Token — .
Gemini “ - ”. ,
— Gemini- . ,Claude-
.

	GPT-5.6	Claude Fable 5	Gemini 3.1 Pro
(SWE-bench)	94.8 74.9%	95.0 64.3% (Pro)	92.1
API	1.05 Token	1 Token (1%-)	2 Token Google

) (2026

:

, ,GPT- Claude Gemini- .
Azure, API , .
, .

. :

2026- “ ” “ ” , “ ” .
“ ” ,2023 2026 , ,
.5%- : , .

DeepSeek V4: -

DeepSeek . 2025 DeepSeek-V3
“ GPT- ” , “ AI- ” — ,
. 2026 DeepSeek V4-Pro: MoE 1.6
, 49B- , Token — - .
DeepSeek V4 Agent- — DeepSWE 12.8- 62.7- (+390%),
NL2Repo- 38.5- .61.5- API- , Responses API API- Anthropic,
Codex Claude- Code. : API- Claude API- DeepSeek
Claude Code — , .

		) Token)	**) Token)**	
DeepSeek V4 Pro		3 US-)$0.43)	6 US-)$0.87)	1 Token
DeepSeek V4 Flash		1 US-)$0.14)	2 US-)$0.28)	1 Token

DeepSeek V4) ,2026 (

: Token + Token , DeepSeek V4 Flash 0.42- , GPT-5.6 Sol 35- , Claude- Fable 5 60- . 80 .140

: DeepSeek

6- 2026 DeepSeek “ API- ” , . ,30 14- . , DeepSeek V4 — . .

Qwen3.7 Max (Tongyi Qianwen): Alibaba Cloud

Qwen (Tongyi Qianwen) Alibaba Cloud Alibaba Cloud . Qwen3.7 Max 2026 : Dense 235B , 256K 1M. Qwen:) , 7B- 235B), , Alibaba Cloud.

	) Token)	
qwen-turbo	0.6	- , ,
qwen-long	2.0	10M
qwen3.7-flash	7.2	,
qwen3.7-plus	12.0	) + ,(
qwen3.7-max	36.0	,

Qwen –Tongyi Qianwen) (2026

Qwen —qwen-turbo- 0.6) Token (qwen3.7-max (36 Token ,(— - . Alibaba Cloud, Qwen .

GLM-5.1 Zhipu: SWE-bench-

GLM Zhipu AI .2026- GLM-5.1 SWE-bench: MoE-Dense 744B Token . : — . GLM-5.1: 6 Token 24- Token — .

Kimi K3:

Kimi Moonshot AI . Kimi K3, ,2026 2.8- . Kimi — , “ ” . Kimi 12-20-) Token ,(. -

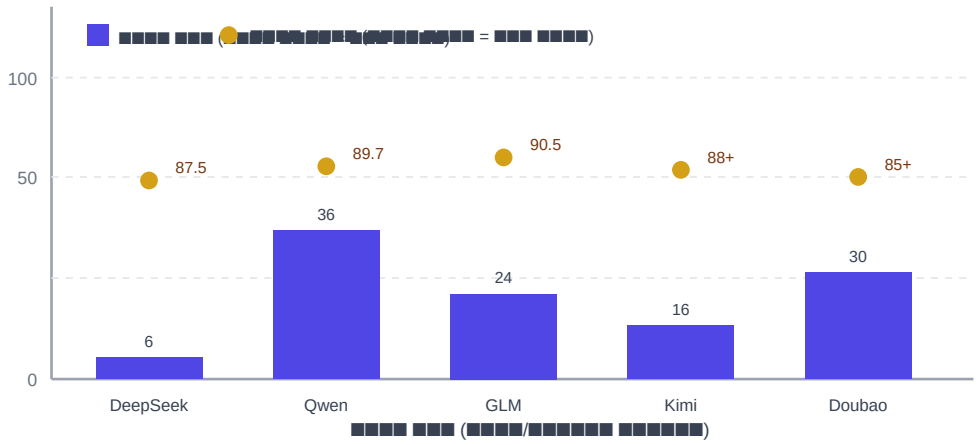
Doubao: - ByteDance

Doubao ByteDance .2026- Doubao Seed 2.1 Pro Volcano Engine, 256K , . Doubao: Token , Volcano Engine, . AI, Doubao .

	DeepSeek V4	**Qwen3.7 Max**	**GLM-5.1**	**Kimi K3**	**Doubao 2.1 Pro**
	87.5	89.7	90.5 - SWE	88+	85+
	1	256K-1M	1		256K
)	6	36	24	12-20	30
(	MIT				

) (2026

: “ ” GPT Claude- . ,
GPT-5- . , Claude —
“ ”, “ ” . ,
, .



:4) — ; — (

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— ? “ ” — ,
.

?

|

?

Claude DeepSeek

|

500<) ?(

Gemini Kimi

|

?

DeepSeek Qwen-turbo

|

GPT-5.6 Claude

:5

:

— . : DeepSeek Qwen) , ,(Claude) , ,(Gemini-) .(, “ ” .

. API:

AI , AI — — API. API (Application Programming Interface, (. ,API- .

Token

API, “Token” . Token — “ AI- ” . Token 0.75- 4) ,(0.5-1- . 1000 1500-2000- Token.

Token . Tokenizer .Token- Token- —, “ ” 3 Token GPT- 2- .DeepSeek- API- Token- , .API-

Token

1.5- Token. + DeepSeek V4 Flash 0.42- 3-) ;(Claude Fable 5 60- 434-) .(
.

API Key

API- API Key — . API ,
API Key. API- :

	API		
DeepSeek	platform.deepseek.com	API Key	
Qwen (Tongyi Qianwen)	bailian.aliyun.com) Bailian)	Alibaba Cloud Key	
GLM (Zhipu)	open.bigmodel.cn	API Key	
Kimi	platform.moonshot.cn	API Key	
Doubao	volcengine.com (Volcano Engine)	Ark API Key	Token
OpenAI GPT	platform.openai.com	API Key	) (
Anthropic Claude	console.anthropic.com	API Key	

API- Key-) (2026

: API Key =

API Key — Key- . : API Key) ;(Git Key;
Key. Key- — .

API-

API- Key, API- . Python- —
DeepSeek .

```
#     -SDK: pip install openai
# DeepSeek        OpenAI SDK,
from openai import OpenAI
import os
```

```
# 1.          )    -API            (
client = OpenAI(
    api_key=os.environ.get("DEEPSEEK_API_KEY"),
    base_url="https://api.deepseek.com"
)

# 2.
response = client.chat.completions.create(
    model="deepseek-chat",  #
    messages=[
        {"role": "system", "content": "              ."},
        {"role": "user", "content": "              ."}
    ]
)

# 3.
print(response.choices[0].message.content)

# 4.
print(f"      : {response.usage.prompt_tokens}")
print(f"      : {response.usage.completion_tokens}")
print(f"       : {response.usage.total_tokens}")
```

:) API- Key ((system , user (. OpenAI — api_key base_url. OpenAI SDK: ,

.

— , :

```
# DeepSeek
client = OpenAI(
    api_key=os.environ.get("DEEPSEEK_API_KEY"),
    base_url="https://api.deepseek.com"
)
# model="deepseek-chat"

# Qwen (     Bailian)
client = OpenAI(
    api_key=os.environ.get("DASHSCOPE_API_KEY"),
    base_url="https://dashscope.aliyuncs.com/compatible-mode/v1"
)
# model="qwen-plus"

# GLM   Zhipu
client = OpenAI(
    api_key=os.environ.get("GLM_API_KEY"),
```

```
    base_url="https://open.bigmodel.cn/api/paas/v4"
)
# model="glm-4"

# OpenAI GPT
client = OpenAI(
    api_key=os.environ.get("OPENAI_API_KEY"),
    # base_url
)
# model="gpt-5.6-terra"
```

AI- — .
(Streaming) AI- , . —
, .

```
#         :
stream = client.chat.completions.create(
    model="deepseek-chat",
    messages=[
        {"role": "user", "content": "                  "}
    ],
    stream=True  #
)

#
for chunk in stream:
    if chunk.choices[0].delta.content is not None:
        print(chunk.choices[0].delta.content, end="", flush=True)
#              ,
```

. Token Plan

: (Pay-as-you-go) (Token Plan / Subscription).
.

	(Pay-as-you-go)	**(Subscription)**
	Token- , , API-	, 50%-20%- , OpenAI Plus/Pro, Claude Pro, Cursor Pro

— , .)
50 ,(.

ChatGPT Free			
ChatGPT Plus	US$20	GPT-5.6 Luna GPT-5.6 Sol, , ,	
ChatGPT Pro	US$200	,	
Claude Free		Claude	
Claude Pro	US$20	,Claude-	
Cursor Free		AI	
Cursor Pro	US$20	AI , 500	
Cursor Business	US$40	, ,	

AI) (2026

. DeepSeek —
. Qwen Bailian) Token .(
Doubao- Volcano- Engine . , : DeepSeek
) (+ Doubao) .(

API

, API . API Key
, — .

OpenRouter (openrouter.ai)	— Key	,	
SiliconFlow (siliconflow.cn)	— API	30%-20%-	,
Azure OpenAI) Microsoft)	API- OpenAI Microsoft Azure	Batch API 50%	OpenAI ,
Baidu Qianfan (qianfan.baidubce.com)	AI- Baidu, ERNIE (Wenxin)		Baidu

API-) (2026

: "Key "

"API Key " Taobao Xianyu —Key ,
, . API Key .

; " " . :

.1 . " " —) ,(
. DeepSeek- 1/60 .
.

.2 — . : DeepSeek V4 Flash) 2-
Token), Claude-) 181- Token). 80%
.

.3 . API- Token — , . ,
max_tokens AI- " " .

.4 Batch- API .)
,(Batch API .50% OpenAI Azure .

.5 .Token- API- . ,
Token . Prompt) " " " 3- "
(50% .Token-

Batch API Prompt	98% 50%-80% 30%-60% 50% 20%-50%		max_tokens Prompt-

) (2026

. : API-

— “ AI” : , API-
. API: , API, .

```
# ai_summarizer.py -           AI
import os
from openai import OpenAI

# 1.     )    -API           ,            (
client = OpenAI(
    api_key=os.environ.get("DEEPSEEK_API_KEY"),
    base_url="https://api.deepseek.com"
)

def summarize(text: str, max_length: int = 200) -> dict:
    """

    Args:
        text:
        max_length:           )      (

    Returns:
        dict:
    """
    try:
        response = client.chat.completions.create(
            model="deepseek-chat",
            messages=[
                {
                    "role": "system",
                    "content": "          ."
                               "                          ."
                               "           3    ."
```

```
                },
                {
                    "role": "user",
                    "content": f"          \:n\n{text}"
                }
            ],
            max_tokens=max_length,  #
            temperature=0.3  #          =
        )

        return {
            "success": True,
            "summary": response.choices[0].message.content,
            "input_tokens": response.usage.prompt_tokens,
            "output_tokens": response.usage.completion_tokens,
            "total_tokens": response.usage.total_tokens
        }

    except Exception as e:
        return {
            "success": False,
            "error": str(e)
        }

# 2.
if __name__ == "__main__":
    long_text = """
    (           (...
    """

    result = summarize(long_text)

    if result["success"]:
        print(f"   : {result['summary']}")
        print(f"        : {result['total_tokens']}")
        #       : DeepSeek V4 Flash
        cost = result['total_tokens'] * 0.000004  # $\approx$ 0.4       1000
        print(f"       : {cost:.4f}   ")
    else:
        print(f"   : {result['error']}")
```

API: API- Key ;
; max_tokens ; temperature- (0.3) ;
; Token- . AI- , —
.

: ,” ” .” ” ,

. API- –” ” ;

.

: API-

) DeepSeek Zhipu — ,(API Key,
) ,(! “ API- ” . AI-
— “ AI- ” “ AI” .

- : GPT-5.6) ,(- Claude) + 1% ,(Gemini) + 2 (
- : DeepSeek V4) , - ,(Qwen3.7) ,(GLM-5.1) ,(SWE- Kimi K3) ,(Doubao) (
- : ? ? ? ?
- Token : 1.5-2- Token; 1.5- Token
- API Key = : , Git-
- OpenAI- SDK: api_key base- _url
- : , , , Batch API,

:43 —
AI

AI. : “ ” “ ,” — *AI*
, “ ” . : . ,
“ ” .*AI-*

. Prompt

AI- , . , Prompt
.

	AI-	“ React, - TypeScript” “ ” AI- “ / — ”	

Prompt

Prompt-) CLAUDE.md), .
— : “ ”, “ ” “ ” .
— AI- , .

:

“ + + ” . : “ 25
console.log-) ,( ) ,(
handleError) (” . “ ” .

. : AI-

AI. ,bug- AI-
. :

1. AI- “ ” , . AI- @ ,
, “ ” . AI- .

.2 “ ” “ ” . “ ,( )
( )” — “ ” .

3. . “ , , ” — AI-
, .

.4 . “ , 200 ” — AI-
.

```
,
```

```
. : , /home :
, : 1. admin 123456 2. “
” .3 : - , 200 - onClick -
LoginButton : @src/components/LoginButton.tsx @src/pages/Login.tsx
```

. :

AI , . .
“ ” ’ . : AI- ,
. : , ; : , . :
’ , , .AI-
“ ” . : , AI-
. : —AI- ; :
— AI- , . : , , .

	’ bug 5-10) (’ , ’ ,	1-3) (, ,

. , : “ ” AI- ,
“ ” . .

•

, AI- “ ” .
.
, “ ” . , . “:
,localStorage- ” — , AI- .
, “ - ” . , AI- “ ” ,
. .
, “ ” . . , ,
. .
, . AI- “ ” “ ” —
AI- , : . .

: “ ” AI-

bug) .(, “ ” , AI-
: , , . AI- —
“ ” .

- Prompt:) ,( ) + + ,( ) +
+ (
- : , , ,
- : , ’ ; : ,
- : , , - ,

:44 0- 1- —

— . “ ”, “ , ,
” . .

.

. :1) (

: , 3-4- . - , .
, “ ” .
: Who — . What — . Why —
. How — HTML+CSS+JS, . Constraint — 30 , .

Think Before Coding, AI- ,
:

```
                    :
1. HTML       )CSS  -JS     (,
.2          ,
.3   4           :
   -
   -
   -
   -        QR
4.           ,
.5
.6           ,
.7
             -HTML    .
```

AI- , . — :
? 4 ? ? — : ?
? — : ? CSS- style ?
?

) ,(—
, , . “ ” .

. :2) (

PRD-

: , , - - . ,
.40

PRD. 3 ,40 , , ’ , .
Simplicity First — React Redux ;
Node.js ; - SQLite PostgreSQL) .(

5- : 1 — API) Express + SQLite + CRUD); 2 —
) + ;(3 — -) API + ;(4 — ; 5
— . “ ” .

, . — AI-
.

```
#    :      full-stack
##
                                  ,              .
##
- Frontend: React + TypeScript + Vite
- Backend: Node.js + Express + TypeScript
-       : SQLite (   better-sqlite3)
-
##     API
-       : /api
-        : { success: boolean, data: any, error: string }
-     RESTful
##
Record { id, amount, type, category, note, date }
##
-    Strict  -TypeScript
-      2
```

```
-            JSDoc
-          commit
```

, . :

```
            -API   .
   :
1.   Express,    3001
2.       SQLite,       records
3.       records:
   id (TEXT PRIMARY KEY)
   amount (REAL)
   type (TEXT: 'income'|'expense')
   category (TEXT)
   note (TEXT)
   date (TEXT: YYYY-MM-DD)
4.       CRUD:
   GET /api/records -
   POST /api/records -
   PUT /api/records/:id -
   DELETE /api/records/:id -
.5
                  .
```

:) (;
) AI- ,(;AI- Goal-Driven Execution — “ ”
“ ” .

AI- , : curl Postman . ,
2) .(“ - - ” .

. :3 AI) (

: API- — “ AI” : , AI- ,
. : ,AI- API,
Prompt .

: “ AI- AI- ” , .Prompt-
Prompt- API- —,
.

:

- API: — ;
- : , ;
- Token: Token, ;
- Prompt: Prompt , .

AI- , “ ” — ?
? Token? “ ” AI- .

. - ’

, - ’ AI. ,
.

1.		Think Before Coding
2.		Simplicity First
3. PRD		Goal-Driven
4.	CLAUDE.md	Harness Engineering
5.	: - -	Surgical Changes
6.		Iterate & Verify
7.	,	Iterate & Verify
8.		Iterate & Verify

- ’ AI

: “ 1”

“ ” 1 44 AI- : 5 , AI-
) ,(. 30 — AI- .
— “ ” “ ” .

- 1 :( ) — , 30
- 2 :( ) —PRD + + 5- , “ ”
- 3) :(AI — AI ,AI- Prompt

- : PRD
- “ - - ”

:45 — AI-

AI : “ ” — , , . “ ” “ ” . *AI* , . *AI* — , — “ ” “ ” .

. : - AI-

AI. : AI- — , “ ” .

- AI : AI- ; , . , . AI- . : , — , .

```
                calculate_discount.
              :
1.       :
   - 100       20%    -> 80
   - 200       50%    -> 100
2.      :
   - 0       20%    -> 0
   - 100           -> 100 (     (
   - 100            -> 100 (     (
3.        :
   -         ->
   -         < 1 ->
   -           ->
          Vitest                .
```

: ,) , (AI- “ ” . AI “ ” ; .

•

, AI- . - —
. AI :

	, , SQL Injection , , N+1 , ,	AI , AI catch console.log AI O(n²) AI AI

AI

— AI- . : “ , , . ” AI , ,
.

. : Git AI

Git AI . AI ,
“ ” — .

Git AI “commit , ” .
— commit. , AI , .
“ ” . AI) ,(.
— ; — , . .

.1 AI,) commit). , AI ,
git checkout .

2. AI, — commit. commit- , ”feat:
.”

.3 , : git checkout -b refactor/auth-system. ,
.

4. .diff- AI, git- diff . —
.

•

AI , . .

:

“ AI- , ? ” — . AI- — ,
, — “ ” . . :
, “ ” .

:

“ AI- , , ? ” — . AI “ ” :
, . , , . -bug-
: Simplicity First, AI- “ , , ” .

:

“AI ? ” — . AI , “ ” .
— . : “ ” , . :
, “ ” .AI-

:

“ , AI ? ” — . , AI- .
, . : ,
.

— “ ” ; —AI “ ” ; — “ ”
; — “ ” . , : , , , .

:

1) :((1) AI- ; (2)
) / / / / ;((3) git commit-
. — .

- AI ” ” ≠ ” ” —
- AI- : — / / — ” ”
- : , , , ,
- Git ” :” commit , , , diff-

- :) ,( ) ,( ) ,( ) (

:46 —

37 44 *AI* — *?*

“ *AI*” “ *AI*” . “ *AI-* ” , “ *AI* ” .

. MCP

MCP (Model Context Protocol) 2025- ;2026- AI . : MCP AI- “ ” — , , , -API- .

MCP, AI , . MCP : MCP AI .MCP- , AI- “ ” “ ” .

, MCP : AI- . , , MCP AI- , . “AI ” Agentic Engineering.

. Skills-

AI, “ ” - “Skills” . , — , , . ,Skill- AI- .

“ ” — ,

. : , .

. Cursor Rules- Custom- Instructions, Claude Code , Replit- .Runme- : “ ” AI- .

. Agent-

,2026- Agent- . : AI , AI- . , AI , AI , AI- — ,

.

Agent- “ ” “ ” : Agent ,

. : , , .

: ” ” Agent-

Agent- — “ ” . :

AI- ;) CLAUDE.md) AI- “

” ; AI- “ ” . “ , ”

Agent- , .

•

AI . . :

, . Andrej Karpathy CLAUDE.md- ,

Anthropic OpenAI, — .

— , . .

, . “ ” — .)

API (— .

.

, . - —

. : ? — .

, . , , . — “

” . .

: ” ”

) .md — ,(AI —

“ ” . , — . , ,

.

- MCP AI : AI- , -API-
- / Skills :” ”
- Agent- = + ; ” , ”
- : , , ,

:47

. — .

“ ” : , , — , .

•

Think Before Coding	,	: ?
		? ?
Simplicity First		’ ,
Surgical Changes		, ”
		” AI-
Goal-Driven Execution	” ” ” ”	, AI-
Iterate & Verify	- -	: , ,

:

•

AI, . : “ ” ,

“ ” .

?

, Builder (v0/Bolt/Replit)

, AI IDE- (Cursor/Trae)

, Agent (Claude Code)

Builder Trae SOLO

:6 AI

. (FAQ)

:1 — AI?

. AI 2026 , AI- . : ,
Builder) v0 ,(Bolt- ; , — ,
“ ” AI- , .

:2 ?

. - —JavaScript/TypeScript; AI —Python; - —
Java Go. : AI, “ ” . AI ; “
” — , , . : , .

:3 AI- ?

,2026 . , AI
. , : (1) AI- ; (2)
, ; (3) AI- -
.

:4 AI ?

. AI- “ ” “ - ” .
. AI “ ” “ ” —
. “ ” “ ” .

:5 AI ?

. : — ? ? . : —
, ? . : — ? ?
? . , —
.

.2026 : ; ,
.

	CLAUDE.md- Andrej Karpathy Anthropic (Claude Code) Skills Vibe Coding Cursor - YouTube awesome-claude-code - GitHub r/cursor Reddit- MDN Web Docs (developer.mozilla.org) Python- (docs.python.org)	; 70 , Agent- AI- IDE- Claude Code Cursor - Python

. : ,

: , . ,
,2026 AI — , full-stack
. , — .
—AI ; —AI .
“ ” —AI ; —AI . —
AI ; —AI bug- .
. , “ ”
— ; .
. , AI- . . ,
– AI.

.1 Andrej Karpathy. (2025). Vibe Coding. Social media post introducing the concept of vibe-driven programming with AI.

2. Andrej Karpathy. (2025). CLAUDE.md - Project guidelines for AI-assisted development. GitHub repository, 70+K stars.

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.) 48-58)

:48 —

— “ ” “ ” . , :
? ? ?

. : “ ” “ ”

(Android) 2003- — Andy Rubin Android :
“ ” . Google ,2005- 2007- Open Handset
Alliance (OHA), 2008 —HTC Dream (G1).
: ,2026- , . ,
, — . ?
.

:
(AOSP, Android Open Source Project) — ,
. Xiaomi- MIUI) HyperOS), Huawei- HarmonyOS, One
UI. : “ ”, “ ” — .

: 1.5- (Cupcake), —Doughnut, Eclair, Frozen
Yogurt, Gingerbread... ,10 ,2019- . ,2026 Android
17, “ ” — Android 8 (API 26)
Android- 10 (API 29) — .

	/	
Android 8 (API 26)	Oreo • 2017	, —
Android 10 (API 29)	2019	(Scoped Storage) —
Android 12 (API 31)	2021	Material You —
Android 13 (API 33)	2022	— " "
Android 14-15 (API 34-35)	2023-2024	, 16KB,
Android 16-17 (API 36-37)	2025-2026	, AI) Gemini)

) ,2026 (

. : Java Kotlin?

. Java " " — ; Kotlin 2017- " " Google, 2019- "Kotlin " (Kotlin-first). ,2026- ,Kotlin- .

: , Kotlin .Java- : , Kotlin 40%-30%- - Java) — ;(, —" "(NullPointerException); , , AI) AI?) Kotlin. Java .

Java) (

```
public class MainActivity extends Activity \{
@Override
protected void onCreate(Bundle b) \{
super.onCreate(b);
setContentView(R.layout.activity);
\}
\}
```

Kotlin) (

```
class MainActivity : ComponentActivity() \{
override fun onCreate(b: Bundle?) \{
super.onCreate(b)
setContent \{ App() \}
\}
\}
```

—Kotlin- . 50 ; : “ ”
Kotlin.

. ?

“ ” . , ,
:

) + (

(APK/AAB)
) + (
/) (

:1

:

- APK: (.apk) — , .exe .Windows-
— .

- AAB: Android App Bundle (.aab) — ” ” Google- Play.
, . — .

: SDK (Software Development Kit — .(“ ”
— API , , . Activity Button-
.SDK-

. : “ ”

, 2026- — ;
, :

	(2026)	
	Kotlin 2.x Jetpack Compose + Material 3 MVVM + ViewModel Room) (+ DataStore ( ) Retrofit + OkHttp + Kotlin Coroutines Gradle + Kotlin DSL Android Studio	) (, IDE- ” - ” Google

2026-

. — “ ” ;
, . : Android- Studio, —
.

. iOS: “ ”

. , — “ ”
“ ” :

		iOS
	Windows / Mac / Linux — Android Studio ( ) — APK) (	Mac (Xcode (Xcode ( ) + 99 —1 7 ; / Test-Flight App- Store ,

iOS-) (2026

: Mac, — . Mac,
— , , .

: ” ”

[ ]] [ ) “ ” ,(“ ” “Android Version” —
. ,

. “ ” .

- ;2008- 2026- ,
- Android 17; Android- ,8-10

- :Kotlin- +30% , ,
- APK , AAB ; SDK-
- :2026 Kotlin + Compose + MVVM + Room + Retrofit + Gradle + Android Studio

:49 —Android Studio

— . “ ” : *Android Studio,* ,
, . ,
— .

. Android Studio

Android Studio (IDE) Google —, : ,
, — . :

.1 developer.android.com/studio) 2026 —
Narwhal; ;(Windows/Mac/Linux-

.2 ” ” — Android SDK ;

.3 SDK-) ; GB ;(

.4 — , ” ” .

:

Android Studio “ ”: 16GB (8GB , ,(20GB
,) VT-x/AMD-V .(BIOS- ,
“ ”) 4 (— .

.

New Project, Empty Activity) (:

Name	My First App	—
Package name	com.example.myfirstapp	; ” ” —
Minimum SDK	API 26(Android 8.0)	— +95%

Finish, Android Studio Gradle)
; .(, — “Hello Android!” .

:

(package name) “ ” — , .
com.example.xxx ;) com.< >.< .(<
com.example — .

. :

— “ ” . :

(Emulator)

. Android- Studio Device Manager Create Device,
) Pixel)) Google APIs) . Run ,
“ ” .

) (

— , , ; .
:

.1 [ ]] ,[7 ” ” (Build number) ” ;”

.2] [USB (USB debugging);

3. , ” USB”;

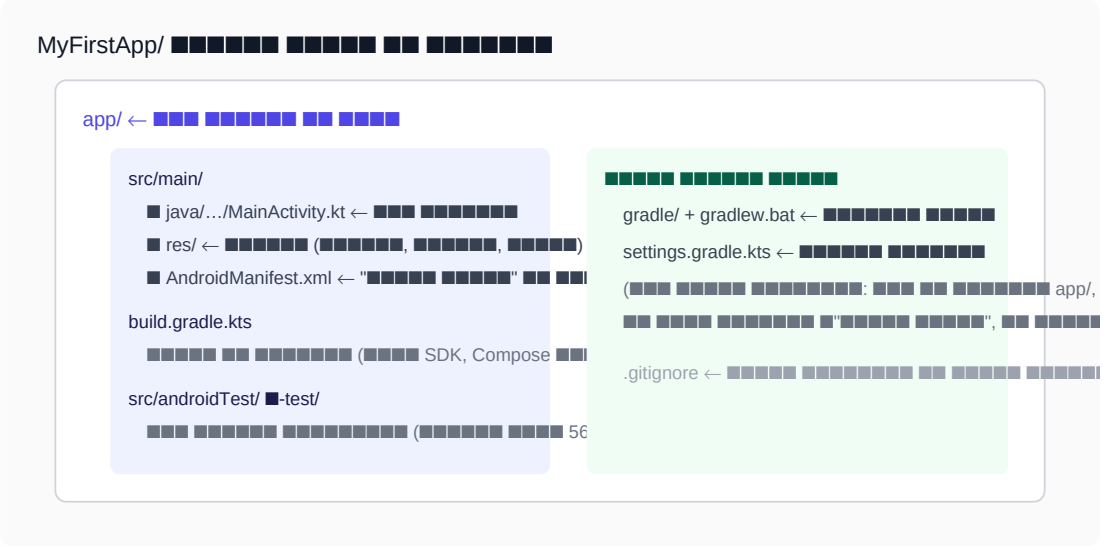
4. Android- Studio — ; Run.

: USB

USB “ ” . —
USB, . — .

. :

, . — , 90% :



:2) : app/)

- MainActivity.kt: ” ” . , .
”Hello Android!”.
- AndroidManifest.xml: ” ” — , ,
,) .(
- res/: — , , .
) .(
- build.gradle.kts: —) Compose).

. : - “Hello”

MainActivity.kt Compose . Text(“Hello Android!”
) , Text(“, ! ” ,(Ctrl+R :Mac-)
Cmd+R). .

```
setContent {
    MyFirstAppTheme {
        Surface(modifier = Modifier.fillMaxSize()) {
            // v
            Text("     !               ")
        }
    }
}
```

— “ ” . . —
: , — “ ” .

. : “ ”

, . — :

Gradle sync failed /	SDK-	Gradle-) maven Aliyun/Tencent Cloud); ”gradle mirror” Settings- SDK Manager
SDK		
/		VT-x/AMD-V ;BIOS- ;

:

— “android studio” . “ , ? ”; “Gradle sync failed: Could not resolve com.android.tools.build” . — .

: ” ”

(1) Android Studio ; (2) ; (3) Hello- ; (4) . :
) .(

- Android Studio = IDE- ; 16GB + 20GB +
- : ,) ; com.example), SDK) API- 26)
- :) ( ) USB — (
- : app/ —MainActivity.kt , Manifest , /res-
- = Run ,

:50 Kotlin —

. —*Kotlin* “ ” , : *20%-80%* . . : < , < .

. :

“ + ” . . Kotlin- :

```
val name = "  "        // val:            )          (
var age = 18             // var:             )          (

age = 19                 // <i class="fas fa-check-circle mr-1"></i> var
// name = "  "          // <i class="fas fa-times-circle mr-1"></i> val          ,


val score: Int = 100     //
val pi = 3.14            //
```

— :

Int		42
Double		3.14
String		” ”
Boolean	/	true / false
List< T>		listOf(1, 2, 3)

Kotlin-

: val

Kotlin: ,val- var- . var- val, IDE- . bug- “ ” — , .

. : “ ”

, —) (. Kotlin-
fun:

```
fun greet(name: String): String {
    return "  , $name!"     //    $
}

println(greet("  "))        //  :     !

//
fun add(a: Int, b: Int) = a + b

println(add(3, 5))           //  : 8
```

:) , , ;(;
$ Kotlin — “ + ” .Java-

. : “ ”

“ ” : (if/when) (for/while).

```
//   : if / else
val age = 17
if (age >= 18) {
    println("   ")
} else {
    println("  ")
}

//       : when (           -switch)
val score = 85
val level = when {
    score >= 90 -> "   "
    score >= 60 -> "  "
    else -> "    !"
}

//   :    for
val fruits = listOf("  ", "  ", "  ")
for (fruit in fruits) {
    println("         $fruit")
}
```

when Kotlin- ;) .(

. : “ ” Kotlin

bug- Java NullPointerException) (— “ ”
. , . Kotlin : .

```
var name: String = "  "   //       :              null
// name = null                // <i class="fas fa-times-circle mr-1"></i>
    :

var nickname: String? = null //          :     null
nickname = "  "

//                null          :
println(nickname?.length)    //       :   null     null,
val len = nickname?.length ?: 0   //      Elvis:            -null
println(len)
```

: ? ?-

String? “ ” , a- ?: b “ a , b- ”) ,
Elvis). “ null” , - — .

. : “ ”

“ ” , “ ” . —, , — .
Kotlin- data class) ,(“ ” — :Java-

```
//       :
class Dog(val name: String) {
    fun bark() = println("$name:     !")
}

val dog = Dog("Rex")      //
dog.bark()                 // Rex:     !

// data class:          )   equals/copy   '     (
data class User(val id: Int, val name: String, val email: String)

val me = User(1, "  ", "zhangsan@example.com")
val twin = me.copy(name = "  ")   //
```

, data class: data class Record(val amount: Double, val category: String, val date: String). ? .

. lambda : “ ” Kotlin

}- ... } — “ ”, lambda) .(onClick- = { count++ } Button- lambda — “ ” .

```
//       :
fun double(x: Int) = x * 2
val result = listOf(1, 2, 3).map(::double)

//    lambda:        ,
val result2 = listOf(1, 2, 3).map { it * 2 }   // it =

//    +        )               (
val bills = listOf(12.5, 88.0, 5.0, 200.0)
val bigBills = bills.filter { it > 50 }.sorted()
// [88.0, 200.0]——"       50,     "
```

“ ” : filter ,( ) map ,( ) sorted ,( ) sum .( ) 90% . Kotlin: — .

• ,

. , — :

	val x = 1 var x = 1 fun :( ) } ... } if ( ) {...} else {...} when { -> { for (x in (} ... } , ” $name” x?.length ?: 0 data class X(val a: Int, val b: String)

Kotlin) (

: 10 Kotlin-

play.kotlinlang.org) Kotlin), ,
. 10- “ ” .51

- val, var- ; ,
- = ; $
- if/when , for ; when ” - ”
- Kotlin: null; - ?- / ?:
- data class —
- 20% 80% : ,

:51 — Jetpack- Compose

. *Compose-* , *XML-* — ; *2021-* *Google* *Jetpack Compose:* *Kotlin,* ,
. *,2026-* *Compose* . “ ” .

. Compose:

Compose : = . ,) , (“ ” (Composable) .Composable@- — .

```
@Composable
fun Greeting(name: String) {
    Text(text = "   , $name!")     // "      "
}

//        :
@Composable
fun HomeScreen() {
    Column {                          //
        Greeting(name = "  ")
        Greeting(name = "   ")
        Button(onClick = { */          */ }) {
            Text("     ")
        }
    }
}
```

“ ” @Composable — “ ” . —
Compose.

. : , ,

Compose “ ” ; :

Text		text) (
Button		onClick) (
TextField		value, onValueChange) (
Image		painter) (

Compose

```
@Composable
fun BasicWidgets() {
    Column {
        Text(text = "     ")
        Button(onClick = { println("  !") }) {
            Text("   ")
        }
        var input by remember { mutableStateOf("") }
        TextField(
            value = input,
            onValueChange = { input = it },
            label = { Text("     ") }
        )
    }
}
```

remember { mutableStateOf(“ ”) } — “ ” Compose; 5 . — .

. : Column, Row, Box

— ? “ ” :

- Column:) ;(
- Row:) ;(
- Box:) ; .(

```
Column {
    Text("      ")
    Row {
        Text("  ")
        Text("  ")
    }
    Box {
```

```
        Text("        ")
        Text("        ")
    }
}
```

? Modifier ( ) — “ ” :

```
Text(
    text = "     ",
    modifier = Modifier
        .padding(16.dp)          //
        .fillMaxWidth()          //
        .background(Color.Blue)  //
        .clickable { */     */ }//
)
//       :          ,       )   padding     ;              (
```

: Modifier-

padding(16.dp).background(Blue)- background(Blue).padding(16.dp)- —
“ , ”) ,(“ , ”) .(
“ ” . ? — .

. : “ ”

. 1- , (State). Compose:
; — .

```
@Composable
fun Counter() {
    // remember + mutableStateOf:
    var count by remember { mutableStateOf(0) }

    Column {
        Text(text = "   $count    ")
        Button(onClick = { count++ }) {   //
            Text("   1+")                //
        }
    }
}
```

— . Compose : “ ” —
. : remember) (mutableStateOf-) Compose-
“ — ” .(

(count++)

Compose

(recomposition)

:3 - “ ” Compose

’ :

MyFirstAppTheme { ... } . Material 3

, . :

.1 –) ColorScheme);

2. , isSystemInDarkTheme() ;

.3) MaterialTheme.colorScheme.primary) .

: Material 3

Android 12, “Material You” . Compose

(dynamicColor). “ ” – .

’ , : “ , ” Compose

“ , ” Compose – , :

- @Preview: @Preview Composable, Android- Studio – , ;
- (Live Edit): 2026- Android Studio ” ” ;
  ;
- API : animateFloatAsState AnimatedVisibility- – ; ” ” .

```
@Composable
fun AnimatedBox() {
    var big by remember { mutableStateOf(false) }
    //        ,
    val size by animateDpAsState(if (big) 120.dp else 60.dp)

    Box(Modifier
        .size(size)
        .background(MaterialTheme.colorScheme.primary)
        .clickable { big = !big }
    )
}
```

: ,

: “ , ”. ,
– “ ”: , .

: ” ”

: , , . :
when- . – “, ” .

- Compose: = @Composable,
- : Text / Button / TextField / Image
- : Column , Row , Box
- Modifier :” ” padding, fillMaxWidth, background, clickable –
- : remember + mutableStateOf –
- Material 3 + ; –

:52 Activity

Compose “ ” , “ ” . ,
; , *Activity,* “ ” .
“ , ” .

’ :

, :

Activity		–
Service	” ”	– , ;
BroadcastReceiver		– ” ” ” ”
ContentProvider		–

95%- Activity; Service BroadcastReceiver- ;
ContentProvider) .(: “ ”
AndroidManifest.xml- – , .

’ **Activity: “ ”**

: , ,WeChat- ...
Activity :
onCreate :) (
onStart ( )
onResume ,) (
onPause) (

onStop) (

onDestroy ( )

onRestart / onStart / onResume: ,

:4 Activity) : – –; – (

Callback		
onCreate	Activity-) (	
onStart		
onResume		,
onPause		,
onStop		,
onDestroy		

Callback-

: “ ” **Activity-**

: . : Activity-)
onDestroy onCreate). : rememberSaveable) remember- ,(Compose-
. 6 .7-

’ Intent: “ ”

Intent ( ) “ ” . , – “
” :

```
//   :1 Intent     -
val intent = Intent(this, DetailActivity::class.java)
intent.putExtra("name", "  ")     //    "    "    )    (
startActivity(intent)

//   :2 Intent   -    -
val web = Intent(Intent.ACTION_VIEW, Uri.parse("https://developer.android.com"))
startActivity(web)   //              )          (
```

: – ; –)
, , .(Compose, Composable Context- LocalContext.current,
startActivity.

’

. Compose- Navigation Compose)
androidx.navigation):

```
val navController = rememberNavController()

NavHost(navController, startDestination = "home") {
    composable("home") { HomeScreen(navController) }
    composable("detail/{name}") { entry ->
        val name = entry.arguments?.getString("name")
        DetailScreen(name ?: "    ")
    }
}

//            :
navController.navigate("detail/  ")     //
navController.popBackStack()              //
```

: ?

,) AppNavGraph.kt) – “ ”
. “ ” “ ” .

’ : Fragment) (

Fragment- – “ ” Activity,
- . Compose- Composable + Navigation,
Fragment. .

: Fragment- ; , “
Activity” . 2023- –“ ” .

: ” ”

:51 (1) remember rememberSaveable- ; (2)
“ ” Navigation ; (3) (Log.d) onCreate/onPause-
Logcat. , .

- : Activity ,( ) Service ,( ) BroadcastReceiver ,( ) ContentProvider)
(
- Callback- : onCreate onStart onResume onPause onStop onDestroy
- :Activity- rememberSaveable
- Intent ” :” – ; –
- Navigation Compose: NavHost + composable

:53

“ ” ; . “
” : *DataStore*) ,(*Room*) (*Retrofit-*) ,(
“ ” –. .

’ :

DataStore	) ,	; -	54
	(		
Room	) ,	SQLite	54
	(	,	
Retrofit	) –		55
	(	HTTP	

: “ ,DataStore- ,Room- Retrofit- ”
. SharedPreferences- ,DataStore- SQL SQLite- Room- –
, .

’ DataStore:

“ ” “ ” , DataStore-) -
Preferences):

```
val Context.dataStore by preferencesDataStore(name = "settings")

//    :
suspend fun saveTheme(isDark: Boolean) {
    context.dataStore.edit { prefs ->
        prefs[THEME_KEY] = isDark
    }
}
```

```
//    :          )Flow              (
val themeFlow: Flow<Boolean> = context.dataStore.data
    .map { it[THEME_KEY] ?: false }
```

Flow – “ ” . , .
) (; .54

’ Room: “ ”

, – . Room
SQLite, “ ” Kotlin. :

```
// 1. Entity:       )     = data class   (
@Entity(tableName = "records")
data class Record(
    @PrimaryKey(autoGenerate = true) val id: Long = 0,
    val amount: Double,
    val category: String,
    val note: String,
    val date: String
)

// 2. Dao:        )   ,    ,    ,    (
@Dao
interface RecordDao {
    @Query("SELECT * FROM records ORDER BY date DESC")
    fun getAll(): Flow<List<Record>>

    @Insert
    suspend fun insert(record: Record)

    @Delete
    suspend fun delete(record: Record)

    @Query("SELECT SUM(amount) FROM records WHERE category = :c")
    suspend fun sumByCategory(c: String): Double
}

// 3. Database:
@Database(entities = [Record::class], version = 1)
abstract class AppDatabase : RoomDatabase() {
    abstract fun recordDao(): RecordDao
}
```

“ ” – . Flow,
– .

’ Retrofit:

, , ... Retrofit “ ”
. :

```
// 1.          :        HTTP
interface WeatherApi {
    @GET("v1/weather")
    suspend fun getWeather(@Query("city") city: String): WeatherResponse
}

// 2.
val api = Retrofit.Builder()
    .baseUrl("https://api.example.com/")
    .addConverterFactory(GsonConverterFactory.create())
    .build()
    .create(WeatherApi::class.java)

// 3.                )       (
val result = api.getWeather("    ")
```

:

(1) ’ – (NetworkOnMainThreadException).
(viewModelScope.launch). (2) HTTP : 9
HTTPS. http://, Manifest- usesCleartextTraffic
– .HTTPS-

’ JSON: “ ”

JSON – “ ”. Retrofit “ ” Kotlin.
:2026-

Gson / Moshi kotlinx.serialization	,Reflection- ;	,) (2026-

JSON-

```
//          @Serializable -data class
@Serializable
data class WeatherResponse(
    val weather: List<Weather>,
    val main: Main,
```

```
    val name: String
)

//     -Retrofit    -Converter     )                Retrofit 2.11+):
// .addConverterFactory(json.asConverterFactory("application/json".toMediaType()))
```

, : JSON- , . “ ”
– JSON- , .

’ : “ ”

’ : “ ”

suspend- launch- Kotlin (Coroutine) –
. : “ ”,) (.

```
//             -ViewModel
viewModelScope.launch {
    val records = dao.getAll().first()  // suspend:
    //            ...
}

//              :
// suspend   —— "        ,          "
// launch    —— "   ,     "!
// viewModelScope —— "       "          ;
```

– ; “ ” . 90%-
.

: ” ”

API) hitokoto ,(Retrofit “ ”,
“ ”) .(“ ” Room.
/DataStore/Room/Retrofit- –, 7 8- .

- : ,DataStore- ,Room- Retrofit-
- DataStore Flow: ?
- Room: Entity@ ( ) + Dao@ ( ) + Database@) (–
- :Retrofit-
- : ’ ; – HTTPS
- = ; viewModelScope

:54 1 –) (

“ ” . , : , ,
, . “ ” . – ;
.

’ :

, “ ” :

Who) (	,
What) (	, , ,
Why ( )	+
How ( )	Kotlin + Compose + Room + MVVM,
Constraint ( )	; Android- 8; –

: (1) Room-) ;((2))
;((3)) ;((4) ; (5) . , “
” .

’ : “ ”

– , :

```
@Entity(tableName = "records")
data class Record(
    @PrimaryKey(autoGenerate = true) val id: Long = 0,
    val type: String,        // "income"     | "expense"
    val amount: Double,      //    )  (
    val category: String,    //    :   ,    ,    …
    val note: String = "",   //    )  (
```

```
    val date: String          // "2026-08-15"
)
```

Dao- “ ” – :

```
@Dao
interface RecordDao {
    @Query("SELECT * FROM records ORDER BY date DESC, id DESC")
    fun getAll(): Flow<List<Record>>

    @Insert
    suspend fun insert(record: Record)

    @Delete
    suspend fun delete(record: Record)

    @Query("SELECT COALESCE(SUM(amount),0) FROM records WHERE type='income' AND
    date LIKE :month || '%'")
    suspend fun monthIncome(month: String): Double    // month      "2026-08"

    @Query("SELECT COALESCE(SUM(amount),0) FROM records WHERE type='expense' AND
    date LIKE :month || '%'")
    suspend fun monthExpense(month: String): Double
}
```

: Double?

BigDecimal- Int- “ ”)
0.1+0.2≠0.3). Double ; , –
.

’ MVVM: “ ”

? , . MVVM : View
( ) , ViewModel) (, Model (Room) .
) Compose)
ViewModel) (
Model) Room)

:5 MVVM – ,

```
class RecordViewModel(private val dao: RecordDao) : ViewModel() {

    //       -Flow
    val records = dao.getAll()
```

```
        .stateIn(viewModelScope, SharingStarted.WhileSubscribed(), emptyList())

    //        :           suspend
    fun addRecord(type: String, amount: Double, category: String, note: String,
    date: String) {
        viewModelScope.launch {
            dao.insert(Record(type = type, amount = amount,
                category = category, note = note, date = date))
        }
    }

    fun deleteRecord(record: Record) {
        viewModelScope.launch { dao.delete(record) }
    }
}
```

(Compose) : val vm: RecordViewModel = viewModel(), val list by vm.records.collectAsState() - , . MVVM + Flow + Compose.

' : +

LazyColumn-) (- , :

```
@Composable
fun RecordListScreen(vm: RecordViewModel) {
    val records by vm.records.collectAsState()

    Column {
        //             )    -     =   (
        SummaryCard(records)

        //
        LazyColumn {
            items(records, key = { it.id }) { record ->
                RecordRow(record) { vm.deleteRecord(record) }
            }
        }

        //       "      "
        FloatingActionButton(onClick = { showAddDialog = true }) {
            Text("+")
        }
    }
}
```

```
@Composable
fun RecordRow(record: Record, onDelete: () -> Unit) {
    Row(Modifier.fillMaxWidth().padding(12.dp)) {
        Text(record.category, fontWeight = FontWeight.Bold)
        Spacer(Modifier.weight(1f))
        Text(if (record.type == "income") "+${record.amount}" else "-${record.
    amount}",
             color = if (record.type == "income") Color(0xFF27AE60) else Color(0
    xFFC0392B))
    }
}
```

AlertDialog) :(- + / + ,
.(...)vm.addRecord-

:

: , “abc”) toDoubleOrNull()).
“ ” . : .

, : “ ”

. :

```
//              -Dao:
@Query("UPDATE records SET amount=:amount, category=:category, note=:note WHERE id
    =:id")
suspend fun update(id: Long, amount: Double, category: String, note: String)

@Query("DELETE FROM records WHERE id=:id")
suspend fun deleteById(id: Long)

//    :                    "   /   " ( -Compose: DropdownMenu   AlertDialog)
//                  — "       "
```

:) ,( )
.(- .

, : +

, : +

, “ ” : , ,
;( ) 0 , ;( ) ,
.( )

, “ ”:

1. :) 2026-08 v), ;
2. : Canvas- Compose , - ” ” – ” .”

:

“ ViewModel ”, . ? AI-) , “AI ” ? “ , , ” .(– , .

- ” ,”
- : Record + Dao) SQL (+ Database
- :MVVM- View , ViewModel , Model
- Flow + stateIn + collectAsState: -
- LazyColumn ; AlertDialog
- (/ /) +) , (

:55 2 –) (

“ ”; “ ” – . :
API- , JSON, “ / / ” .API-
“ ” –, , ; .

’ API-

OpenWeatherMap QWeather – , .
OpenWeatherMap:

1. Key (apikey) API keys;
2. Current Weather) :(api.openweathermap.org/data/2.5/weather?q=beijing&appid=YOUR_KEY&units=metric;
3. JSON- – ” ” API.

:
API Key “ ”; ,GitHub- .
: local.properties-) gitignore) buildConfig Gradle.
“ API Key Android- ” .

’ : Retrofit + JSON

“ ” JSON- API- ; :

```
// JSON: {"weather":[{"description":"  "}],"main":{"temp":26.5,"humidity":40},"name
    ":"    "}
data class WeatherResponse(
    val weather: List<Weather>,
    val main: Main,
    val name: String
)
data class Weather(val description: String)
data class Main(val temp: Double, val humidity: Int)
```

```
interface WeatherApi {
    @GET("data/2.5/weather")
    suspend fun getWeather(
        @Query("q") city: String,
        @Query("appid") key: String,
        @Query("units") metric: String = "metric",
        @Query("lang") lang: String = "zh_cn"
    ): WeatherResponse
}
```

Gson JSON- data- class – , .

’ : / /

“ ”. , , ...

sealed class) :(

```
sealed class UiState {
    object Loading : UiState()                    //
    data class Success(val data: WeatherResponse) : UiState()  //
    data class Error(val message: String) : UiState()          //
}

// -ViewModel:
fun search(city: String) {
    viewModelScope.launch {
        _state.value = UiState.Loading
        _state.value = try {
            UiState.Success(api.getWeather(city))
        } catch (e: Exception) {
            UiState.Error("                ,        ?")
        }
    }
}

//   :
when (val s = state) {
    is UiState.Loading -> CircularProgressIndicator()   //
    is UiState.Success -> WeatherContent(s.data)        //
    is UiState.Error -> Text(s.message)                 //
}
```

Loading/Success/Error “ - ” ; ,

.

’ :

–) :(

```
@Composable
fun WeatherContent(data: WeatherResponse) {
    Column(
        Modifier.fillMaxSize().padding(24.dp),
        horizontalAlignment = Alignment.CenterHorizontally
    ) {
        Text(data.name, fontSize = 28.sp, fontWeight = FontWeight.Bold)
        Text(
            "${data.main.temp}°C",
            fontSize = 72.sp,
            fontWeight = FontWeight.Light
        )
        Text(
            data.weather.firstOrNull()?.description ?: "    ",
            fontSize = 20.sp,
            color = Color.Gray
        )
        Spacer(Modifier.height(16.dp))
        Row {
            Text("    ${data.main.humidity}%")
            Spacer(Modifier.width(24.dp))
            Text("    ${data.main.temp}°C")
        }
    }
}
```

: ()firstOrNull) ,(:?- “ ”
– null . 72.sp , “ ” (Light)
– .

’

’

- : TextField ;) ,” ” Enter- ;(
- : ,(26.5°) ,( ) , ;
- : ,DataStore- – ;6
- : Compose- PullToRefreshBox-) Material 3), .” ”

:

emoji () when(description) - , .
, emoji .

:

(1) ; (2) -) ;(! (3)
- ; (4) - ; (5) .
- .

- API- ,JSON- - ” ”
- API Key local.properties,
- Retrofit + data class: -JSON-
- Loading/Success/Error
- , , -

:56 –, Google- Play

– ? “ ” “ ” : ,
, *Google- Play* . –
.

’ : Logcat, UI-

Logcat Layout Inspector		: Log.d(”TAG”,” =$x”), TAG- – ; Debug Tools Layout Inspector,

: ?

Logcat- (Stacktrace);)
com.example.yourapp) – . “ Exception- ”) ? ,(?
“at ... : ” . 80% .

’ :

– . “ ” “ ” .
, :

(Instru- mented)	,) (	app/src/test/ app/src/androidTest/ ——	

```
//             "             "
fun parseAmount(input: String): Double? {
    val v = input.trim().toDoubleOrNull()
    return if (v != null && v > 0) v else null
}

//       )JUnit4)
@Test
fun emptyInputReturnsNull() {
    assertEquals(null, parseAmount(""))
}
@Test
fun negativeReturnsNull() {
    assertEquals(null, parseAmount("-5"))
}
@Test
fun normalValueParses() {
    assertEquals(12.5, parseAmount("12.5"))
}
```

. : test Run. –
; .

, : **“ ”**

– “ ” . Android
Studio debug; release . :

.1 Build Generate Signed Bundle / APK;

2. keystore) (– ;

.3 release AAB – .

: keystore =

, . keystore- ,
– . keystore-) +
(– .

, Google- Play

AAB-

) (

:6 Google- Play

- : - 25 ; Google ;
- : ,(512×512)) ,(2) ;(
- : – ” ;”
- : , ; – .

: Google Play , (Huawei, Xiaomi, MyApp .(, – AAB/APK .
– .

, :

“ ” . :

- : ,512×512 . ? AI ;
  ;
- : ,2 ,4-6 ; ”) ”
  ” ;(”
- : ” + ;” ” , ” –
  , .

: A/B

, : ? ?
? – ; .

,

, :

.1 :LazyColumn- Column- – ;

.2 : Coil ; 5MB ;

.3 ’ : , – ;

.4 Profiler: Android- Studio , CPU – , ; .

. Firebase Crashlytics) , :(, – “ , , ” .

.

: ” ”

(1) 3 parseAmount; (2) Logcat- ; (3) APK release) “ ” ;((4) – . .

- : Logcat , , Layout Inspector
- : ,
- :) – / /(
- : keystore- ; – ” ”
- : AAB

:57 –,

. “ ” “ ” : ,
, , -,2026- AI.
.

, : ,

, , , ... “ ”) Android-
6). : ,(Manifest-) ,( ) – . Compose-
rememberPermissionState Accompanist:

```
//          -Manifest:
// <uses-permission android:name="android.permission.CAMERA" />

@Composable
fun CameraButton() {
    val permission = rememberPermissionState(
        permission = android.Manifest.permission.CAMERA
    )
    Button(onClick = {
        when {
            permission.status.isGranted -> openCamera()
            permission.status.shouldShowRationale -> showExplainDialog()
            else -> permission.launchPermissionRequest()
        }
    }) { Text("  ") }
}
```

: ?

“ ” , launchPermissionRequest- .
) Intent) “ ” .
– .

’ :

, “ ”. : ,
, . :

```
// -Android 13+                POST_NOTIFICATIONS (          (

// 1.            )    -Android 8+)
val channel = NotificationChannel(
    "reminder", "    ", NotificationManager.IMPORTANCE_DEFAULT
)
val manager = getSystemService(NotificationManager::class.java)
manager.createNotificationChannel(channel)

// 2.
val notification = NotificationCompat.Builder(this, "reminder")
    .setSmallIcon(R.drawable.ic_notification)
    .setContentTitle("           ")
    .setContentText("             ")
    .setPriority(NotificationCompat.PRIORITY_DEFAULT)
    .build()
if (ActivityCompat.checkSelfPermission(this,
        Manifest.permission.POST_NOTIFICATIONS) == PackageManager.
    PERMISSION_GRANTED
) {
    manager.notify(1, notification)
}
```

) “ 20:00- ”) WorkManager - “ ”
, . - .

’ :

, “ ” :

- : API- - Hilt) :(”
- ;”
- : - (app / core / feature-xxx),
.

:

Hilt , . + .
, - . ,
- “ ” AI.

’ :2026 AI

2026- AI- . - :

	AI	
	Gemini Android- Studio AI AI- AI- ” / / ” + , AI-	

AI

- AI- , - 11- ,
“AI ”) .(37-47 : , AI - .

’ Compose-

Compose , “ ” . :

- : @Immutable ,derivedStateOf-
;
- key: ,items(list key = { it.id }) - key,
;
- Compose Layout- Inspector: Composable;
, .

, Compose- (Metrics) “ ” - ,
.

: ” ”

(1) “ 20:00- ” (WorkManager + + Android 13); (2) API- Hilt; (3) AI “ ” , . , .

- : Manifest- + +
- : , , ; + -
- - WorkManager;

- Hilt , –
- AI: , , , ; –) 37-47)

:58

. – .

“ ” : , , – .

’

/	var x by remember { mutableStateOf(initialValue) }
	rememberSaveable { mutableStateOf(initialValue) }
	`Column { }/Row { }`
	LazyColumn { items(list) { ... } }
	navController.navigate(”route”)
	`viewModelScope.launch { ...}`
	`val list by flow.collectAsState()`
	context.dataStore.edit { it[KEY] = value }
	Room: Entity@ / @Dao / @Database
	Retrofit + suspend
	NotificationChannel + NotificationCompat.Builder
	`Log.d("TAG", "message")`

) (2026

’ **(FAQ)**

Q1: – ?

: 1-2 Kotlin; 3-4 Compose ; 5-6 ; 7-8 . – .

1-2 , .

Q2: Java?

. , Kotlin. Java “ ” .
Kotlin, – Java ; .

Q3: ?

) , ,(–, , , ;
. , .

Q4: ?

:2026- “ ” , “ Kotlin + +
” . , .

Q5: – ?

: (1) StackOverflow- ; (2)
; (3) – “ + + + ” AI-
. .

,

	developer.android.com Kotlin (kotlinlang.org) Google Codelabs Stack Overflow) android) Juejin- / CSDN Google Developers ”AI ”) -37 47)	; - Compose/Kotlin, + ; – ;) (AI

) (2026

’ : Gradle

,) AI CI):

`./gradlew assembleDebug`	APK Windows-) (/.
`./gradlew assembleRelease`	) (
`./gradlew test`	
`./gradlew lint`	:
`./gradlew clean`	) ” (”
`./gradlew :app: dependencies`	) (

Gradle

: , clean-

- “ ” . ./gradlew clean . — Android Studio, File Invalidate Caches.

: N

: N

: “ Activity”, , Kotlin, Compose, , , — .

: , , AI- “ ”.

: . , . ,

.

” .” . ,

– .

.1 Android Developers. (2026). Android Developer Guides and Jetpack Compose Documentation. developer.android.com.

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5. AndroidX Release Notes. (2026). Room / DataStore / Navigation / WorkManager releases.

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. iOS) 59-69)

:59 iOS —

— “ *iPhone-* ” “ *iPhone-* ” . ,
: iOS ? “ ” ? ,
?

iOS:

2007 iPhone- . — 2008-
App- Store iPhone SDK .iPhone- “
” : App- Store , .
: iPhone OS, 2010- ;iOS- iOS 17 (2023),
2025- —iOS 26 ,2025- 2026 iOS 27.
, : iOS- —
iOS- .Android-

App- Store	2008	
Swift	2014	iOS
SwiftUI	2019	—
iOS 26) (	2025	; Apple Intelligence
iOS 27	2026	) (

iOS) (2026

: “ ” “ ”

“ ” (Walled Garden) — :
, Xcode, App- Store. ?
:

- : iPhone, — 5- 95% Android-) ;(
- : 70–80% , API- ” ;”
- : App- Store ;
- , : , .

:

iOS- : (Xcode ,(macOS- 99-) , .(— : , .60

: Objective-C Swift?

iOS “ ”. Objective-C — 2008- ;2014 Swift ,2014- Swift 6) 2024- .(,2026- .Swift-

: Swift, Objective-C — . Swift:) - C), Optional- “ ”, SwiftUI, Swift Playgrounds iPad- .

Objective-C) (

```
NSString *name = @"  "; [name stringByAppendingString:@"  "]; //          ,       -
      :
```

Swift) (

```
var name = "  " name += "  " //
```

?

, iOS . “ ”:

) + (

(Swift + SwiftUI)

) + (

Apple

App- Store

:1 iOS) “ ” (

:

- (Signing): iOS ” ” — ” .”
  Xcode Apple- ID ; . .67
- (Simulator): iPhone , .
  — Android.
- TestFlight: ” ” — 10,000- ,
  TestFlight. iOS.

iOS: “ ”

	(2026)	
	Swift 6 SwiftUI + (HIG) MVVM + @Observable SwiftData) (+ UserDefaults ( ) URLSession + async/await + Codable Xcode Build System / xcode- build Xcode) (macOS-	) (

iOS- 2026-

? Android — “ ” . ,
: — .

: ” ”

iPhone- [ ] [ ] ,[ ] : ,iOS- . “ ”
App- Store — , “ ” “ ” .

- 2008- App- Store ” ;” 2026- iOS 27
- : , , ,
- : ; 99- ;
- Swift; Objective-C

- Android: ; TestFlight
- :2026- Swift 6 + SwiftUI + MVVM + SwiftData + URLSession + Xcode

:60 —Xcode

, . “ ” : *Xcode,* ,
iOS- , . ,
— .

: iOS- macOS. ? :) Mac mini MacBook Air ,(, “ Android- ”) .(Windows-

		( )
(RAM)	Apple Silicon M1-) (8GB 40GB (Xcode (macOS	M2 16GB 256GB

) (2026

:

Swift Playgrounds iPad- Swift) ;(.iPhone-
61 Playgrounds- , — . “ ”
— .

Xcode

Xcode , . :

.1 App- Store, ”Xcode” — GB, ;

.2) ;(

.3 Xcode Settings Locations Command- Line Tools —
.

: ” ”

Xcode macOS , . App- Store Xcode-) .(— iOS .

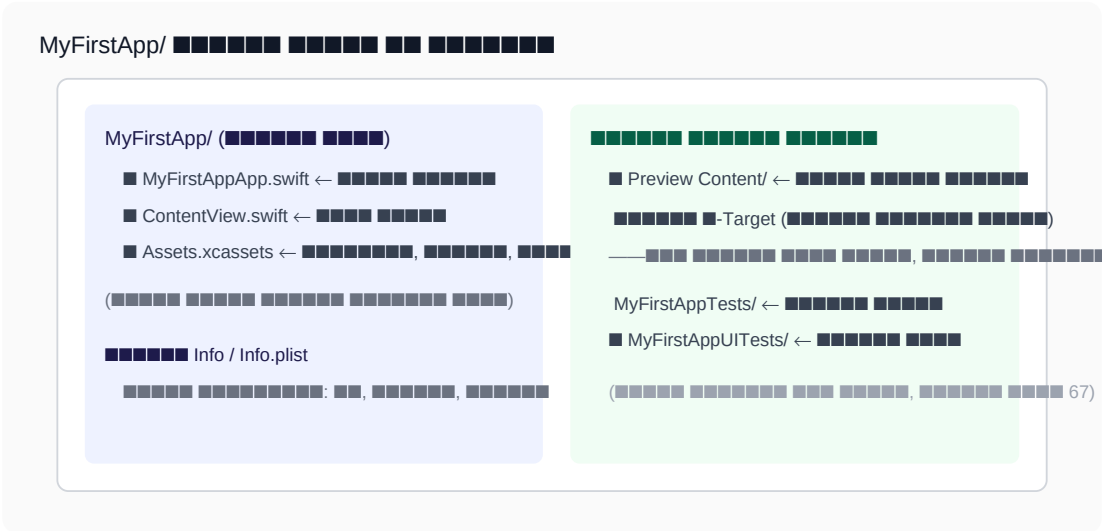
Xcode Create New Project iOS App, :

Product Name	MyFirstApp	
Team	Apple- ID ( )	;
Organization Identifier	com.example	Product- Name
Interface	SwiftUI	
Language	Swift	

Create, Xcode .) iPhone 17) <
— “Hello, world!” . iOS- .

:

; :



:2 Xcode-

- MyFirstAppApp.swift: ; SwiftUI- —, ” ;”
- ContentView.swift: ; ;

- Assets.xcassets: — , ;
- :Target- ; , iOS- — .

.

: iPhone-

:Xcode- “ ”

, “ ” :

/	CmdR OptionCmd CtrlI CmdO Cmd/	” ”) (!

Xcode-) (

:) iPhone iPad- ( )
Snippets- , (— , .

: iPhone-

, . Apple ID :

.1 iPhone-) Wi-Fi);

2. Xcode- ; Apple ID ;

.3 :iPhone- [ ] [ ]] VPN [;

.4 ,< .

: ” ”

Apple ID 7) .(,
; 99) (— .67

+

+

ContentView.swift,)Text- “Hello, world!”) CmdR —
. : “ ” .

Signing / Xcode /	Team App- Store	Target- Signing ; Device Erase All Content; ; .xip

: ” ”

(1) Xcode ; (2) ; (3) Hello- ; (4) Target- “ ” .
.

- : + Xcode; ? Swift- Playgrounds iPad-
- Xcode App- Store; —
- : , Team, , SwiftUI, Swift
- : MyFirstApp/; Target-
- Apple ID 7 ;
- = CmdR ,

:61 Swift —

. —*Swift* , *:*
20%- *80%* . . *:*
, .

:

“ + ” . . Swift- let :var-

```
let name = "  "        // let:    )          (
var age = 18             // var:     )           (

age = 19                 // <i class="fas fa-check-circle mr-1"></i> var
// name = "   "         // <i class="fas fa-times-circle mr-1"></i> let          ,


let score: Int = 100     //
let pi = 3.14            //

print("   \(name)")   //        :        +
```

:

Int Double String Bool [Int] ( )	/ 	42 3.14 ” ” true / false ,1] 2, 3]

Swift-

: let

Swift: let, var- . var- let, Xcode
. “ ” — , .

(Optional): “ ” Swift

,Swift- . : “ ” , Swift —
“ ” .

```
var name: String = "  "   //       :              nil
// name = nil                // <i class="fas fa-times-circle mr-1"></i>          :


var nickname: String? = nil  //          :     nil (   optional)
nickname = "   "

//      optional     "   "    :
if let n = nickname {        // if let:
    print("    \(n)")
} else {
    print("         ")
}

let len = nickname?.count ?? 0   // ??           )     coalescing)
print(len)
```

: ! ” ”

nickname!) (“ nil” — , .
: if- let / ?? .!- Xcode ! — .

: “ ”

, . (Closure) “ ” , :

```
//     :      ;
func greet(name: String) -> String {
    return "   , \(name)!"
}
print(greet(name: "  "))     //     ,

// closure:        ,
let numbers = [1, 2, 3, 4]
let doubled = numbers.map { $0 * 2 }   // $0 =
// [2, 4, 6, 8]

let big = numbers.filter { $0 > 2 }    //
// [3, 4]
```

SwiftUI- : Button(“ ” ({ ... } . “ ”

—map, filter, sorted — 90% .

: “ ” Swift

Swift : (struct), (class) . “ ” — , :

```
// -struct:
struct Record {
    var id = UUID()
    var amount: Double
    var category: String
    var date: Date

    // -struct   "memberwise initializer"   :
}
let r = Record(amount: 25.5, category: "  ", date: Date())

//     :      "              "
protocol Identifiable {
    var id: UUID { get }
}
struct Record: Identifiable { ... }   //           =
```

SwiftUI- struct (View .(— , - ;62 Swift struct.

/	let x = 1/var x = 1 , ” \(name)” func :) (-> } ... } if } ... } else { ... } for x in } ... } if let x = } ... } x ?? struct X { var a: Int; var b: String } { $0 * 2 }

: 15 Swift Playgrounds-

? Xcode Playground. ? Swift- Playgrounds iPad- .
, , Xcode . 15 “ ”
.62

- let var; ,
- Swift: , if let / ?? !-
- ; = , SwiftUI-
- struct (Protocol); SwiftUI- struct
- 20% 80% : ,

:62 — SwiftUI-

. *SwiftUI* *2019-* — “ ”
, , . *,2026- SwiftUI*
- . “ ” .

SwiftUI: ,

“ ”: “ (20,30)- ”, . SwiftUI “ ”:
“ ”, — .

```
import SwiftUI

struct ContentView: View {
    var body: some View {      // body:    "         "
        VStack {               //
            Text("  ,   !")
                .font(.largeTitle)      //      :
            Button("    ") {
                print("  !")
            }
            .buttonStyle(.borderedProminent)  //
        }
        .padding()
    }
}
```

: = View; body ; “ ”.
SwiftUI.

: , ,

Text
Button
TextField
Image
		Text(" ")
		Button(" (" } {
		TextField(" ", text: ($
		`Image(systemName: "star.fill")`

SwiftUI

Image(systemName:)- — SF Symbols) ,(“star.fill” . iOS: .

: VStack, HStack, ZStack

- VStack:) ;(
- HStack:) ;(
- ZStack:) ; .(

```
VStack(alignment: .leading, spacing: 12) {
    HStack {
        Text("   ")
        Spacer()                  //       :
        Text("-25.5   ").foregroundColor(.red)
    }
    Divider()                     //
    Text("  :        ")
}
.padding(16)                      //
.background(Color(.systemGray6)) //
.cornerRadius(12)                 //

//             :    padding    background;
```

:

( )padding(16).background- padding(16).( )background- — “ ”) ,(“ ”) .(“ ” . — , .

: “ ”

. SwiftUI: ; — .

```
struct CounterView: View {
    @State private var count = 0   // @State:

    var body: some View {
        VStack {
            Text("   \(count)    ")
            Button("   1+") {
                count += 1          //
            }
        }
    }
}
```

“ - ” — $ “ ” :

```
struct InputView: View {
    @State private var text = ""

    var body: some View {
        TextField("     ", text: $text)  //      -      $
            .textFieldStyle(.roundedBorder)
        Text("   : \(text)")   //
    }
}
```

(count += 1)

SwiftUI

(body (

:3 - “ ” SwiftUI

, : “ ” SwiftUI

- #Preview: #Preview { ContentView() }, — ;
- : (Color(.systemGray6), .primary) , ;
- : withAnimation { count += 1 } .animation(.default, value: count) — .

```
struct AnimatedHeartView: View {
    @State private var liked = false

    var body: some View {
        Button {
            withAnimation(.spring(duration: 0.4)) {
                liked.toggle()
            }
        } label: {
            Image(systemName: liked ? "heart.fill" : "heart")
                .font(.system(size: 48))
                .foregroundColor(liked ? .red : .gray)
                .scaleEffect(liked ? 1.3 : 1.0)
        }
    }
}
```

: ,

“ , ” . ,
— , . : “ ” .

: ” ”

:

(/)	.padding(16) .cornerRadius(12) .background(Color(.systemGray6)) .font(.title.bold())/.font(. system(size: 20)) .foregroundColor(.red) .frame(maxWidth: .infinity) .frame(width: 100, height: 44) VStack(alignment: .leading) .opacity(0.6) .shadow(radius: 5) .onTapGesture { ...} .keyboardType(.decimalPad)

SwiftUI-

? — . : padding, , .
: , . :

) + ,(,State@- (index + 1) % 3. ,
— “, ” .

- SwiftUI : ” ,”
- = View; body ; ” ”
- : Text / Button / TextField / Image; SF Symbols
- : VStack , HStack , ZStack ; Spacer
- @State ; $ -
- #Preview ” ;” ; withAnimation

:63 ,

. “ ” : ,( )
) .(: “ ” .

“ ”

SwiftUI- .State@- “ ” :

@State @Binding @StateObject / @Observable @EnvironmentObject / .environment		(ViewModel)	, , ,

SwiftUI- (2026; @Observable (

```
//           iOS 17+:     @Observable (            (
import Observation

@Observable
class CounterModel {                 //
    var count = 0
    func increment() { count += 1 }
}

struct CounterView: View {
    @State private var model = CounterModel()   //

    var body: some View {
        VStack {
            Text("   \(model.count)    ")
            Button("    ") { model.increment() }
        }
```

```
    }
}
```

: —@State; — @Observable.
@StateObject/@ObservableObject — ; , .

NavigationStack: “ ”

iOS- 16 NavigationStack — : , : -

```
struct HomeView: View {
    var body: some View {
        NavigationStack {
            List {
                NavigationLink("       ") {
                    DetailView(name: "       ")   //
                }
                NavigationLink("       ") {
                    DetailView(name: "       ")
                }
            }
            .navigationTitle("       ")   //
        }
    }
}

struct DetailView: View {
    let name: String
    var body: some View {
        Text("   \(name)")
            .navigationTitle(name)
            .navigationBarTitleDisplayMode(.inline)
    }
}
```

NavigationLink “ ”; navigationTitle
.) “ ” (\)Environment@- .dismiss) —
90% .

: sheet TabView-

“ ” :

```
// 1. sheet       ,          )          (
struct HomeView: View {
    @State private var showAdd = false
```

```
    var body: some View {
        Button("      ") { showAdd = true }
            .sheet(isPresented: $showAdd) {
                AddRecordView()     //
            }
    }
}

// 2.           ")  "       (
struct MainTabView: View {
    var body: some View {
        TabView {
            RecordListView()
                .tabItem { Label("  ", systemImage: "book") }
            StatsView()
                .tabItem { Label("    ", systemImage: "chart.pie") }
            SettingsView()
                .tabItem { Label("   ", systemImage: "gear") }
        }
    }
}
```

: ?

: NavigationStack) ,(sheet ,) , ,(—TabView-) .(.

List: “ ” iOS

“ ” iOS- List — , , :

```
struct RecordListView: View {
    @State private var records = ["    25.5", "   6.0", "    8000+"]

    var body: some View {
        List {
            ForEach(records, id: \.self) { r in
                Text(r)
            }
            .onDelete { indexSet in          //
                records.remove(atOffsets: indexSet)
            }
        }
    }
}
```

+ NavigationLink + sheet + TabView — : “
+ ” . .

: ” ”

:

Activity- ,Android- iOS “, , ” . SwiftUI-
scenePhase:

```
@main
struct MyApp: App {
    @Environment(\.scenePhase) private var phase

    var body: some Scene {
        WindowGroup {
            ContentView()
        }
        .onChange(of: phase) { _, newPhase in
            switch newPhase {
            case .active:    print("      :          ")
            case .inactive:  print("    )              (")
            case .background: print("     :        ")
            @unknown default: break
            }
        }
    }
}
```

: , . iOS — ( )
, ; ,68 “
” .

: TabView (/ /) ,List-
(NavigationStack) “+” .sheet-
— ; , 65 .

- : @State) ,(@Binding ,( ) @Observable ,( ) .environment ( )
- @Observable; @ObservableObject
- NavigationStack ; @Environment(\.dismiss)
- , ,sheet- TabView
- List ” :”

:64

“ ” ; .
“ ” *iOS: UserDefaults*) ,(*SwiftData*) (*URLSession-*)
,(*Codable* *JSON* *async/await.*
.

:

UserDefaults	,) (	, -	;65 66
SwiftData	,) (	; @Model	65
URLSession	) —	,	66
	(		

: “ ,UserDefaults- ,SwiftData-
URLSession” . Core Data (SwiftData ,(SQLite
— .

UserDefaults:

```
//   :
UserDefaults.standard.set(true, forKey: "isDarkMode")
UserDefaults.standard.set("    ", forKey: "favoriteCity")

//   :      )   ,          (
let isDark = UserDefaults.standard.bool(forKey: "isDarkMode")
let city = UserDefaults.standard.string(forKey: "favoriteCity") ?? "    "

//   :
UserDefaults.standard.removeObject(forKey: "isDarkMode")
```

— SwiftData; UserDefaults-
) .(

SwiftData: “ ”

SwiftData iOS- 17, “ , , ” :
@Model , :

```
import SwiftData

// 1. @Model:                    "   "
@Model
class Record {
    var amount: Double
    var category: String
    var note: String
    var date: Date

    init(amount: Double, category: String, note: String, date: Date) {
        self.amount = amount
        self.category = category
        self.note = note
        self.date = date
    }
}

// 2.
@main
struct MyApp: App {
    var body: some Scene {
        WindowGroup {
            ContentView()
        }
        .modelContainer(for: Record.self)   //
    }
}

// 3.  -View        @Query        @Environment
struct RecordListView: View {
    @Query(sort: \Record.date, order: .reverse) var records: [Record]
    @Environment(\.modelContext) private var context

    var body: some View {
        List(records) { r in
            Text("\(r.category) \(r.amount)    ")
        }
```

```
    }

    func add(amount: Double, category: String) {
        let r = Record(amount: amount, category: category, note: "", date: Date())
        context.insert(r)     //
        try? context.save()   //
    }
}
```

“@Model + modelContainer + @Query” — .
, @Query — .

Codable: “ ” JSON-

JSON. Codable Swift “JSON ” —
, :

```
// JSON: {"name":"    ","main":{"temp":26.5},"weather":[{"description":"  "}]}
struct WeatherResponse: Codable {
    let name: String
    let main: Main
    let weather: [Weather]
}
struct Main: Codable { let temp: Double }
struct Weather: Codable { let description: String }

//     JSON
let decoder = JSONDecoder()
let data = jsonString.data(using: .utf8)!
let result = try decoder.decode(WeatherResponse.self, from: data)
print(result.weather[0].description)   // "  "
```

: JSON- , . “ ”
) “26.5” Double) — JSON- , .

URLSession + async/await:

iOS- URLSession- , async/await Swift) (— “
” “ ” :

```
struct WeatherService {
    //          : async    "          ", throws    "      "
    func fetchWeather(city: String) async throws -> WeatherResponse {
        let url = URL(string:
            "https://api.example.com/weather?city=\(city)")!
        let (data, _) = try await URLSession.shared.data(from: url)
```

```
        return try JSONDecoder().decode(WeatherResponse.self, from: data)
    }
}

//         )    .task):
.task {
    do {
        let weather = try await service.fetchWeather(city: "     ")
        //          …
    } catch {
        print("   : \(error)")
    }
}
```

:

(1) async/await, — . Swift , try await .do/catch- (2) ATS HTTP : iOS HTTPS. http:// Info.plist- (NSAppTransportSecurity), HTTPS — .

: ” ”

“ ” :

, , ,404 ...500 :

```
func friendlyMessage(_ error: Error) -> String {
    if let urlError = error as? URLError {
        switch urlError.code {
        case .notConnectedToInternet: return "    ,          ?"
        case .timedOut:               return "       ,             ?"
        case .cannotFindHost:         return "       "
        default:                      return "     : \(urlError.code)"
        }
    }
    if let d = error as? DecodingError {
        return "           )     -API     ("
    }
    return "   : \(error.localizedDescription)"
}
```

“ ” “ ” (print error- .(er- ror.localizedDescription — . API) hitokoto ,(URLSession + async/await “ ” , “ ” . “ ” SwiftData. UserDefaults/SwiftData/Codable/URLSession — , 7 8- .

- : ,UserDefaults- ,SwiftData- URLSession
- SwiftData: @Model , modelContainer , @Query context.insert-
- Codable JSON- : ,
- URLSession + async/await: ; do/catch
- : ; HTTPS (ATS)

:65 1 —) (

“ ” . , : , ,
. , “ ” . — ;
.

:

Who) (	;
What) (	, , ,
Why ( )	+
How ( )	Swift + SwiftUI + SwiftData + MVVM,
Constraint ( )	; iOS- 17) SwiftData)

“ ” : (1) SwiftData-) ;((2)
) ;((3)) ;((4) ; (5) . , “
” .

: “ ”

— , :

```
import SwiftData

@Model
class Record {
    var type: String      // "income"     | "expense"
    var amount: Double    //    )  (
    var category: String  //     :   ,    ,    ...
    var note: String      //    )  (
    var date: Date        //
```

```
    init(type: String, amount: Double, category: String, note: String = "", date:
    Date = Date()) {
        self.type = type
        self.amount = amount
        self.category = category
        self.note = note
        self.date = date
    }
}
```

: Double?

Decimal-) ( ) 0.1+0.2 ≠ 0.3). Double ; , Decimal- - “ Int ” — .

MVVM:

? , . MVVM : View ( ) , ViewModel) (, Model- (SwiftData) .

) SwiftUI)

ViewModel) @Observable)

Model) SwiftData)

:4 MVVM — ,

```
import Observation

@Observable
class RecordViewModel {
    var records: [Record] = []
    private let context: ModelContext

    init(context: ModelContext) { self.context = context }

    func load() {
        //       #Predicate   SwiftData +
        let descriptor = FetchDescriptor<Record>(
            sortBy: [SortDescriptor(\.date, order: .reverse)]
        )
        records = (try? context.fetch(descriptor)) ?? []
    }

    func add(type: String, amount: Double, category: String, note: String) {
```

```
        let r = Record(type: type, amount: amount, category: category, note: note)
        context.insert(r)
        try? context.save()
        load()                          //       ,
    }

    func delete(_ record: Record) {
        context.delete(record)
        try? context.save()
        load()
    }
}
```

@State private var vm = RecordViewModel(...) vm.records-
— , .

: +

```
struct RecordListView: View {
    @State private var vm: RecordViewModel
    @State private var showAdd = false

    var body: some View {
        NavigationStack {
            List {
                Section("      ") {          //
                    SummaryRow(records: vm.records)
                }
                Section("   ") {
                    ForEach(vm.records) { r in
                        RecordRow(record: r)
                    }
                    .onDelete { idx in
                        idx.forEach { vm.delete(vm.records[$0]) }
                    }
                }
            }
            .navigationTitle("          ")
            .toolbar {
                Button { showAdd = true } label: {
                    Image(systemName: "plus")
                }
            }
            .sheet(isPresented: $showAdd) {
                AddRecordView(vm: vm)          //
            }
```

```
        }
    }
}
```

: .keyboardType(.decimalPad)) ,(/ —
(Picker), — , “ ” (...)vm.add- dismiss().

:

: , “abc” —) Double(text);
nil). “ ” . :
.

,

- : .onDelete List +)
  ;(
- :) ;(
- : ” / / ” filter + reduce :

```
func monthSummary(_ records: [Record]) -> (income: Double, expense: Double) {
    let cal = Calendar.current
    let monthRecords = records.filter {
        cal.isDate($0.date, equalTo: Date(), toGranularity: .month)
    }
    let income = monthRecords
        .filter { $0.type == "income" }
        .reduce(0) { $0 + $1.amount }
    let expense = monthRecords
        .filter { $0.type == "expense" }
        .reduce(0) { $0 + $1.amount }
    return (income, expense)
}
```

: +

, “ ”: , ,
;( ) 0 ;( ) ,
.( )

, “ ” :

1. : ” ” / Swift Charts —
; ;

.2 iCloud: iCloud SwiftData- — iPhone ,iPad- ” ” .

:

“ ViewModel ”, . ? AI-) “ AI” ? “ , ” .(, — , .

- ” ,”
- : @Model + modelContainer + FetchDescriptor
- :MVVM- View , ViewModel @Observable , SwiftData-
- : List + Section + toolbar + sheet
- (/ /) + (Swift Charts, iCloud)

:66 2 —) (

“ ”, “ ” — . :
API , JSON, — “ / / ” — *.API-*
, “ ” —, , ; .

API

OpenWeatherMap QWeather (HeFeng) — ,
. OpenWeatherMap:

1. (apikey) API keys;
2. Current- Weather: api.openweathermap.org/data/2.5/weather?q=beijing&appid= - &units=metric;
3. JSON- — ” ” API.

:
API “ ”: ,GitHub- ,
. :iOS- Config.xcconfig-) gitignore- (Info.plist.
“ iOS xcconfig” .

: URLSession + Codable

JSON- API- , :

```
// JSON: {"weather":[{"description":"  "}],"main":{"temp":26.5,"humidity":40},"name
    ":"    "}
struct WeatherResponse: Codable {
    let weather: [Weather]
    let main: Main
    let name: String
}
struct Weather: Codable { let description: String }
struct Main: Codable { let temp: Double; let humidity: Int }
```

```
struct WeatherService {
    let key: String   //    -xcconfig

    func fetchWeather(city: String) async throws -> WeatherResponse {
        var comps = URLComponents(string: "https://api.openweathermap.org/data/2.5/
    weather")!
        comps.queryItems = [
            URLQueryItem(name: "q", value: city),
            URLQueryItem(name: "appid", value: key),
            URLQueryItem(name: "units", value: "metric"),
            URLQueryItem(name: "lang", value: "zh_cn")
        ]
        let (data, _) = try await URLSession.shared.data(from: comps.url!)
        return try JSONDecoder().decode(WeatherResponse.self, from: data)
    }
}
```

: / /

“ ”. , , ...

enum + switch:

```
enum LoadState {
    case loading                  //
    case success(WeatherResponse) //
    case error(String)            //
}

@Observable
class WeatherViewModel {
    var state: LoadState = .loading
    private let service = WeatherService(key: Config.apiKey)

    func search(city: String) async {
        state = .loading
        do {
            let data = try await service.fetchWeather(city: city)
            state = .success(data)
        } catch {
            state = .error("                ,        ?")
        }
    }
}

//   :
```

```
switch vm.state {
case .loading: ProgressView("  …")
case .success(let d): WeatherContent(data: d)
case .error(let msg): Text(msg).foregroundColor(.secondary)
}
```

- loading/success/error “ - ” : ,
.

:

```
struct WeatherContent: View {
    let data: WeatherResponse

    var body: some View {
        VStack(spacing: 12) {
            Text(data.name)
                .font(.title.bold())
            Text("\(Int(data.main.temp))°C")
                .font(.system(size: 76, weight: .thin))
            Text(data.weather.first?.description ?? "    ")
                .font(.title3)
                .foregroundColor(.secondary)
            HStack(spacing: 24) {
                Label("   \(data.main.humidity)%", systemImage: "humidity")
                Label("   \(Int(data.main.temp))°C", systemImage: "thermometer")
            }
        }
        .padding()
    }
}
```

: first?) ,(??- “ ” ;
76 ; Label- “ + ” , HStack-
.

- : .searchable , ;vm.search-
- : ,UserDefaults- ;
- : .refreshable { await vm.search(...) } — ;
- : Image(systemName:) (sun.max / cloud.rain / snowflake),
.

: ” ”

OpenWeatherMap : “ ” “beijing” .
“ / ” .
/ - — .

:

: ,

, “ ” “ ” . :
.1 : String(data: data, encoding: .utf8) JSON- —
, ;

.2 : Proxyman Charles —
;

.3 : Features Network Link Conditioner) (
3G — ” .”

: ?ATS-

“App Transport Security”, http:// .
HTTPS;) NSAllowsLocalNetworking ,(Info.plist-
.

(1) ; (2) —) ;(!
(3) ; (4) ; (5) .
— “ ” .

- API- ,JSON- — ” ”
- API- xcconfig,
- Codable + URLSession async/await: , JSON-
- loading/success/error -
- , , —

:67 —,
App- Store

— ? “ ” “ ” : ,
, App- Store . —
.

: print,

print + SwiftUI +		print(” =\(x)”) Xcode ; - Preview#- + View-Hierarchy

: ?

Swift- , “Fatal error” , Xcode- .
Xcode- Window Organizer Crashes) .(, “
+ swiftui” — .

:

— . “ ” “ ” . iOS-
XCTest, :

		Target	
UI	,) (	MyFirstAppTests MyFirstAppUITests ——	

iOS-) Test Target)

```
import XCTest

//            "            "
func parseAmount(_ input: String) -> Double? {
    guard let v = Double(input.trimmingCharacters(in: .whitespaces)),
          v > 0 else { return nil }
    return v
}

final class AmountTests: XCTestCase {
    func testEmptyInputReturnsNil() {
        XCTAssertNil(parseAmount(""))
    }
    func testNegativeReturnsNil() {
        XCTAssertNil(parseAmount("-5"))
    }
    func testNormalValueParses() {
        XCTAssertEqual(parseAmount("12.5"), 12.5)
    }
}
```

: CmdU. — .

.

: **“ ”**

iOS — “ ” . :

- : Apple ID + Xcode (Personal Team) — , 7- ;
- : 99) (— .

: Xcode Product Archive ;( ) Organizer, Distribute App App Store Connect . : Generic iOS Device) (— .

: **=**

, — “ ” .) .p12 (. , “ ”) Automatically manage signing (Xcode- — .

App- Store

Apple 1-7) (

:5 App- Store

- : developer.apple.com, 99- ;
- : ,1024×1024) — ,( ) iPhone), ;
- : — ” ;”
- : 1–7 ; , - ” ” — , ;
- TestFlight:) 10,000 ,(— .

:

“ ” . :

- : ,1024×1024 . ? AI; ;
- : 2) iPhone), ; ”) ” ” ;(”
- : ” + ;” ” , ” — .

: ” ”

:

, :

. Xcode- Organizer) ;(SDK) Firebase Crashlytics).

.

.1 LazyVStack / List: VStack- — ;

.2 : AsyncImage-) (— 5MB ;

.3 : , — async/await;

4. Instruments: Xcode- (Time Profiler, (— , ; .

(1) 3 parseAmount) CmdU ;((2) ; (3) Product Archive TestFlight; (4) — . .

- : print, , SwiftUI
- XCTest: (CmdU) + UI + —
- :) 7 ;(
- : Generic iOS Device + Archive + Distribute;
- : ; TestFlight

:68 —,

. “ ” “ ” : ,
, , —2026- — iOS AI- .
.

: ,

, , , ... “ ” iOS .
: “ + + ” :

```
// 1.      -Info.plist          )      -        (!
// Privacy - Location When In Use Usage Description: "                    "

// 2.
import CoreLocation

@Observable
class LocationModel {
    private let manager = CLLocationManager()

    func request() {
        switch manager.authorizationStatus {
        case .notDetermined: manager.requestWhenInUseAuthorization()
        case .denied, .restricted: */                */
        default: /*   ,     */
        }
    }
}

// 3.               :
// UIApplication.shared.open(URL(string: UIApplication.openSettingsURLString)!)
```

: iOS- ” - ”

iOS- — . : “
”) ,(.) “

” (— .

:

— “ ”. iOS- UserNotifications,

: , , . :

```
import UserNotifications

// 1.            )        (
UNUserNotificationCenter.current()
    .requestAuthorization(options: [.alert, .sound, .badge]) { granted, _ in
        print(granted ? "  " : "  ")
    }

// 2.     "            -20:00     "
func scheduleDailyReminder() {
    let content = UNMutableNotificationContent()
    content.title = "           "
    content.body = "              "
    content.sound = .default

    var comps = DateComponents()
    comps.hour = 20
    let trigger = UNCalendarNotificationTrigger(dateMatching: comps, repeats: true)

    let request = UNNotificationRequest(
        identifier: "daily-reminder",
        content: content,
        trigger: trigger
    )
    UNUserNotificationCenter.current().add(request)
}
```

: iOS- “ ”, —

; .

: MVVM

65 ;MVVM- :

- : ViewModel- protocol — ” ”
  ; ;
- : ViewModel) ;(—;

- : ” @Observable + .” VIPER Clean Architecture — . , .

: ” ” SwiftUI

iOS “ ” : (WidgetKit)) “ ” ,((App Intents) Siri . , , — .

:2026 iOS AI

,2026- AI . — :

	AI	
	Xcode- + AI AI- AI- XCTest : ” / / ” + , AI-	

iOS AI

— ,AI- , — 11- , “ AI”) 37–47); : Swift , AI — .

: ” ”

:Swift- @MainActor Task-

Swift 6 :

- : SwiftUI @MainActor , (await MainActor.run { ... } ViewModel- ;(MainActor@-
- Task: .task { } — ;SwiftUI- Task body.

```
struct WeatherScreen: View {
    @State private var vm = WeatherViewModel()

    var body: some View {
        Group {
            switch vm.state {
```

```
            case .loading: ProgressView()
            case .success(let d): WeatherContent(data: d)
            case .error(let m): Text(m)
            }
        }
        .task {                          //
            await vm.search(city: "     ")
        }
    }
}
```

“ ” Swift 6 — ,
. — .
(1) “ 20:00- ”) + ;((2)
ViewModel- ; (3) “ ” .
, “ ” .

- : Info.plist-) — ,(,
- iOS- : ” ”
- : , , ; UserNotifications
- MVVM : ,( ) ,
- : WidgetKit- + App- Intents
- AI: , , , ;) 37–47)

:69

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“ ” : , — .

/	@State private var x =
	`@Observable class VM { var x = 0 }`
	`VStack { }/HStack { }`
	`Spacer()`
	List { ForEach( ) { ... } }
	`NavigationStack + NavigationLink`
	.sheet(isPresented: $show) { {
	`@Environment(\.dismiss) var dismiss`
	`try await URLSession.shared.`
	`data(from: url)`

JSON	`struct X: Codable+JSONDecoder().`
	`decode`
	UserDefaults.standard.set(, forKey: (” ”
	`@Model+modelContainer+@Query`
	(.primary /
	Color(.systemGray6))
	CmdU

iOS) (2026

(FAQ)

Q1: — iOS?

, . Swift Playgrounds iPad/iPhone- ;61
Xcode macOS- , . , Mac mini

, .

Q2: iOS?

: 0 (Xcode + Apple ID .(: 99- 700-) .(— . — App- Store — .

Q3: —SwiftUI UIKit?

SwiftUI. ,2026- , SwiftUI, .
UIKit , — SwiftUI- UIKit)
,(. .

Q4: iOS?

:2026- “ ” , “ Swift- + SwiftUI +
” . iOS . ,
.

Q5: — ?

, — .) , ;(
, - App- Store Connect . : ,
, - -) (/ . .

Xcode-

/	CmdR / CmdU
	OptionCmd
	CmdO
	CmdF
	CtrlI
/	Cmd/
	CmdK
	Cmd +
) , (	Refactor Rename

Xcode-

: ” ” ?

“ ” — . CmdK
. - ? Xcode) ,(— “
DerivedData” . .iOS-

	developer.apple.com SwiftUI- Tutorials Swift (swift.org) Hacking with Swift Stack Overflow) swiftui) Apple Developer) WWDC) (HIG) " AI") 37–47)	— Swif- tUI/Swift, , Paul Hud- son, " " iOS AI

iOS) (2026

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: " SwiftUI", Xcode, Swift, SwiftUI,
, , — .
: , , AI- " ".
: . , . ;
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" ." iPhone. ,
– .iOS-

.1 Apple Inc. (2026). SwiftUI Framework Documentation. developer.apple.com.

2. Apple Inc. (2026). Swift Programming Language Guide. swift.org.

3. Apple Inc. (2026). SwiftData Framework Documentation. developer.apple.com.

4. Apple Inc. (2026). App Store Review Guidelines. developer.apple.com/app-store/review.

5. Apple Inc. (2026). Human Interface Guidelines (HIG). developer.apple.com/design.

6. Apple Inc. (2026). Xcode 27 Release Notes and App Distribution Guide.

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“ ” :

- : /IMEI- .
- : , .
- **.2 ” ” :**

“ ” :

- : Apple- ID, Huawei, Xiaomi .’
- : ” ,” .
- **eSIM**: eSIM-) iPhone, ,(eSIM , SIM . eSIM ” – eSIM .
- : , . iPhone – ” ,” – .
- **.3 ” :”**

“ ” :

- : , ” ”?
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B:

- : ,WeChat- [ ] - [ ] - [ ] -] .[
- : , iPhone- [ ]- - [ ] -] .[
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	SoC / Chip	
	RAM ( )	” ”
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,	ROM ( )	” ”
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,	OLED / LCD	
. 128G, 256G –		
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OLED – ,	NFC	
; LCD – ,		
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” ”	OTA Update	
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” ” ,[ ]	Root	-
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Root	BL Lock	Bootloader
, .	NPU	
AI, -	On-device AI	AI
CPU, .		
AI ,	Qi2	
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, -	eSIM	SIM
MagSafe, .		
eSIM	LTPO	
LTPO	:’ ?	:

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- : ” .”
- : Apple –iTunes, Huawei –Huawei Phone Assistant, Xiaomi –Mi Mover.
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.2 ’ WeChat?

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1.6

- ” ,” ” ”) , + ,
72 , EXIF).
- ” ” / ” .”
- AI-) / / (.
- ”) (” .
-) VoiceOver/TalkBack, , ,AI- ,
RTT/TTY).
- ,” - - - ” 3 .
- : :1) , zRAM/swap,
, , .(
- :20 Termux — Linux Android) , , ,
, .(
- :23 GitHub- —) Git ,GitHub- ,
, Pull Request, , .(

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- :29 — , , , 20-20-20 ,
, , ” ” , .

- :30 — AirPlay / Miracast / DLNA, , Samsung DeX, CarPlay / Android Auto / HiCar / CarLife, .
- :31 — , 5- , ,) , , ,(,CapCut- , 3” .”
- :32 — 30 , 72 , , , , .

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- :27 — , , (MeeTime / Sunflower), WeChat , SOS- / / .
- :28 — , ” ,” , , , .

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- : , / , / 0.5x–2.5x, .
- : , .
- : R / , ESC .
- : , .

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,) ,(, 1.6-) ,(1.7- 5-) (6- . -

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- : Snapdragon 8 Elite Gen5 / Dimensity 9500) .(2026
- : iPhone 17 Pro, Samsung S26 Ultra, Xiaomi 17 Ultra, Huawei Pura 80 Ultra, vivo X300 Pro.
- AI: DeepSeek- R1, ChatGPT-5, 80 TOPS NPU .
- : 4,000–5,000 .
- : .2026
- : Wi-Fi 7 / 5G-A / HarmonyOS / LE Audio — .

. :) (Hero-

- : , SVG.
- : 8 ,) , - , , WeChat, , , - .(
- : ,localStorage- .
- : — ” / / / .”

1.8

“ AI”) 37–47, 11 :(Agentic- Engineering, AI, (SDD), , AI ,2026- API, , , (MCP/ (. ,) ,(. 2026 “ AI: ” .

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- 13 ” :” Qingzhuo, , , , AdGuard- — ” / / .”
- : iOS, Xiaomi/Redmi, Huawei/Honor, OPPO/realme/OnePlus, vivo/iQOO .Samsung-
- :8 , , SMS , iOS- — .13
- : - + + , .

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- 13 :) + + ,( ) : ,(/) ≠ .(
- : Xiaomi, Huawei/Honor, OPPO, vivo, iPhone .Samsung-

- 32 ” :” , , / SOS- / ; .13
- : ” .”

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- :13 (Sunflower, ToDesk) ; - AI (Trae / WorkBuddy / Dumate) ; miHoYo (HoYo Play).
- : LocalSend, Feige Chuanshu, AirDrop; KDE Connect, , .
- - (2FA): Aegis, Microsoft Authenticator, .
- TickTick : ; .

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F:

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. , . **:**) (

: , . : Huawei Mate X, Samsung Galaxy Z Fold, Xiaomi MIX Fold, vivo X Fold. : “ = + ” .

-) (: , . : Huawei Pocket, Samsung Galaxy Z Flip, OPPO Find N Flip, Xiaomi MIX Flip. : ” + .”
- **:2024-2025- 500,000- 100) 13- ,(UTG , , 10,000 5,000-6,000 .**

2024-2026: ,500,000 UTG, 4000-5000.

2. ?

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: PDF, . 7.5- PDF A4 — .

.1 / : . PPT, — .

.2 ” ”: 90- — /) ,(- , — .

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- : 230-260 , 230- ; 12-14 , " 12- "
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(Huawei, Samsung) “ ” , ,
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.7 (2025)
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: , HarmonyOS, ,
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- **Samsung Galaxy Z Fold/Flip**: , , Z Flip
. — .
- **Xiaomi MIX Fold/Flip**: , , .
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- **vivo X Fold**: , .
- **OPPO Find N**: , , .
- **Huawei Pocket**: ,
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- **: —Huawei Mate X; —Samsung Z Fold; —Xiaomi/vivo; —Huawei Pocket Samsung- Z Flip.**

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(v1.9)

- : Apple Watch, Huawei (Oura/Galaxy Ring)
- : eSIM , , ,
- : , , ,
- AI: CapCut- AI, Kling AI, AI
- : ,

, .2026 ,AI- , . , ; , .